

Hybrid Modelling through Simplified Physics and Data-Driven Correction

A Comparative Modelling Study of Lithium Diffusion in a Spherical Particle

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Executive Summary

The increasing use of high-fidelity physical models in engineering provides detailed descriptions of complex systems, but often at the cost of substantial computational effort and limited suitability for real-time prediction, optimization and control. Conversely, simplified physical models are faster and easier to interpret, but their assumptions may introduce systematic prediction errors when relevant physical mechanisms are neglected.

This work investigates whether data-driven learning can compensate for such model discrepancies while retaining the structural information provided by a simplified physical description. The selected case study concerns lithium diffusion inside a spherical electrode particle, represented through the Single Particle Model. The problem provides a controlled setting in which the effects of physical simplification can be isolated, quantified and corrected.

A conservative finite-volume formulation is first developed for solid-phase lithium diffusion with a constant diffusion coefficient. The implementation is validated through comparison with the Guo–White approximate analytical solution and through a mesh-convergence study. This preliminary model defines the numerical and physical baseline for the subsequent analysis.

The reference formulation is then extended by introducing two mechanisms deliberately omitted from the simplified model:

1. a solid-state diffusion coefficient depending on both local lithium concentration and particle temperature;
2. a lumped thermal balance accounting for current-induced heating and heat exchange with the surrounding environment.

The resulting thermo-diffusive model acts as the high-fidelity reference system. It generates synthetic concentration and temperature data under time-dependent operating conditions. The simplified and reference models therefore describe the same dominant diffusion process but differ in their constitutive, thermal and boundary physics. This controlled mismatch provides the central learning problem investigated in the report.

The first data-driven strategy consists of a Physics-Informed Neural Network constrained by the simplified constant-diffusivity equation. The PINN combines observations generated by the full reference simulator with initial, boundary and governing-equation constraints derived from the reduced model. Its purpose is not simply to reproduce the solution of the simplified PDE, but to determine whether observational data can compensate implicitly for the physics omitted from that equation.

The standalone PINN recovers the principal spatial and temporal evolution of the concentration field, but its accuracy deteriorates during the operating phases in which the mismatch between simplified and reference physics becomes dominant. In particular, the constant-flux boundary representation cannot fully describe the ramp-down and rest phases of the reference current profile. On the complete structured grid, the standalone PINN obtains a relative L^2 error of

$$\varepsilon_{L^2, \text{PINN}} = 5.0674 \times 10^{-2}.$$

This result demonstrates that simplified physics and observational data can produce a meaningful approximation of the reference process, while also showing that the imposed reduced equation introduces a persistent modelling bias.

Two complementary residual-learning strategies are subsequently introduced. In the first, a feedforward neural network learns the discrepancy between the full reference simulator and an independently executable simplified finite-volume model. In the second, the trained PINN is frozen and used as the baseline of a post-training Hybrid-PINN:

$$C_{\text{Hybrid}}(\rho, \tau) = C_{\text{PINN}}(\rho, \tau) + \widehat{\Delta C}(\rho, \tau).$$

The correction network learns the residual concentration field without modifying the parameters or physical loss of the original PINN. This modular formulation separates the physics-informed baseline from the empirical correction and avoids repeating the expensive PINN training stage whenever the residual model is updated.

The post-training correction reduces the full-field relative error to

$$\varepsilon_{L^2, \text{Hybrid}} = 9.5238 \times 10^{-4},$$

corresponding to an error reduction of

$$98.12\%.$$

The improvement is particularly significant during ramp-down and rest, where the standalone PINN remains affected by its simplified boundary representation. The corrected model also improves centre concentration, surface concentration, volume-average concentration and representative radial profiles.

Predictive accuracy, however, is not treated as sufficient evidence of physical consistency. Adding an unconstrained residual can alter properties originally imposed by the PINN. The Hybrid-PINN is therefore evaluated through dedicated diagnostics covering concentration bounds, initial-condition consistency, centre symmetry, global mass balance, full-reference surface-flux consistency, full and simplified PDE residuals, numerical regularity and derivative magnitudes.

The initial condition and centre-symmetry condition are retained exactly through the architecture of the residual correction. The corrected concentration also remains within the normalized physical interval for the considered trajectories. Global conservation and full-physics PDE consistency, however, are not guaranteed by the residual-learning objective. They improve substantially relative to the standalone PINN but remain empirical outcomes rather than exact mathematical properties. This distinction is essential: a residual model may reproduce reference observations with high accuracy without automatically inheriting all the governing constraints of the reference system.

The thermal dynamics omitted from the simplified concentration model are investigated independently through two alternative data-driven approaches: Sparse Identification of Nonlinear Dynamics with Control and a feedforward neural network. Eight reference trajectories with different peak currents and operating-phase durations are generated. Five trajectories are used for training or identification, one for validation, one for interpolation testing and one for extrapolation testing.

SINDYc searches for a sparse temperature equation in a second-order polynomial library and identifies the two-term model

$$\frac{d\theta}{dt} = -2.4685252 \times 10^{-3}\theta + 2.4685255 \times 10^{-3}I^2, \quad \theta = T - T_{\text{amb}}.$$

The two terms have a direct physical interpretation: the first represents heat rejection towards the ambient condition, while the second represents current-induced heat generation.

The FFN approximates the same mapping from temperature state and applied current to the temperature derivative without requiring a predefined candidate library. It achieves accurate interpolation, with a temperature RMSE of

$$6.9088 \times 10^{-3} \text{ K.}$$

Its extrapolative error increases to

$$7.9459 \times 10^{-2} \text{ K,}$$

although the resulting trajectory remains stable and physically plausible. SINDYc exhibits stronger generalization, obtaining extrapolative temperature RMSE

$$4.9426 \times 10^{-4} \text{ K,}$$

with only two active terms, compared with 1185 trainable FFN parameters. SINDYc is also faster in both model identification and rollout for the present scalar problem.

This result does not establish a universal superiority of sparse identification over neural networks. The reference thermal dynamics are simple, and their correct terms are explicitly contained in the adopted candidate library. Consequently, the experiment represents a particularly favourable setting for SINDYc. The FFN remains a relevant alternative when the appropriate functional library is unknown, incomplete or unable to represent more complex thermal phenomena.

Overall, the study leads to four principal conclusions:

1. simplified governing equations can provide an effective structural baseline even when they do not contain all the mechanisms present in the reference process;
2. observational data allow a PINN to compensate partially for physical misspecification, but the imposed simplified physics continues to influence the standalone prediction;
3. explicit residual learning can remove most of the remaining systematic discrepancy without retraining the original physics-informed model;
4. improved agreement with reference data does not automatically imply exact physical consistency, and residual-corrected models must therefore be accompanied by explicit conservation, boundary and PDE diagnostics.

The proposed framework should be interpreted as a controlled proof of concept rather than a complete electrochemical cell model. Nevertheless, it demonstrates a general hybrid-modelling principle: simplified physics can capture the dominant structure of a system, while data-driven components can learn the effects of unresolved or unavailable mechanisms. The balance between accuracy, physical consistency, computational cost and interpretability ultimately determines whether a sparse equation, a neural correction or a fully physics-informed formulation is the most appropriate representation.

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1 Introduction

Physics-Informed Neural Networks (PINNs) have recently emerged as a promising framework for solving ordinary and partial differential equations by embedding physical laws directly into the optimization process [Cuomo et al., 2022; Khanra et al., 2026]. Their ability to combine observational data with governing equations has motivated numerous applications in computational mechanics, fluid dynamics, heat transfer and, more recently, electrochemical battery modelling [Niraimathi et al., 2026].

Most existing works formulate PINNs using the same governing equations employed to generate the reference solution. In this setting, the objective is typically to replace a numerical solver while preserving the accuracy of the original physical model. Although this approach has demonstrated the effectiveness of PINNs as surrogate solvers, it assumes that the governing equations accurately describe the underlying physical process and are entirely available during training. When the embedded equations are misspecified, the resulting bias can substantially reduce predictive accuracy [Zou et al., 2024].

The present work investigates a different scenario that is particularly relevant in engineering practice. High-fidelity physical models are often available but are computationally expensive and contain a level of detail that may not be necessary for all applications. Instead of enforcing the complete mathematical description within the PINN, this work deliberately adopts a simplified physical model while generating the training data from a more accurate reference simulator. This combination of mechanistic structure and data-driven flexibility follows the general motivation of physics-guided and grey-box modelling [Jia et al., 2020; Tu et al., 2024].

The central research question addressed in this work is therefore:

Can a Physics-Informed Neural Network constrained by simplified governing equations accurately reproduce the behaviour of a more accurate reference model when trained on data generated by that model?

The proposed methodology first investigates this question through a standalone PINN. The network combines observational data from the reference model with simplified physical constraints, allowing it to compensate implicitly for part of the inaccuracies introduced by the reduced mathematical description. This standalone result is retained as an independently evaluated baseline in Section 16.

The discrepancy with respect to the reference solution is subsequently learned explicitly through two complementary residual-learning realizations. The first corrects an independently executable simplified finite-volume model and is presented in Sections 17–18. The second freezes the trained PINN and adds a coordinate-based feedforward correction network; this post-training Hybrid-PINN extension is presented in Section 21. In both cases, a simplified physics-based baseline captures the dominant system behaviour while a data-driven component accounts empirically for the missing physical effects.

1.1 Research Objectives

The study addresses the following research questions.

1. Can a PINN constrained by simplified governing equations accurately reproduce the solution generated by a more detailed reference model?
2. How does the use of observational data compensate for the inaccuracies introduced by the simplified physics?
3. Which physical simplifications have the largest impact on the prediction error?
4. Can the discrepancy between the simplified and reference models be explicitly learned through a data-driven correction model, both from a simplified finite-volume baseline and from a frozen PINN baseline?
5. Which physical properties are retained exactly, recovered only empirically, or lost when an unconstrained residual correction is added to the PINN prediction?

1.2 Methodological Strategy

The proposed methodology is organized as a two-stage computational principle with two complementary implementations of its corrective stage.

The central idea is to exploit a deliberately simplified physical model while using synthetic data generated by a more accurate reference model. Rather than attempting to reproduce the complete governing equations, the objective is to investigate whether a PINN constrained by simplified physics can still approximate the behaviour of the reference model.

1.2.1 Stage 1 – PINN with Simplified Physics

The first stage focuses on the development of a Physics-Informed Neural Network whose physical constraints are intentionally simplified with respect to the reference model.

The reference diffusion model is used exclusively to generate synthetic observations for training and validation. During optimization, however, the PINN is constrained only by the simplified governing equations.

The objective of this stage is to evaluate whether the combination of observational data and simplified physical constraints is sufficient to accurately reproduce the solution generated by the reference model.

1.2.2 Stage 2 – Data-Driven Model Correction

Once the simplified PINN framework has been validated, the second stage explicitly models the discrepancy between a reduced baseline and the reference solution.

Instead of expecting a single model to compensate implicitly for all missing physics, a dedicated data-driven component is trained on the residual error left by the reduced description. Treating model-form error as a learnable corrective contribution is a recurring strategy in hybrid modelling [Garg et al., 2022; Ghnatios and Chinesta, 2025].

The resulting hybrid framework combines:

- a simplified physics-based model, responsible for describing the dominant physical mechanisms;
- a data-driven correction, responsible for compensating for the missing physical effects.

The final prediction can therefore be interpreted as

$$u_{\text{hybrid}} = u_{\text{physics}} + \delta_{\text{data}},$$

where u_{physics} denotes the prediction obtained from the simplified physical model and δ_{data} represents the correction learned from the available data.

Two versions of this principle are implemented. The standalone FVM-FNN hybrid model uses a simplified numerical solver as its baseline, whereas the post-training Hybrid-PINN uses the previously trained and frozen PINN. The two models share an additive residual-learning idea but do not share the same inputs, output representation or conservation mechanism. Their formulations are therefore reported separately in Sections 17 and 21.

1.3 Reference Model

As a first case study, the physical system under investigation is a representative active material particle belonging to one electrode (either the anode or the cathode).

Lithium transport inside the particle is described by the solid-state diffusion equation. This model is adopted as the reference (ground truth) for generating synthetic data and evaluating the predictive performance of the PINN.

The objective of this work is therefore methodological rather than electrochemical. The single-particle diffusion problem provides a controlled benchmark that captures one of the fundamental transport mechanisms occurring inside lithium-ion batteries while remaining sufficiently simple to allow a systematic investigation.

The diffusion equation describes the transport of lithium ions inside the active material particle, whereas the boundary condition models the electrochemical reactions occurring at the particle surface. The intercalation flux represents the exchange of lithium ions between the electrolyte and the solid particle during charging and discharging. In this way, the proposed toy model captures the essential microscopic mechanisms governing lithium transport while avoiding the complexity of a complete battery model.

1.3.1 Simplified Physics

Rather than enforcing the complete mathematical description of the reference model, simplified versions of the governing equations are considered.

Possible simplifications include:

- assuming a constant diffusion coefficient;
- prescribing a simplified intercalation flux at the particle surface;
- neglecting secondary physical phenomena;
- simplifying the boundary conditions.

These simplified formulations intentionally introduce modelling inaccuracies that the neural network is expected to compensate for using the available data.

1.4 Research Roadmap

The study is organized into three successive stages. The first validates the numerical and PINN ingredients on benchmark diffusion problems. The second investigates a PINN constrained by deliberately simplified physics. The third introduces explicit residual-learning models and evaluates their accuracy and physical consistency against the same full-physics reference trajectory.

Stage 1 – Benchmark Validation

The first stage aims at validating the correctness of the PINN implementation before introducing battery-related physics.

The one-dimensional heat equation is adopted as a benchmark because:

- it is governed by a diffusion equation similar to that encountered in electrochemical systems;
- analytical solutions are available, allowing a rigorous validation of the PINN;
- it provides a controlled environment for verifying the implementation of automatic differentiation, PDE residuals, boundary conditions and initial conditions.

At the end of this stage, the PINN implementation is expected to accurately reproduce the analytical solution of the benchmark problem.

Stage 2 – PINN with Simplified Physics

The second stage constitutes the main methodological investigation of this internship.

Rather than considering an entire lithium-ion battery, the study focuses on lithium diffusion within a representative active material particle, one of the fundamental transport processes occurring inside battery electrodes.

Synthetic observations are generated using a high-fidelity reference diffusion model, while the PINN is trained by enforcing a deliberately simplified mathematical description of the same physical process. The simplified model retains the dominant diffusion mechanisms while neglecting or approximating selected physical effects, such as the boundary conditions, the intercalation flux, or other secondary phenomena.

The objective of this stage is to investigate whether the combination of observational data and simplified physical constraints enables the PINN to accurately reproduce the behaviour of the reference model despite the intentionally simplified governing equations.

The trained PINN constitutes the physics-informed baseline for the post-training residual correction developed in Section 21. It is frozen before the corrective network is trained, so the standalone PINN results remain unchanged and directly comparable with the corrected prediction.

Stage 3 – Residual-Learning Extensions

The third stage learns the discrepancy with respect to the thermo-diffusive reference model. A simplified FVM-FNN branch learns a complete radial correction from reduced physical states, while a second branch learns a scalar correction to the frozen PINN at each space-time coordinate. The latter produces the Hybrid-PINN studied in Section 21. Both branches are evaluated on the common structured grid and are accompanied by checks of accuracy, mass balance, physical bounds and differential consistency.

1.5 Expected Outcome

The outcome of this work is the development and validation of a hybrid modelling framework based on simplified physics, Physics-Informed Neural Networks and explicit residual learning.

More specifically, the study assesses whether a PINN constrained by a deliberately simplified physical model can reproduce the behaviour of a more detailed reference model, and whether its remaining structured error can be reduced by a separately trained feedforward correction network.

Although the investigation is carried out on a simplified particle problem, the methodology demonstrates how a frozen physics-informed baseline can be reused without retraining and augmented through a modular data-driven correction.

The residual-learning stage is implemented and evaluated in this report. The resulting comparisons distinguish dense-grid reconstruction from generalization to unseen operating trajectories and distinguish exact architectural constraints from physical properties that are assessed only a posteriori.

2 Physical Model of Lithium Diffusion in a Spherical Particle

The spherical solid-diffusion formulation and its principal validation benchmark follow the model presented by Guo and White [Guo and White, 2012].

2.1 Physical Domain and State Variable

The physical system consists of a single spherical active material particle belonging to one of the electrodes of a lithium-ion cell. The particle is assumed to have radius R_p . This representative spherical-particle description is also central to Single Particle Model formulations and to the inference of solid-state diffusivity [Gümrükçüoğlu et al., 2024].

Due to spherical symmetry, the lithium concentration depends only on the radial coordinate r and on time t . The concentration field is therefore represented as

$$c_s = c_s(r, t),$$

where

$$r \in [0, R_p], \quad t \geq 0.$$

The subscript s indicates that the concentration refers to lithium in the solid phase of the active material particle.

The concentration $c_s(r, t)$ is measured in

$$\text{mol m}^{-3}.$$

2.2 Modelling Assumptions

The following assumptions are introduced:

1. the active material particle is perfectly spherical;

2. the lithium concentration is spherically symmetric, so that no angular dependence is considered;
3. lithium transport inside the particle occurs exclusively through solid-state diffusion;
4. the solid-phase diffusion coefficient D_s is spatially uniform and constant in time;
5. the particle radius R_p remains constant;
6. electrochemical intercalation and deintercalation occur only at the external surface of the particle;
7. the molar flux at the particle surface may vary with time and is denoted by $j(t)$.

Under these assumptions, the three-dimensional diffusion problem is reduced to a one-dimensional problem along the radial coordinate.

2.3 Mass Conservation

Consider a spherical shell bounded by the radial coordinates r and $r + \Delta r$.

The volume of the shell is approximately

$$\Delta V = 4\pi r^2 \Delta r.$$

Let $N_r(r, t)$ denote the radial molar flux of lithium through the solid particle. The molar flow rate crossing a spherical surface of radius r is

$$\dot{n}(r, t) = 4\pi r^2 N_r(r, t).$$

The accumulation of lithium inside the spherical shell is equal to the difference between the incoming and outgoing molar flow rates:

$$\frac{\partial c_s}{\partial t} \Delta V = \dot{n}(r, t) - \dot{n}(r + \Delta r, t).$$

Substituting the expressions for the shell volume and molar flow rate gives

$$\frac{\partial c_s}{\partial t} 4\pi r^2 \Delta r = 4\pi r^2 N_r(r, t) - 4\pi (r + \Delta r)^2 N_r(r + \Delta r, t).$$

Dividing by $4\pi r^2 \Delta r$ and taking the limit $\Delta r \rightarrow 0$ yields the local mass conservation equation

$$\frac{\partial c_s}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_r).$$

2.4 Fick's Law

The radial molar flux inside the solid particle is described by Fick's first law:

$$N_r(r, t) = -D_s \frac{\partial c_s}{\partial r},$$

where D_s is the solid-phase diffusion coefficient, measured in

$$\text{m}^2 \text{s}^{-1}.$$

The negative sign indicates that diffusion occurs from regions of higher concentration toward regions of lower concentration.

Substituting Fick's law into the mass conservation equation gives

$$\frac{\partial c_s}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(-r^2 D_s \frac{\partial c_s}{\partial r} \right).$$

Assuming that D_s is constant, it can be taken outside the spatial derivative:

$$\boxed{\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right)}$$

for

$$0 < r < R_p, \quad t > 0.$$

This is the solid-phase diffusion equation in spherical coordinates.

2.5 Expanded Form of the Diffusion Equation

The spatial differential operator can be expanded using the product rule:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = 2r \frac{\partial c_s}{\partial r} + r^2 \frac{\partial^2 c_s}{\partial r^2}.$$

Dividing by r^2 gives

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r}.$$

The governing equation can therefore also be written as

$$\boxed{\frac{\partial c_s}{\partial t} = D_s \left(\frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r} \right)}$$

for $0 < r < R_p$.

The additional term

$$\frac{2}{r} \frac{\partial c_s}{\partial r}$$

is a consequence of the spherical geometry and would not appear in a one-dimensional Cartesian diffusion problem.

2.6 Initial Condition

The initial lithium concentration inside the particle is prescribed as

$$c_s(r, 0) = c_{s,0}(r).$$

A common simplifying assumption is a spatially uniform initial concentration:

$$\boxed{c_s(r, 0) = c_{s,0}} \quad 0 \leq r \leq R_p.$$

This means that, before the application of the external flux, lithium is uniformly distributed inside the particle.

2.7 Boundary Condition at the Particle Centre

At the centre of the spherical particle, no preferential radial direction exists. Spherical symmetry therefore requires the concentration gradient to vanish:

$$\boxed{\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0}$$

for $t > 0$.

This is a homogeneous Neumann boundary condition.

Physically, it implies that there is no net molar flux through the centre of the particle:

$$N_r(0, t) = 0.$$

2.8 Boundary Condition at the Particle Surface

At the external surface of the particle, the diffusive flux is related to the intercalation or deintercalation process.

The boundary condition is written as

$$\boxed{-D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t)}$$

where $j(t)$ is the prescribed molar flux at the particle surface, measured in

$$\text{mol m}^{-2} \text{s}^{-1}.$$

The sign convention must be defined consistently. According to the formulation adopted above:

- $j(t) > 0$ corresponds to an outward lithium flux;
- $j(t) < 0$ corresponds to an inward lithium flux.

An inward flux increases the average lithium concentration inside the particle, whereas an outward flux decreases it.

2.9 Complete Initial-Boundary Value Problem

The complete mathematical problem is therefore

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right), \quad 0 < r < R_p, \quad t > 0,$$

subject to the initial condition

$$c_s(r, 0) = c_{s,0}(r), \quad 0 \leq r \leq R_p,$$

and the boundary conditions

$$\begin{aligned} \left. \frac{\partial c_s}{\partial r} \right|_{r=0} &= 0, \\ -D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} &= j(t). \end{aligned}$$

The solution of this problem provides the spatial and temporal evolution of the lithium concentration inside the active material particle.

2.10 Quantities of Interest

Although the complete solution is the concentration field $c_s(r, t)$, two additional quantities are particularly relevant.

The surface concentration is defined as

$$c_{s,\text{surf}}(t) = c_s(R_p, t).$$

The volume-averaged concentration inside the particle is

$$\bar{c}_s(t) = \frac{1}{V_p} \int_{V_p} c_s dV,$$

where the particle volume is

$$V_p = \frac{4}{3} \pi R_p^3.$$

Using spherical symmetry,

$$\bar{c}_s(t) = \frac{3}{R_p^3} \int_0^{R_p} c_s(r, t) r^2 dr.$$

Therefore,

$$\bar{c}_s(t) = \frac{3}{R_p^3} \int_0^{R_p} c_s(r, t) r^2 dr$$

The surface concentration is important because the electrochemical reaction takes place at the particle surface, while the average concentration measures the total amount of lithium stored inside the particle.

3 Non-Dimensional Formulation

3.1 Motivation

The dimensional diffusion model involves quantities characterized by very different orders of magnitude. For example, the particle radius is typically of the order of micrometres, the solid-state diffusion coefficient may be of the order of $10^{-14} \text{ m}^2 \text{ s}^{-1}$, and the lithium concentration may be of the order of 10^4 mol m^{-3} .

Working directly with these dimensional variables may lead to poorly scaled numerical equations. This issue is particularly relevant for Physics-Informed Neural Networks, since the different terms entering the loss function may have considerably different magnitudes.

For this reason, the governing equations are rewritten in terms of dimensionless variables.

3.2 Characteristic Scales

The characteristic length scale of the problem is the particle radius

$$r_{\text{ref}} = R_p.$$

The characteristic concentration scale is chosen as the maximum lithium concentration in the solid particle,

$$c_{\text{ref}} = c_{s,\text{max}}.$$

The characteristic diffusion time is obtained by balancing the temporal and spatial terms of the diffusion equation.

Starting from

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right),$$

the temporal term scales as

$$\frac{\partial c_s}{\partial t} \sim \frac{c_{\text{ref}}}{t_{\text{ref}}},$$

while the diffusion term scales as

$$D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) \sim D_s \frac{c_{\text{ref}}}{R_p^2}.$$

Balancing the two terms gives

$$\frac{c_{\text{ref}}}{t_{\text{ref}}} = D_s \frac{c_{\text{ref}}}{R_p^2}.$$

Therefore,

$$\boxed{t_{\text{ref}} = \frac{R_p^2}{D_s}}$$

is the characteristic diffusion time of the particle.

This quantity represents the typical time required for lithium to diffuse over a distance comparable to the particle radius.

3.3 Dimensionless Variables

The following dimensionless variables are introduced:

$$\rho = \frac{r}{R_p}$$

$$\tau = \frac{D_s t}{R_p^2}$$

$$C(\rho, \tau) = \frac{c_s(r, t)}{c_{s, \max}}$$

where

$$\rho \in [0, 1], \quad \tau \geq 0, \quad 0 \leq C \leq 1.$$

The dimensional variables can consequently be written as

$$r = R_p \rho,$$

$$t = \frac{R_p^2}{D_s} \tau,$$

$$c_s(r, t) = c_{s, \max} C(\rho, \tau).$$

3.4 Transformation of the Time Derivative

Since

$$\tau = \frac{D_s t}{R_p^2},$$

the chain rule gives

$$\frac{\partial \tau}{\partial t} = \frac{D_s}{R_p^2}.$$

The dimensional concentration is

$$c_s = c_{s, \max} C.$$

Therefore,

$$\frac{\partial c_s}{\partial t} = c_{s, \max} \frac{\partial C}{\partial \tau} \frac{\partial \tau}{\partial t}.$$

Substituting the derivative of τ gives

$$\boxed{\frac{\partial c_s}{\partial t} = c_{s,\max} \frac{D_s}{R_p^2} \frac{\partial C}{\partial \tau}}$$

3.5 Transformation of the Radial Derivatives

Since

$$\rho = \frac{r}{R_p},$$

the chain rule gives

$$\frac{\partial \rho}{\partial r} = \frac{1}{R_p}.$$

The first radial derivative becomes

$$\frac{\partial c_s}{\partial r} = c_{s,\max} \frac{\partial C}{\partial \rho} \frac{\partial \rho}{\partial r}.$$

Hence,

$$\boxed{\frac{\partial c_s}{\partial r} = \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho}}$$

The quantity inside the spherical diffusion operator is therefore

$$r^2 \frac{\partial c_s}{\partial r} = (R_p \rho)^2 \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho}.$$

Simplifying,

$$r^2 \frac{\partial c_s}{\partial r} = R_p c_{s,\max} \rho^2 \frac{\partial C}{\partial \rho}.$$

Taking the derivative with respect to r ,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{1}{R_p} \frac{\partial}{\partial \rho} \left(R_p c_{s,\max} \rho^2 \frac{\partial C}{\partial \rho} \right).$$

Since R_p and $c_{s,\max}$ are constant,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = c_{s,\max} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

Finally, since

$$r^2 = R_p^2 \rho^2,$$

we obtain

$$\boxed{\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{c_{s,\max}}{R_p^2} \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)}$$

3.6 Dimensionless Governing Equation

Substituting the transformed derivatives into the dimensional equation

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right)$$

gives

$$c_{s,\max} \frac{D_s}{R_p^2} \frac{\partial C}{\partial \tau} = D_s \frac{c_{s,\max}}{R_p^2} \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

The common factor

$$\frac{D_s c_{s,\max}}{R_p^2}$$

can be cancelled from both sides, resulting in

$$\boxed{\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)}$$

for

$$0 < \rho < 1, \quad \tau > 0.$$

The diffusion coefficient and the particle radius no longer appear explicitly in the dimensionless PDE. Their influence is incorporated into the definition of the dimensionless time τ .

3.7 Expanded Dimensionless Equation

Using the product rule,

$$\frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) = 2\rho \frac{\partial C}{\partial \rho} + \rho^2 \frac{\partial^2 C}{\partial \rho^2}.$$

Therefore,

$$\boxed{\frac{\partial C}{\partial \tau} = \frac{\partial^2 C}{\partial \rho^2} + \frac{2}{\rho} \frac{\partial C}{\partial \rho}}$$

for $0 < \rho < 1$.

As in the dimensional formulation, the expanded equation appears singular at $\rho = 0$. The centre of the particle must consequently be treated using spherical symmetry and an appropriate limiting expression.

3.8 Dimensionless Initial Condition

The dimensional initial condition is

$$c_s(r, 0) = c_{s,0}(r).$$

Dividing by $c_{s,\max}$ gives

$$C(\rho, 0) = \frac{c_{s,0}(R_p \rho)}{c_{s,\max}}.$$

Defining

$$C_0(\rho) = \frac{c_{s,0}(R_p \rho)}{c_{s,\max}},$$

the dimensionless initial condition becomes

$$\boxed{C(\rho, 0) = C_0(\rho)}$$

If the initial concentration is spatially uniform,

$$c_{s,0}(r) = c_{s,0},$$

then

$$\boxed{C(\rho, 0) = C_0 = \frac{c_{s,0}}{c_{s,\max}}}$$

3.9 Dimensionless Boundary Condition at the Centre

The dimensional symmetry condition is

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0.$$

Using

$$\frac{\partial c_s}{\partial r} = \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho},$$

we obtain

$$\frac{c_{s,\max}}{R_p} \left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0.$$

Therefore,

$$\boxed{\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0}$$

3.10 Dimensionless Surface Flux

The dimensional surface condition is

$$-D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t).$$

Substituting the transformed radial derivative gives

$$-D_s \frac{c_{s,\max}}{R_p} \left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = j(t).$$

Rearranging,

$$-\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \frac{j(t)R_p}{D_s c_{s,\max}}.$$

The dimensionless surface flux is therefore defined as

$$\boxed{\delta(\tau) = \frac{j(t)R_p}{D_s c_{s,\max}}}$$

where dimensional and dimensionless time are related by

$$t = \frac{R_p^2}{D_s} \tau.$$

The surface boundary condition becomes

$$\boxed{-\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \delta(\tau)}$$

or, equivalently,

$$\boxed{\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = -\delta(\tau)}$$

3.11 Complete Dimensionless Problem

The complete dimensionless initial-boundary value problem is

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right), \quad 0 < \rho < 1, \quad \tau > 0,$$

subject to

$$C(\rho, 0) = C_0(\rho), \quad 0 \leq \rho \leq 1,$$

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0,$$

and

$$-\left.\frac{\partial C}{\partial \rho}\right|_{\rho=1} = \delta(\tau).$$

This formulation is equivalent to the dimensional model but is more suitable for numerical implementation and Physics-Informed Neural Network training.

3.12 Dimensionless Quantities of Interest

The dimensionless surface concentration is

$$C_{\text{surf}}(\tau) = C(1, \tau)$$

The dimensionless volume-averaged concentration is

$$\bar{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho.$$

Therefore,

$$\bar{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho$$

The corresponding dimensional quantities can be recovered through

$$c_{s,\text{surf}}(t) = c_{s,\text{max}} C_{\text{surf}}(\tau),$$

and

$$\bar{c}_s(t) = c_{s,\text{max}} \bar{C}(\tau).$$

3.13 Interpretation of the Dimensionless Time

The dimensionless time is defined as

$$\tau = \frac{D_s t}{R_p^2}.$$

Consequently:

- $\tau \ll 1$ corresponds to times much shorter than the characteristic diffusion time;
- $\tau \approx 1$ corresponds to times comparable to the diffusion time across the particle;
- $\tau \gg 1$ corresponds to times much longer than the characteristic diffusion time.

The dimensional time can always be recovered as

$$t = \frac{R_p^2}{D_s} \tau$$

Thus, the same dimensionless solution can represent particles with different radii and diffusivities, provided that dimensional time is rescaled accordingly.

3.14 Operational Summary

The previous sections established the complete mathematical formulation of the lithium diffusion problem inside a spherical particle. Although the derivation is analytical, its ultimate purpose is the numerical implementation of a reference solver that will be used to generate the training data for the Physics-Informed Neural Network.

More specifically, the mathematical formulation provides all the ingredients required to construct the numerical model:

- the governing partial differential equation (PDE), describing the evolution of the lithium concentration inside the particle;
- the initial condition, defining the concentration profile at the beginning of the simulation;
- the symmetry boundary condition at the particle centre;
- the flux boundary condition at the particle surface.

Rather than solving the PDE analytically, the objective is to solve it numerically. The implementation will therefore follow the pipeline

$$\boxed{\text{PDE} \rightarrow \text{Spatial Discretization} \rightarrow \text{System of ODEs} \rightarrow \text{Numerical Integration}}$$

The role of the dimensionless formulation is equally important from a computational perspective. By introducing the variables

$$\rho = \frac{r}{R_p}, \quad \tau = \frac{D_s t}{R_p^2}, \quad C = \frac{c_s}{c_{s,\max}},$$

the computational domain becomes fixed,

$$\rho \in [0, 1],$$

while the governing equation no longer explicitly depends on the physical parameters R_p , D_s and $c_{s,\max}$.

Consequently, the numerical implementation will operate entirely on the dimensionless problem

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right),$$

subject to the corresponding initial and boundary conditions.

In Python, this mathematical model will not be implemented directly as a continuous PDE. Instead, the spatial domain will first be discretized into a finite number of radial nodes,

$$\rho_0, \rho_1, \dots, \rho_N,$$

so that the unknown concentration field becomes the finite-dimensional vector

$$\mathbf{C}(\tau) = [C_0(\tau), C_1(\tau), \dots, C_N(\tau)]^T.$$

The spatial derivatives appearing in the diffusion equation will then be approximated through finite-difference formulas, transforming the original PDE into a coupled system of ordinary differential equations (ODEs). This ODE system will finally be integrated in time using SciPy's `solve_ivp` routine, producing the numerical evolution of the concentration field.

The resulting solver will constitute the reference physical model of the project. Its numerical solutions will subsequently be employed to generate the synthetic datasets used for training and validating the Physics-Informed Neural Network.

4 Spatial Discretization

The governing equation is defined over the continuous spatial domain

$$0 \leq \rho \leq 1.$$

Since no analytical solution is sought, the continuous domain is replaced by a finite set of discrete spatial locations.

This procedure, known as spatial discretization, converts the original partial differential equation into a finite-dimensional dynamical system, which can subsequently be integrated in time.

4.1 Uniform Radial Grid

The interval

$$[0, 1]$$

is divided into N equal subintervals, generating $N + 1$ computational nodes. The spacing between two consecutive nodes is

$$\Delta\rho = \frac{1}{N}$$

The coordinates of the nodes are therefore

$$\rho_i = i\Delta\rho, \quad i = 0, \dots, N.$$

Consequently,

$$\rho_0 = 0,$$

corresponds to the particle centre, while

$$\rho_N = 1$$

corresponds to the particle surface.

4.2 Discrete Unknowns

The continuous concentration field

$$C(\rho, \tau)$$

is no longer represented at every point of the spatial domain.
Instead, only its values at the computational nodes are retained.
The unknown becomes

$$C_i(\tau) = C(\rho_i, \tau), \quad i = 0, \dots, N.$$

The complete concentration profile is therefore represented by the vector

$$\mathbf{C}(\tau) = \begin{bmatrix} C_0(\tau) \\ C_1(\tau) \\ \vdots \\ C_N(\tau) \end{bmatrix}$$

Instead of computing an unknown function over a continuous domain, the numerical solver computes the evolution of this finite-dimensional vector.

4.3 Connection with the Python Implementation

This discretization directly determines the data structure used in the numerical solver.

The computational grid will be generated as

$$\rho_i = i\Delta\rho,$$

which in Python corresponds to

$$\text{rho} = \text{np.linspace}(0.0, 1.0, N+1)$$

The unknown concentration field will be stored as

$$\mathbf{C} = \text{np.zeros}(N+1)$$

where

$$C[i] = C_i(\tau).$$

Each entry of the vector represents the lithium concentration at one specific radial position inside the particle.

4.4 From a PDE to a Finite-Dimensional System

At this stage, the governing equation

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)$$

is still continuous.

The only modification introduced so far is that the solution will be evaluated only at the discrete locations

$$\rho_0, \rho_1, \dots, \rho_N.$$

The spatial derivatives appearing in the governing equation have not yet been approximated.

The next step consists of replacing these derivatives with finite difference formulas, thereby transforming the partial differential equation into a coupled system of ordinary differential equations.

5 Conservative Finite-Volume Discretization

5.1 Motivation

The dimensionless diffusion equation is written in conservative form as

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

Instead of expanding the spherical diffusion operator and approximating its derivatives separately, the finite-volume method integrates the conservation equation over a finite control volume surrounding each computational node.

This approach directly preserves the physical balance

$$\boxed{\text{accumulation} = \text{incoming contribution} - \text{outgoing contribution}}$$

and is therefore particularly suitable for diffusion problems.

5.2 Computational Nodes and Control-Volume Interfaces

The computational nodes remain those previously introduced:

$$\rho_i = i\Delta\rho, \quad i = 0, \dots, N,$$

with

$$\Delta\rho = \frac{1}{N}.$$

Each node ρ_i is associated with a spherical control volume bounded by two interfaces:

$$\rho_{i-\frac{1}{2}}, \quad \rho_{i+\frac{1}{2}}.$$

For internal nodes, the interfaces are positioned halfway between consecutive nodes:

$$\rho_{i+\frac{1}{2}} = \frac{\rho_i + \rho_{i+1}}{2} = \left(i + \frac{1}{2}\right) \Delta\rho$$

and

$$\rho_{i-\frac{1}{2}} = \frac{\rho_{i-1} + \rho_i}{2} = \left(i - \frac{1}{2}\right) \Delta\rho.$$

At the physical boundaries,

$$\rho_{-\frac{1}{2}} = 0, \quad \rho_{N+\frac{1}{2}} = 1.$$

Therefore, the control volumes associated with the centre and surface nodes are half control volumes.

5.3 Dimensionless Control-Volume Measure

The dimensional volume of a spherical shell is

$$\Delta V_i = \frac{4\pi}{3} \left(r_{i+\frac{1}{2}}^3 - r_{i-\frac{1}{2}}^3 \right).$$

In the dimensionless formulation, the common geometrical factor 4π cancels from the conservation equation. It is therefore sufficient to introduce the dimensionless volume weight

$$\mathcal{V}_i = \int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 d\rho$$

which gives

$$\mathcal{V}_i = \frac{\rho_{i+\frac{1}{2}}^3 - \rho_{i-\frac{1}{2}}^3}{3}.$$

This quantity accounts for the fact that spherical control volumes increase in size as the radial coordinate increases.

Consequently, equally spaced radial nodes do not represent equal physical volumes.

5.4 Integration of the Governing Equation

The governing equation is integrated over the control volume associated with node i :

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 \frac{\partial C}{\partial \tau} d\rho = \int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) d\rho.$$

Assuming that the nodal value $C_i(\tau)$ represents the average concentration inside the corresponding control volume, the left-hand side is approximated as

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 \frac{\partial C}{\partial \tau} d\rho \approx \mathcal{V}_i \frac{dC_i}{d\tau}.$$

Applying the fundamental theorem of calculus to the right-hand side gives

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) d\rho = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}}.$$

The semi-discrete conservation equation is therefore

$$\boxed{\mathcal{V}_i \frac{dC_i}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}}}$$

This equation states that the rate of concentration accumulation inside a control volume is determined by the difference between the diffusive contributions at its outer and inner interfaces.

5.5 Interface Approximation

At an internal interface $\rho_{i+\frac{1}{2}}$, the radial concentration gradient is approximated using the two neighbouring nodal values:

$$\boxed{\frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} \approx \frac{C_{i+1} - C_i}{\Delta \rho}}.$$

Similarly,

$$\boxed{\frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}} \approx \frac{C_i - C_{i-1}}{\Delta \rho}}.$$

The corresponding geometrically weighted interface contributions are

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} \approx \rho_{i+\frac{1}{2}}^2 \frac{C_{i+1} - C_i}{\Delta \rho},$$

and

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}} \approx \rho_{i-\frac{1}{2}}^2 \frac{C_i - C_{i-1}}{\Delta \rho}.$$

5.6 Finite-Volume Equation at an Internal Node

Substituting the interface approximations into the conservation equation gives

$$\mathcal{V}_i \frac{dC_i}{d\tau} = \rho_{i+\frac{1}{2}}^2 \frac{C_{i+1} - C_i}{\Delta \rho} - \rho_{i-\frac{1}{2}}^2 \frac{C_i - C_{i-1}}{\Delta \rho}.$$

Therefore,

$$\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta \rho} \left[\rho_{i+\frac{1}{2}}^2 (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 (C_i - C_{i-1}) \right]$$

for

$$i = 1, \dots, N - 1.$$

This is the conservative finite-volume equation for the internal nodes.

The time derivative at node i depends only on

$$C_{i-1}, \quad C_i, \quad C_{i+1},$$

but the two contributions are weighted by the areas of the corresponding spherical interfaces.

6 Treatment of the Particle Centre

6.1 Centre Control Volume

Unlike the internal control volumes, the control volume associated with the particle centre extends only from

$$\rho = 0$$

to the first interface,

$$\rho_{\frac{1}{2}} = \frac{\Delta \rho}{2}.$$

Consequently, the centre control volume occupies only half of the width of an internal control volume.

Its dimensionless volume measure is

$$\mathcal{V}_0 = \int_0^{\rho_{\frac{1}{2}}} \rho^2 d\rho.$$

Evaluating the integral gives

$$\mathcal{V}_0 = \frac{\rho_{\frac{1}{2}}^3}{3} = \frac{\Delta \rho^3}{24}.$$

6.2 Symmetry Boundary Condition

The governing conservation equation for the centre control volume is

$$\mathcal{V}_0 \frac{dC_0}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=0}.$$

The second contribution vanishes.

Indeed, spherical symmetry requires

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0,$$

while the surface area of a sphere also approaches zero as

$$\rho \rightarrow 0.$$

Therefore,

$$\boxed{\rho^2 \left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0.}$$

Only the flux through the outer interface contributes to the mass balance.

6.3 Semi-Discrete Equation at the Centre

The concentration gradient at the first interface is approximated using the two neighbouring nodal values,

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho_{\frac{1}{2}}} \approx \frac{C_1 - C_0}{\Delta \rho}.$$

Since

$$\rho_{\frac{1}{2}} = \frac{\Delta \rho}{2},$$

the interface contribution becomes

$$\rho^2 \left. \frac{\partial C}{\partial \rho} \right|_{\rho_{\frac{1}{2}}} = \left(\frac{\Delta \rho}{2} \right)^2 \frac{C_1 - C_0}{\Delta \rho}.$$

Substituting this approximation into the conservation equation gives

$$\mathcal{V}_0 \frac{dC_0}{d\tau} = \frac{\Delta \rho}{4} (C_1 - C_0).$$

Finally, substituting

$$\mathcal{V}_0 = \frac{\Delta \rho^3}{24}$$

yields

$$\boxed{\frac{dC_0}{d\tau} = \frac{6(C_1 - C_0)}{\Delta \rho^2}.$$

This equation replaces the internal-node equation at the particle centre.

6.4 Physical Interpretation

The concentration inside the centre control volume changes only because of diffusion through its external interface.

If

$$C_1 > C_0,$$

lithium diffuses towards the particle centre and the concentration at the centre increases.
Conversely, if

$$C_1 < C_0,$$

lithium diffuses away from the particle centre and the concentration decreases.
When

$$C_1 = C_0,$$

no concentration gradient exists across the interface and therefore

$$\frac{dC_0}{d\tau} = 0.$$

The resulting discrete equation therefore preserves the expected diffusive behaviour while remaining fully consistent with the finite-volume conservation principle.

7 Treatment of the Particle Surface

7.1 Surface Control Volume

The control volume associated with the particle surface is bounded by the last internal interface,

$$\rho_{N-\frac{1}{2}} = 1 - \frac{\Delta\rho}{2},$$

and by the physical particle surface,

$$\rho_{N+\frac{1}{2}} = 1.$$

Therefore, similarly to the centre control volume, the surface control volume occupies only half of the width of a standard internal control volume.

Its dimensionless volume measure is

$$\mathcal{V}_N = \int_{\rho_{N-\frac{1}{2}}}^1 \rho^2 d\rho.$$

Evaluating the integral gives

$$\boxed{\mathcal{V}_N = \frac{1 - \rho_{N-\frac{1}{2}}^3}{3}}.$$

Using

$$\rho_{N-\frac{1}{2}} = 1 - \frac{\Delta\rho}{2},$$

the volume measure can also be written as

$$\mathcal{V}_N = \frac{1 - \left(1 - \frac{\Delta\rho}{2}\right)^3}{3}.$$

Unlike the centre control volume, the surface control volume is located at the largest radial coordinate and therefore represents a comparatively large physical volume.

7.2 Finite-Volume Balance at the Surface

The conservative diffusion equation integrated over the surface control volume gives

$$\mathcal{V}_N \frac{dC_N}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=1} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}}.$$

The first term represents the contribution through the physical surface of the particle, while the second represents diffusion through the last internal interface.

The concentration gradient at the internal interface is approximated by

$$\frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}} \approx \frac{C_N - C_{N-1}}{\Delta\rho}.$$

Therefore,

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}} \approx \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta\rho}.$$

7.3 Incorporation of the Surface Boundary Condition

The dimensionless boundary condition at the particle surface is

$$-\frac{\partial C}{\partial \rho} \Big|_{\rho=1} = \delta(\tau),$$

where $\delta(\tau)$ is the dimensionless intercalation flux.

Equivalently,

$$\frac{\partial C}{\partial \rho} \Big|_{\rho=1} = -\delta(\tau).$$

Since

$$\rho = 1$$

at the particle surface,

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=1} = -\delta(\tau).$$

Substituting the boundary condition and the internal-interface approximation into the finite-volume balance gives

$$\mathcal{V}_N \frac{dC_N}{d\tau} = -\delta(\tau) - \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta\rho}.$$

7.4 Semi-Discrete Equation at the Surface

Dividing by the volume measure of the surface control volume yields

$$\boxed{\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[-\delta(\tau) - \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta\rho} \right]}.$$

Equivalently, the internal diffusion contribution can be written with the opposite concentration difference:

$$\boxed{\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho} - \delta(\tau) \right]}.$$

This equation replaces the internal-node equation at $i = N$.
It depends on

$$C_{N-1}, \quad C_N, \quad \delta(\tau).$$

Unlike the finite-difference ghost-node formulation, the prescribed surface flux is incorporated directly into the conservation balance and no computational node outside the particle is required.

7.5 Physical Interpretation

The evolution of the surface concentration is controlled by two contributions.

The term

$$\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho}$$

represents diffusion between the last internal control volume and the particle surface.

The term

$$-\delta(\tau)$$

represents the prescribed molar exchange through the external particle surface.

Under the adopted sign convention,

$$\delta(\tau) > 0$$

corresponds to a positive outward flux and therefore tends to decrease the concentration inside the surface control volume.

Conversely,

$$\delta(\tau) < 0$$

corresponds to an inward flux and tends to increase the surface concentration.

If

$$\delta(\tau) = 0,$$

the particle surface is impermeable and its concentration evolves only because of diffusion between the surface and the particle interior.

8 Complete Semi-Discrete System

After discretizing the spatial domain using the finite-volume method, the original diffusion partial differential equation is transformed into a finite-dimensional system of ordinary differential equations.

The concentration field is represented by the vector

$$\mathbf{C}(\tau) = [C_0(\tau) \ C_1(\tau) \ \cdots \ C_N(\tau)]^T,$$

whose components evolve according to different equations depending on their position within the computational domain.

8.1 Centre Node

The centre node is governed by

$$\boxed{\frac{dC_0}{d\tau} = \frac{6(C_1 - C_0)}{\Delta\rho^2}}.$$

This equation results from the finite-volume balance over the half control volume adjacent to the centre and naturally incorporates the symmetry boundary condition.

8.2 Internal Nodes

For the internal control volumes,

$$i = 1, \dots, N - 1,$$

the governing equations are

$$\boxed{\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta\rho} \left[\rho_{i+\frac{1}{2}}^2 (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 (C_i - C_{i-1}) \right]}.$$

Each internal control volume exchanges mass only with its two adjacent neighbours through the corresponding spherical interfaces.

8.3 Surface Node

The surface control volume satisfies

$$\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho} - \delta(\tau) \right].$$

Unlike the internal nodes, the surface node is directly affected by the prescribed intercalation flux imposed through the boundary condition.

8.4 Compact Vector Form

Collecting all equations together, the semi-discrete model can be written compactly as

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}).$$

Here,

$$\mathbf{C}(\tau) = [C_0 \quad C_1 \quad \cdots \quad C_N]^T$$

contains the nodal concentrations, while the nonlinear operator

$$\mathbf{f}(\tau, \mathbf{C})$$

represents the finite-volume discretization of the spherical diffusion equation together with the imposed boundary conditions.

The original diffusion partial differential equation has therefore been transformed into a conservative system of coupled ordinary differential equations, which can be integrated numerically in time.

9 Numerical Time Integration

The finite-volume discretization transforms the original diffusion partial differential equation into the semi-discrete system

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}),$$

where the spatial derivatives have been replaced by algebraic expressions, while time remains a continuous variable.

This system generally does not admit a closed-form analytical solution. Consequently, the nodal concentrations are obtained through numerical time integration.

9.1 Initial Value Problem

The semi-discrete system is solved together with the dimensionless initial condition

$$\mathbf{C}(0) = \mathbf{C}_0,$$

where

$$\mathbf{C}_0 = [C_0(0) \quad C_1(0) \quad \cdots \quad C_N(0)]^T$$

contains the initial concentration at every computational node.

The complete problem therefore consists of the following initial value problem:

$$\begin{cases} \frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}), \\ \mathbf{C}(0) = \mathbf{C}_0. \end{cases}$$

9.2 Numerical Integration

The resulting system is integrated using a standard numerical solver for ordinary differential equations.

In the present work, the integration is performed using the `solve_ivp` routine provided by the SciPy scientific computing library.

At each integration step, the solver performs the following operations:

1. evaluates the current concentration vector

$$\mathbf{C}(\tau),$$

2. computes the corresponding time derivative

$$\mathbf{f}(\tau, \mathbf{C}),$$

by evaluating the finite-volume equations derived in the previous sections,

3. estimates the solution at the next time instant,
4. repeats the procedure until the prescribed final simulation time is reached.

The integration algorithm automatically selects the intermediate time steps required to satisfy the prescribed numerical tolerances, thus providing an accurate approximation of the transient diffusion process.

9.3 Simulation Output

The numerical integration produces the complete evolution of the dimensionless concentration field

$$C(\rho, \tau),$$

at every computational node and at every sampled time instant.

In particular, the surface concentration

$$C(1, \tau) = C_N(\tau)$$

is obtained directly from the last component of the numerical solution vector.

Depending on the intended application, either the complete concentration field or only the surface concentration can be stored for subsequent analysis.

9.4 Role in the Present Work

The numerical solution generated by the finite-volume model constitutes the reference solution employed throughout this work.

The resulting concentration profiles are used to generate synthetic data, which will subsequently be employed for the training and validation of the Physics-Informed Neural Network and the simplified physics FVM developed in the following stages of the project.

10 Python Implementation

The finite-volume formulation developed in the previous sections is implemented in Python to generate the numerical reference solution of the diffusion problem. The implementation follows directly the mathematical derivation and can be divided into four main stages: generation of the computational mesh, evaluation of the geometrical quantities, assembly of the semi-discrete system, and numerical time integration.

10.1 Mesh Generation

The first step consists in discretizing the dimensionless radial domain

$$0 \leq \rho \leq 1$$

using a uniform grid composed of $N + 1$ computational nodes,

$$\rho_i = i\Delta\rho, \quad \Delta\rho = \frac{1}{N}.$$

The interface positions are then computed as

$$\rho_{i+\frac{1}{2}} = \frac{\rho_i + \rho_{i+1}}{2},$$

thereby defining the boundaries of every control volume.

10.2 Evaluation of the Control-Volume Geometry

Once the mesh has been generated, the geometrical quantities required by the finite-volume formulation are evaluated.

For each control volume, the dimensionless volume measure is computed as

$$\mathcal{V}_i = \frac{\rho_{i+\frac{1}{2}}^3 - \rho_{i-\frac{1}{2}}^3}{3},$$

while the interface locations are stored for the subsequent evaluation of the diffusive fluxes.

Since the computational mesh remains fixed throughout the simulation, these quantities are calculated only once before the numerical integration begins.

10.3 Assembly of the Semi-Discrete System

The core of the implementation consists of assembling the right-hand side of the ordinary differential equation,

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}),$$

where

$$\mathbf{C}(\tau) = [C_0 \quad C_1 \quad \cdots \quad C_N]^T.$$

Given the current concentration vector, the algorithm evaluates the time derivative of every nodal concentration according to the equations derived in the previous sections.

The centre node is updated using the symmetry condition, the internal nodes are computed from the diffusive fluxes across the neighbouring interfaces, whereas the surface node additionally incorporates the prescribed intercalation flux through the particle boundary.

As a result, the function implementing $\mathbf{f}(\tau, \mathbf{C})$ returns the complete vector of concentration derivatives required by the ODE solver.

10.4 Numerical Integration

The assembled right-hand side is supplied to the `solve_ivp` routine of the SciPy library together with

- the initial concentration vector;
- the integration interval;
- the desired sampling instants.

The solver repeatedly evaluates the right-hand side and advances the solution in time until the final simulation time is reached.

The numerical output consists of the concentration vector

$$\mathbf{C}(\tau)$$

evaluated at every requested time instant.

11 Experimental Setup and Reference Simulations

The numerical framework presented in the previous sections was implemented in Python to generate the reference dataset used throughout this work. The objective of these simulations is not to reproduce a complete battery cell model, but to provide a physically consistent solution of lithium diffusion inside a single spherical active material particle. This high-fidelity solution will subsequently serve as the reference model against which simplified physical formulations and Physics-Informed Neural Networks will be evaluated.

The physical parameters adopted in the simulations are taken from the positive-electrode model reported by Guo and White [Guo and White, 2012], ensuring that the reference solution is representative of a realistic lithium-ion battery particle. The parameters are summarized in Table 1.

Using these values, the characteristic diffusion time becomes

$$t_D = \frac{R^2}{D_s} \approx 5.96 \times 10^3 \text{ s},$$

which defines the dimensionless time variable

$$\tau = \frac{D_s t}{R^2}.$$

Consequently, a simulation over one diffusion time corresponds approximately to

$$0 \leq \tau \leq 1,$$

or equivalently

$$0 \leq t \leq 6000 \text{ s}.$$

The governing diffusion equation is driven by the intercalation flux imposed at the particle surface,

$$-D_s \left. \frac{\partial c}{\partial r} \right|_{r=R} = j(t),$$

which physically represents the molar flux of lithium ions exchanged between the electrolyte and the active material during electrochemical reactions.

In a complete Single Particle Model, this quantity depends on the applied current and electrode properties according to electrochemical relations involving Faraday's law and the specific surface area. Therefore, the intercalation flux is generally expressed as

$$j(t) = f(I(t), a_s, F, L, \dots),$$

where the exact expression depends on the adopted electrochemical model.

For the purpose of this work, this complete formulation constitutes the reference physics. The subsequent stages of the project will investigate simplified descriptions of the intercalation process, including prescribed flux profiles and constant boundary fluxes, in order to evaluate how much physical information can be removed while maintaining accurate concentration predictions through Physics-Informed Neural Networks.

11.1 Time-Dependent Intercalation Flux

The lithium flux imposed at the particle surface is determined by the externally applied battery current. The dimensional surface boundary condition is written as

$$-D_s \left. \frac{\partial c}{\partial r} \right|_{r=R_p} = j(t),$$

where $j(t)$ is the molar intercalation flux at the particle surface.

For the positive-electrode particle considered in the present work, the molar flux is related to the applied current through Faraday's law,

$$j(t) = -\frac{I(t)}{FS_p},$$

where $I(t)$ is the applied current, F is Faraday's constant, and S_p is the total electroactive surface area.

The negative sign denotes lithium intercalation according to the sign convention adopted in this work. A positive applied current therefore produces a negative outward radial flux, corresponding to lithium entering the solid particle.

After nondimensionalization, the surface flux becomes

$$\delta(\tau) = \frac{R_p}{D_s c_{s,\max}} j(t),$$

and consequently

$$\delta(\tau) = -\frac{I(t)R_p}{D_s FS_p c_{s,\max}}.$$

The dimensionless and dimensional times are related by

$$t = \frac{R_p^2}{D_s} \tau.$$

Therefore, the complete expression of the dimensionless flux is

$$\delta(\tau) = -\frac{R_p}{D_s FS_p c_{s,\max}} I \left(\frac{R_p^2}{D_s} \tau \right).$$

11.2 Applied Current Profile

To obtain a time-dependent reference input while avoiding discontinuous changes at the beginning and end of the loading phase, the applied current is described by a smooth ramp-constant-ramp profile:

$$I(t) = \begin{cases} \frac{I_{1C}}{2} \left[1 - \cos \left(\frac{\pi t}{t_r} \right) \right], & 0 \leq t < t_r, \\ I_{1C}, & t_r \leq t < t_{\text{load}}, \\ \frac{I_{1C}}{2} \left[1 + \cos \left(\frac{\pi(t - t_{\text{load}})}{t_r} \right) \right], & t_{\text{load}} \leq t < t_{\text{load}} + t_r, \\ 0, & t_{\text{load}} + t_r \leq t \leq t_f. \end{cases}$$

Here, t_r denotes the current ramp duration, t_{load} defines the beginning of the ramp-down phase, and t_f is the final simulation time.

The profile satisfies

$$I(0) = 0, \quad I(t_f) = 0,$$

and reaches the nominal 1C current during the central loading interval.

The corresponding dimensionless flux is therefore

$$\delta(\tau) = -\delta_{1C} g(\tau),$$

where

$$\delta_{1C} = \frac{I_{1C} R_p}{D_s F S_p c_{s,\text{max}}} = 0.200452,$$

and $g(\tau)$ is the dimensionless representation of the applied current profile.

11.3 Dimensionless Current Profile

Introducing

$$\tau_r = \frac{D_s t_r}{R_p^2}, \quad \tau_{\text{load}} = \frac{D_s t_{\text{load}}}{R_p^2}, \quad \tau_f = \frac{D_s t_f}{R_p^2},$$

the normalized current profile becomes

$$g(\tau) = \begin{cases} \frac{1}{2} \left[1 - \cos \left(\frac{\pi \tau}{\tau_r} \right) \right], & 0 \leq \tau < \tau_r, \\ 1, & \tau_r \leq \tau < \tau_{\text{load}}, \\ \frac{1}{2} \left[1 + \cos \left(\frac{\pi(\tau - \tau_{\text{load}})}{\tau_r} \right) \right], & \tau_{\text{load}} \leq \tau < \tau_{\text{load}} + \tau_r, \\ 0, & \tau_{\text{load}} + \tau_r \leq \tau \leq \tau_f. \end{cases}$$

Hence, the reference boundary condition is

$$-\frac{\partial C}{\partial \rho} \Big|_{\rho=1} = -\delta_{1C} g(\tau).$$

11.4 Simplified Flux Model

The time-dependent flux defined above constitutes the reference boundary input. In the simplified PINN formulation of Section 15.1, this input is replaced by a constant approximation,

$$\delta_{\text{simp}}(\tau) = -\delta_{1C},$$

throughout the represented time interval.

The discrepancy introduced by this approximation is therefore

$$\Delta\delta(\tau) = \delta(\tau) - \delta_{\text{simp}}(\tau).$$

This simplification removes the transient current dynamics while preserving the nominal 1C operating level. It therefore provides a controlled modification of the boundary physics whose effect on the predicted concentration field can be quantified.

11.5 Generation of the Reference Dataset

The computed concentration profiles constitute the reference solution of the diffusion model.

For each simulation, the solver provides the complete concentration field

$$C(\rho, \tau),$$

from which quantities of interest can be directly extracted. In particular, the surface concentration

$$C(1, \tau) = C_N(\tau)$$

is obtained from the last component of the solution vector.

The generated numerical solutions form the synthetic dataset employed throughout the remainder of the project for the training and validation of the Physics-Informed Neural Network.

The final summary of used parameters value is provided below, as well as definition of boundary inputs table.

12 Experimental Results

The proposed finite-volume solver was evaluated using the experimental setup described in the previous section. The computational domain was discretized using 101 uniformly distributed radial nodes and 201 stored time instants over a physical simulation horizon of 6000 s, corresponding to a final dimensionless time of $\tau_f = 1.007$. The resulting concentration field therefore consists of $101 \times 201 = 20301$ samples, which are subsequently employed to construct the reference dataset for PINN training.

Initially, the particle is assumed to be uniformly lithiated with a dimensionless concentration equal to $C_0 = 0.5$. Consequently, the concentration field is spatially uniform at $\tau = 0$, satisfying the prescribed initial condition. During the charging process, lithium enters the particle through the external surface, generating a concentration gradient directed from the surface towards the particle centre.

Table 1: Parameters adopted in the reference numerical simulations.

Category	Parameter	Symbol	Value
Physical	Particle radius	R_p	$8.5 \times 10^{-6} \text{ m}$
	Solid diffusion coefficient	D_s	$1.213 \times 10^{-14} \text{ m}^2/\text{s}$
	Maximum lithium concentration	$c_{s,\max}$	51410 mol/m^3
	Faraday constant	F	96487 C/mol
	Particle surface area	S_p	1.167 m^2
Simulation	Initial normalized concentration	C_0	0.5
	Nominal current (1C)	I_{1C}	1.656 A
	Current ramp duration	t_r	300 s
	Constant-current duration	t_{load}	3600 s
	Final simulation time	t_f	6000 s
	Characteristic diffusion time	$t_D = \frac{R_p^2}{D_s}$	5956 s
Numerical	Number of radial intervals	N	100
	Number of radial nodes	$N + 1$	101
	Radial spacing	$\Delta\rho$	0.01
	Number of stored time instants	N_τ	201
	Absolute tolerance	atol	10^{-8}
	Relative tolerance	rtol	10^{-6}
	ODE solver		<code>solve_ivp</code> (RK45)

Table 2: Definition of the boundary input.

Quantity	Expression	Description
Applied current	$I(t)$	Ramp–constant–ramp profile
Molar intercalation flux	$j(t) = -\frac{I(t)}{FS_p}$	Faraday’s law
Dimensionless flux	$\delta(\tau) = \frac{R_p}{D_s c_{s,\max}} j(t)$	Surface boundary condition
Dimensionless PDE BC	$-\left. \frac{\partial C}{\partial \rho} \right _{\rho=1} = \delta(\tau)$	Neumann boundary condition

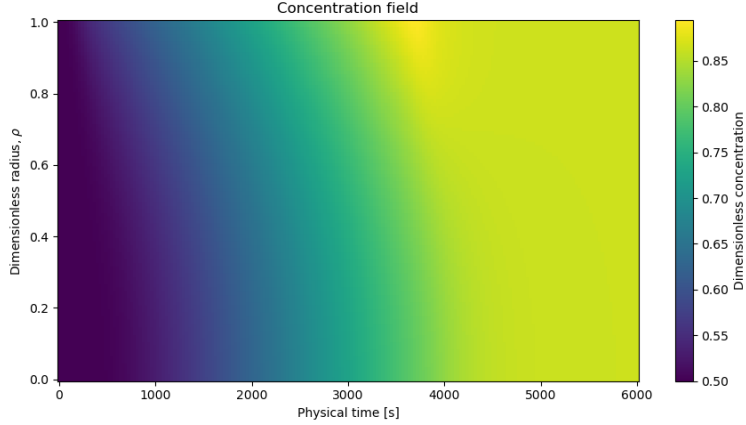


Figure 1: Concentration Map of the Particle

Figure 1 illustrates the spatio-temporal evolution of the concentration field. As expected, the concentration increases progressively throughout the particle while remaining higher near the surface during the charging stage. Once the applied current is removed, diffusion gradually redistributes lithium inside the particle until an almost uniform concentration field is recovered.

Figure 2 reports the temporal evolution of the concentrations at the particle centre and at the particle surface. The surface concentration reacts immediately to the imposed intercalation flux, reaching a maximum value of approximately $C_s = 0.894$ at $t \approx 3690$ s. During the subsequent relaxation stage, the surface concentration decreases slightly, whereas the concentration at the particle centre continues to increase as diffusion redistributes lithium inside the particle. At the end of the simulation both quantities converge towards the same equilibrium value of approximately 0.863, indicating that the internal concentration gradient has almost completely vanished.

The corresponding evolution of the volume-averaged concentration is presented in Figure 3. During the charging stage, the average concentration increases almost linearly, consistently with the nearly constant lithium insertion rate imposed by the operating current profile. After the current is progressively reduced and eventually removed, the average concentration reaches a constant value close to 0.8635, reflecting the conservation of the total lithium content.

The radial concentration profiles at different time instants are shown in Figure 4. At intermediate times the highest concentration is always observed at the particle surface, while the particle centre exhibits a delayed response due to finite diffusion kinetics. As time progresses, the concentration gradients progressively decrease until an almost uniform profile is recovered at the end of the simulation.

From a numerical perspective, the proposed implementation remained stable throughout the entire simulation. No NaN or infinite values were generated, all concentration values remained within the physically admissible interval $0 \leq C \leq 1$, and the maximum mass-balance residual was equal to 4.52×10^{-5} , as reported in Figure 5. These observations indicate that the finite-volume discretization preserves both numerical stability and physical consistency under the selected operating conditions.

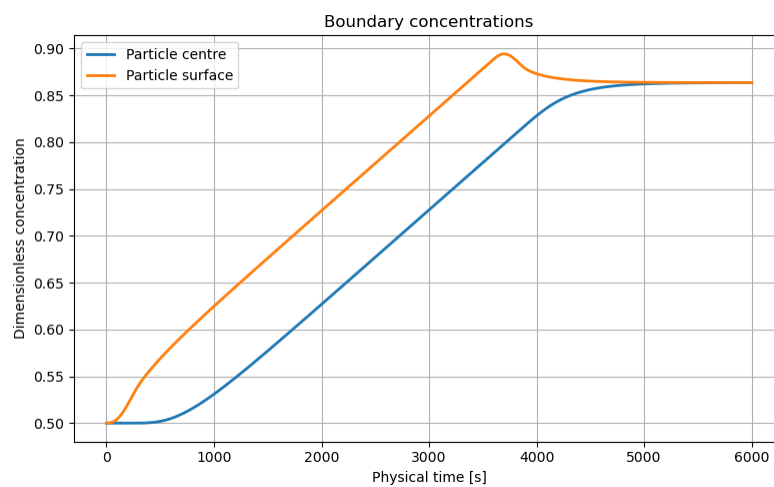


Figure 2: Concentration on Boundary

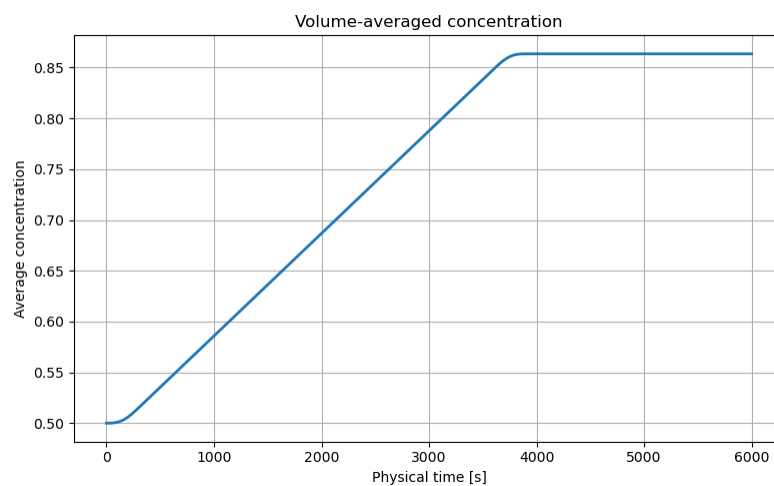


Figure 3: Average Concentration in the Particle

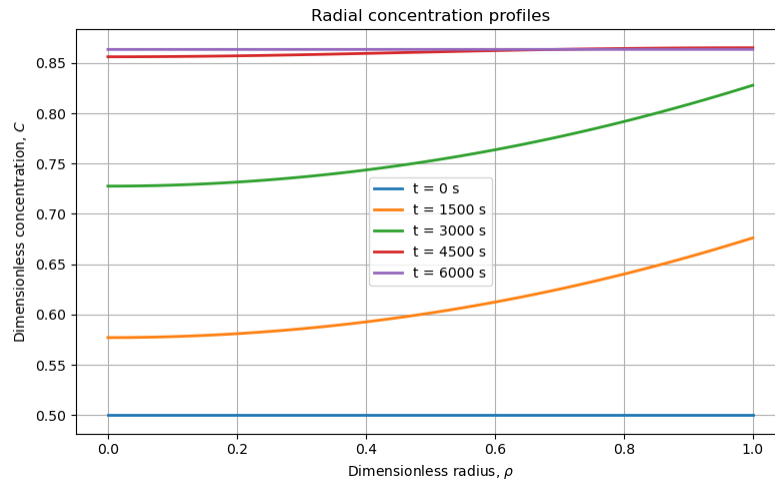


Figure 4: Radial Profiles

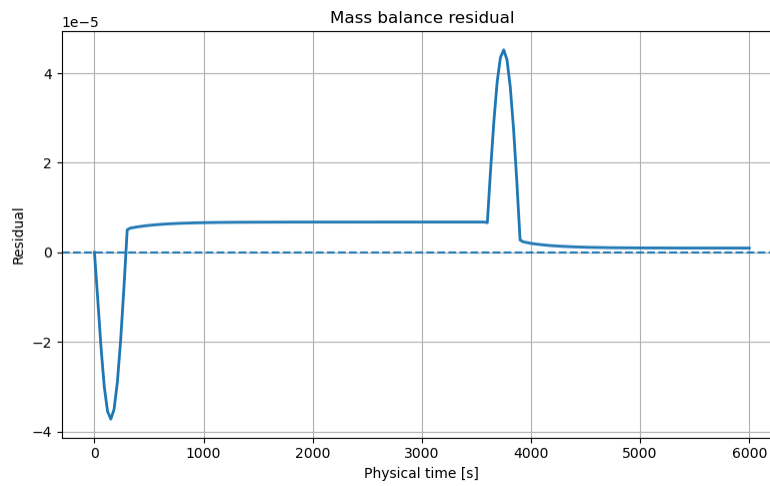


Figure 5: Mass Balance Residual

12.1 Validation

The accuracy of the proposed finite-volume implementation was assessed through two complementary validation procedures. First, the numerical solution was compared against the approximate analytical solution proposed by Guo and White [Guo and White, 2012]. Second, a mesh-convergence study was carried out to verify that the numerical solution is independent of the spatial discretization.

12.1.1 Comparison with the Guo–White Approximate Solution

The finite-volume solution was compared with the approximate surface-concentration solution proposed by Guo and White [Guo and White, 2012] for a constant dimensionless intercalation flux,

$$\delta(\tau) = -0.2,$$

with initial concentration $C_0 = 0.5$.

The reference curve was not obtained by digitizing the published figure. Instead, it was reconstructed directly from the analytical approximation given in Eq. (46) of Guo and White [Guo and White, 2012].

The eigenvalues λ_n were computed numerically as the positive roots of

$$\lambda_n - \tan(\lambda_n) = 0.$$

The first five eigenvalues were retained, consistently with the $N = 5$ approximation adopted in the paper. The following eigenvalue, λ_6 , was additionally used in the truncation-error correction term proposed by the authors.

For a constant flux, the resulting reference surface concentration was evaluated as

$$C_s^{\text{GW}}(\tau) = C_0 - 3\delta\tau - 2\delta \sum_{n=1}^N \frac{1 - \exp(-\lambda_n^2 \tau)}{\lambda_n^2} - 2\delta e_N^{\text{apx}}(\tau),$$

where $e_N^{\text{apx}}(\tau)$ denotes the approximation of the neglected eigenfunction contributions.

The Guo–White solution was evaluated over 300 uniformly distributed dimensionless time instants in the interval $0 \leq \tau \leq 0.68$. Since the finite-volume solution and the reference solution were defined on different temporal grids, the finite-volume surface concentration was linearly interpolated at the reference time instants before computing the error.

Figure 6 shows that the two solutions are nearly indistinguishable over the complete time interval. The resulting root-mean-square error is

$$\text{RMSE} = 6.33 \times 10^{-4},$$

confirming the accuracy of the finite-volume implementation for the constant-flux benchmark.

12.1.2 Mesh Convergence Study

A mesh-convergence analysis was performed using five spatial discretizations,

$$N = \{25, 50, 100, 200, 400\},$$

while keeping all the remaining simulation parameters unchanged.

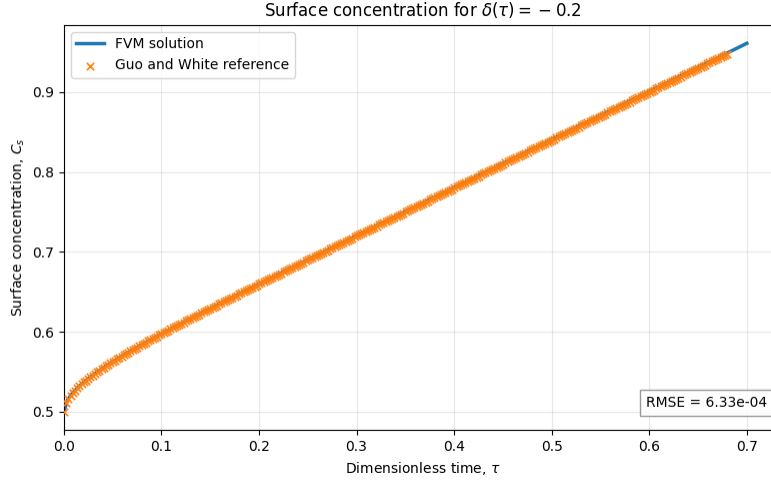


Figure 6: FVM Vs Guo and White Surface Concentration over tau

Figure 7 reports the evolution of the surface concentration, the volume-averaged concentration and the final radial concentration profile obtained with the different meshes. All the solutions rapidly converge towards the same response as the mesh is refined. In particular, the curves corresponding to $N = 100$, $N = 200$ and $N = 400$ are almost perfectly superimposed for all the monitored quantities.

The final radial concentration profile exhibits only negligible differences between the finest discretizations, indicating that the numerical error associated with the spatial discretization is already very small for $N = 100$. Consequently, this mesh resolution was selected throughout the remainder of this work as a suitable compromise between numerical accuracy and computational efficiency.

Table 3: Mesh convergence analysis using the $N = 400$ solution as reference.

Radial nodes (N)	$\ \mathbf{C}_s - \mathbf{C}_{s,\text{ref}}\ _2$	$\ \tilde{\mathbf{C}} - \tilde{\mathbf{C}}_{\text{ref}}\ _2$
25	9.3471×10^{-4}	6.1095×10^{-4}
50	2.3872×10^{-4}	1.4729×10^{-4}
100	5.6692×10^{-5}	3.3493×10^{-5}
200	1.9012×10^{-5}	1.1247×10^{-5}
400	0	0

Table 3 shows Euclidean norm when computing concentration and dimensionless concentration w.r.t. concentration ref, i.e. maximal concentration in the sphere. These results show that independently from the mesh adopted, concentrations obtained as outcome of the model are reasonably the same, which involves into saying that we are able to chose N .

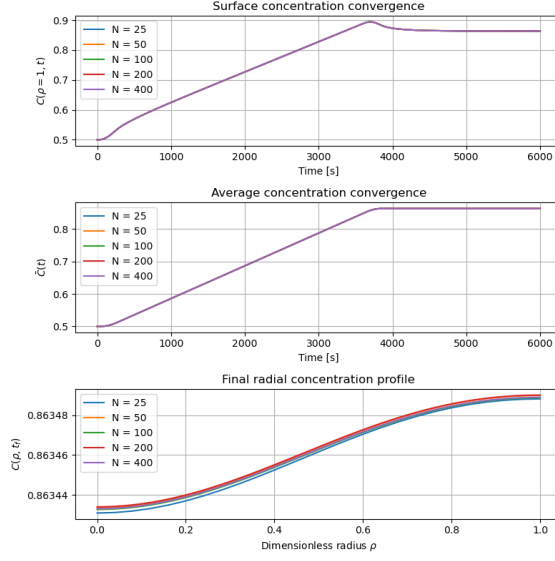


Figure 7: Concentrations behaviors by varying mesh

13 Model Extension

13.1 Motivation

The reference model introduced in the previous sections assumes a constant solid-phase diffusion coefficient and neglects thermal effects. Under these assumptions, the dimensionless governing equation is linear, and the concentration evolution is controlled only by the prescribed intercalation flux and by diffusion within the spherical particle.

Although this formulation provides a suitable benchmark for validating the finite-volume implementation, it represents a relatively simple learning problem for a Physics-Informed Neural Network. In particular, a PINN constrained by the same constant-diffusivity equation used to generate the reference data would be supplied with an almost complete description of the underlying physical process.

The objective of the present extension is therefore to construct a more realistic and physically richer reference model, thereby increasing the difficulty of the prediction task assigned to the PINN constrained by simplified physics. The extended reference model introduces two additional mechanisms:

1. a lumped thermal balance accounting for irreversible heating and convective heat exchange with the environment;
2. a solid-phase diffusion coefficient depending on both the local lithium concentration and the particle temperature.

The resulting formulation establishes a one-way thermo-diffusive coupling. The applied current determines the temperature evolution, while temperature and concentration modify the solid dif-

fusion coefficient. The modified diffusivity then affects the spatial and temporal evolution of the lithium concentration.

The coupling structure can be summarized as

$$\boxed{I(t) \longrightarrow T(t) \longrightarrow D_s(C, T) \longrightarrow C(\rho, \tau)}$$

with an additional direct dependence of D_s on the local concentration field.

It is important to clarify that the thermal extension does not introduce a spatial heat-conduction PDE inside the particle. Instead, the temperature is represented through a lumped thermal state $T(t)$, assumed uniform over the entire particle domain. Convective heat transfer is introduced in the thermal energy balance, whereas the lithium transport problem remains spatially distributed.

This extended thermo-diffusive model is adopted as the high-fidelity reference model used for synthetic data generation. In the subsequent PINN formulation, part of this additional physics will deliberately be removed by imposing a constant diffusion coefficient and neglecting the thermal state.

13.2 Thermally Coupled Diffusion Equation

When the solid diffusion coefficient depends on the system state, it can no longer be extracted from the spatial derivative. The dimensional lithium diffusion equation must therefore be written as

$$\boxed{\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 D_s(c_s, T) \frac{\partial c_s}{\partial r} \right]}$$

for

$$0 < r < R_p, \quad t > 0.$$

The corresponding symmetry condition at the particle centre remains

$$\boxed{\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0.}$$

At the particle surface, the dimensional intercalation-flux condition becomes

$$\boxed{-D_s(c_s(R_p, t), T(t)) \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t).}$$

Consequently, the diffusion coefficient now influences both the internal transport rate and the concentration gradient required to satisfy the prescribed surface flux.

13.3 State-Dependent Solid Diffusion

The constant-diffusivity formulation adopted in the preliminary model provides a linear description of lithium transport inside the active material particle. In real electrode materials, however, the solid-state diffusion coefficient may vary with both the local lithium concentration and the operating temperature. The extended reference model therefore replaces the constant coefficient D_s with a

state-dependent diffusivity. Concentration-dependent solid diffusivity is also a central quantity in experimental parameter-inference studies [Gümrukçüoğlu et al., 2024]:

$$D_s = D_s(C, T),$$

where $C = C(\rho, \tau)$ is the local dimensionless lithium concentration and $T = T(\tau)$ is the particle temperature.

The purpose of this extension is to construct a more realistic and more demanding reference model. The resulting concentration dynamics are no longer governed by a linear diffusion equation with constant coefficients. Instead, the local transport rate changes during the simulation according to the evolving concentration field and thermal state. This introduces a controlled physical mismatch with respect to the simplified model that will subsequently be enforced within the Physics-Informed Neural Network.

13.3.1 Concentration and Temperature Dependence

In the implemented reference model, the solid diffusion coefficient is defined as

$$D_s(C, T) = D_{s,\text{ref}} \exp[\alpha (C - C_0)] \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right],$$

where:

- $D_{s,\text{ref}}$ is the reference solid-state diffusion coefficient;
- C_0 is the initial dimensionless concentration;
- α controls the sensitivity of diffusivity to lithium concentration;
- E_a is the activation energy associated with solid-state diffusion;
- R_g is the universal gas constant;
- T_{ref} is the reference temperature.

The two exponential factors describe distinct physical effects. The first one introduces a concentration dependence,

$$f_C(C) = \exp[\alpha (C - C_0)],$$

while the second corresponds to an Arrhenius-type temperature correction,

$$f_T(T) = \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right].$$

The complete diffusivity can therefore be written compactly as

$$D_s(C, T) = D_{s,\text{ref}} f_C(C) f_T(T).$$

At the initial reference state,

$$C = C_0, \quad T = T_{\text{ref}},$$

both correction factors are equal to one. Consequently,

$$D_s(C_0, T_{\text{ref}}) = D_{s,\text{ref}}.$$

Thus, $D_{s,\text{ref}}$ represents the diffusivity of the particle at the initial concentration and reference temperature.

13.3.2 Effect of Lithium Concentration

The concentration-dependent factor is

$$f_C(C) = \exp[\alpha(C - C_0)].$$

For the value adopted in the present simulations,

$$\alpha = 1,$$

an increase in lithium concentration with respect to the initial state produces an exponential increase in the local diffusion coefficient. In particular,

$$C > C_0 \implies f_C(C) > 1,$$

whereas

$$C < C_0 \implies f_C(C) < 1.$$

Since the concentration is spatially distributed, the resulting diffusivity also varies along the radial coordinate. Therefore, during the charging phase,

$$D_s = D_s(\rho, \tau),$$

even though the temperature is assumed spatially uniform.

Regions with different lithium concentrations are consequently characterized by different local transport rates. In the present charging simulation, the concentration initially increases more rapidly near the particle surface. The corresponding local diffusivity therefore becomes larger near the surface than in the particle interior, modifying the transient redistribution of lithium.

13.3.3 Effect of Temperature

The temperature correction follows the Arrhenius-type relation

$$f_T(T) = \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right].$$

When the particle temperature exceeds the reference temperature,

$$T > T_{\text{ref}},$$

the reciprocal-temperature difference satisfies

$$\frac{1}{T} - \frac{1}{T_{\text{ref}}} < 0.$$

Hence, the exponent becomes positive and

$$f_T(T) > 1.$$

The diffusion coefficient therefore increases as the particle temperature rises. Conversely, temperatures below the reference value reduce the diffusivity.

The activation energy adopted in the reference model is

$$E_a = 3.5 \times 10^4 \text{ J mol}^{-1},$$

while the universal gas constant is

$$R_g = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}.$$

Even a moderate temperature variation can therefore produce a non-negligible modification of the diffusion coefficient.

13.3.4 Dimensional Governing Equation

When the diffusion coefficient depends on the evolving state, it must remain inside the divergence operator. The dimensional governing equation becomes

$$\boxed{\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 D_s(c_s, T) \frac{\partial c_s}{\partial r} \right]}.$$

This equation differs from the constant-diffusivity formulation

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right),$$

because D_s can no longer be extracted from the spatial derivative.

Expanding the variable-coefficient operator gives

$$\frac{\partial c_s}{\partial t} = D_s \left(\frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r} \right) + \frac{\partial D_s}{\partial r} \frac{\partial c_s}{\partial r}.$$

The additional contribution

$$\frac{\partial D_s}{\partial r} \frac{\partial c_s}{\partial r}$$

arises exclusively from the spatial variation of the diffusion coefficient. Since D_s depends on concentration, its radial derivative is indirectly determined by the concentration gradient.

For this reason, the conservative form is retained in the numerical implementation, avoiding the explicit evaluation of $\partial D_s / \partial r$.

13.3.5 Dimensionless Diffusion Coefficient

The dimensionless time variable continues to be defined using the constant reference coefficient,

$$\tau = \frac{D_{s,\text{ref}} t}{R_p^2}.$$

A normalized diffusion coefficient is consequently introduced as

$$\hat{D}(C, T) = \frac{D_s(C, T)}{D_{s,\text{ref}}}.$$

Using the adopted state-dependent relation gives

$$\hat{D}(C, T) = \exp[\alpha(C - C_0)] \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right].$$

At the initial reference state,

$$\hat{D}(C_0, T_{\text{ref}}) = 1.$$

The variable $\hat{D}$ is dimensionless and measures the local diffusivity relative to its reference value.

13.3.6 Dimensionless Governing Equation

After introducing the normalized diffusivity, the extended dimensionless diffusion equation becomes

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C, T) \frac{\partial C}{\partial \rho} \right].$$

The initial condition remains

$$C(\rho, 0) = C_0.$$

At the particle centre, spherical symmetry gives

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0.$$

The surface condition is still scaled using $D_{s,\text{ref}}$. Therefore,

$$-\hat{D}(C, T) \left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \delta(\tau),$$

where

$$\delta(\tau) = \frac{R_p j(t)}{D_{s,\text{ref}} c_{s,\text{max}}}.$$

The complete extended problem is thus

$$\left\{ \begin{array}{ll} \frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C, T) \frac{\partial C}{\partial \rho} \right], & 0 < \rho < 1, \quad \tau > 0, \\ C(\rho, 0) = C_0, & 0 \leq \rho \leq 1, \\ \frac{\partial C}{\partial \rho} \Big|_{\rho=0} = 0, & \tau > 0, \\ -\hat{D}(C, T) \frac{\partial C}{\partial \rho} \Big|_{\rho=1} = \delta(\tau), & \tau > 0. \end{array} \right.$$

Because $\hat{D}$ depends on the unknown concentration and on the temperature state, the resulting diffusion problem is nonlinear and thermally coupled.

13.3.7 Finite-Volume Discretization with Variable Diffusivity

The conservative finite-volume formulation is retained for the extended model. At each numerical evaluation, the normalized diffusivity is first computed at every radial node:

$$\hat{D}_i = \hat{D}(C_i, T), \quad i = 0, \dots, N.$$

Since the diffusive transport terms are evaluated at the control-volume interfaces, the nodal diffusivities are interpolated using the arithmetic mean,

$$\hat{D}_{i+\frac{1}{2}} = \frac{\hat{D}_i + \hat{D}_{i+1}}{2}.$$

The geometrically weighted diffusive contribution at an internal interface becomes

$$\rho^2 \hat{D} \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} \approx \rho_{i+\frac{1}{2}}^2 \hat{D}_{i+\frac{1}{2}} \frac{C_{i+1} - C_i}{\Delta \rho}.$$

Consequently, the equation for an internal control volume is

$$\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta \rho} \left[\rho_{i+\frac{1}{2}}^2 \hat{D}_{i+\frac{1}{2}} (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 \hat{D}_{i-\frac{1}{2}} (C_i - C_{i-1}) \right],$$

for

$$i = 1, \dots, N-1.$$

At the particle centre, the corresponding equation becomes

$$\frac{dC_0}{d\tau} = \frac{6 \hat{D}_{\frac{1}{2}} (C_1 - C_0)}{\Delta \rho^2}.$$

At the particle surface, the prescribed boundary flux is incorporated directly into the finite-volume balance:

$$\boxed{\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \hat{D}_{N-\frac{1}{2}} \frac{C_{N-1} - C_N}{\Delta\rho} - \delta(\tau) \right].}$$

The variable diffusivity is therefore evaluated dynamically throughout the integration. At every ODE-solver evaluation, the current concentration field and temperature are used to update $\hat{D}_i$, after which the interface coefficients and the finite-volume derivatives are recomputed.

13.3.8 Coupled State Representation

The temperature affects lithium diffusion through $\hat{D}(C, T)$, while the temperature itself evolves according to the applied current and the thermal balance introduced in the following subsection.

The complete numerical state is therefore extended from the concentration vector

$$\mathbf{C}(\tau) = [C_0(\tau) \quad \cdots \quad C_N(\tau)]^T$$

to

$$\boxed{\mathbf{y}(\tau) = [C_0(\tau) \quad C_1(\tau) \quad \cdots \quad C_N(\tau) \quad T(\tau)]^T.}$$

The coupled semi-discrete problem can then be written as

$$\boxed{\frac{d\mathbf{y}}{d\tau} = \begin{bmatrix} \mathbf{f}_C(\tau, \mathbf{C}, T) \\ f_T(\tau, T) \end{bmatrix},}$$

where $\mathbf{f}_C$ denotes the finite-volume diffusion operator with state-dependent diffusivity and f_T denotes the lumped thermal balance.

This formulation allows the concentration field and temperature to be integrated simultaneously within the same implicit ODE solver.

13.4 Lumped Thermal Model

The previous formulation assumes that the particle temperature remains constant throughout the entire simulation. Although this assumption is reasonable under mild operating conditions, it neglects the thermal effects generated during battery operation. Since the diffusion coefficient is strongly temperature dependent, even relatively small temperature variations may alter the lithium transport dynamics.

Rather than introducing a full heat-conduction equation inside the particle, the present work adopts a lumped thermal model. The particle is therefore assumed to be thermally homogeneous, so that its temperature depends only on time,

$$\boxed{T = T(\tau).}$$

This approximation is justified by the small dimensions of the active material particle, for which internal thermal gradients are typically much smaller than concentration gradients. Consequently, the thermal state can be represented by a single ordinary differential equation, while the spatial dependence remains confined to the diffusion problem.

13.4.1 Energy Balance

The particle temperature evolves according to the balance between heat generation and heat dissipation,

$$\frac{dT}{dt} = \beta_Q I^2(t) - \beta_H (T - T_{\text{amb}}),$$

where

- β_Q is the heat-generation coefficient, which converts the electrical current into thermal power;
- β_H is the effective convective heat-transfer coefficient;
- T_{amb} denotes the ambient temperature;
- $I(t)$ is the externally applied current.

The first term models irreversible heat generation caused by Joule losses, while the second term represents heat exchange with the surrounding environment according to Newton's cooling law.

13.4.2 Dimensionless Thermal Equation

Since the diffusion problem is solved using the dimensionless time

$$\tau = \frac{D_{s,\text{ref}} t}{R_p^2},$$

the thermal model is rewritten using the same temporal variable. Applying the chain rule,

$$\frac{dT}{d\tau} = \frac{R_p^2}{D_{s,\text{ref}}} \frac{dT}{dt},$$

which yields

$$\frac{dT}{d\tau} = \frac{R_p^2}{D_{s,\text{ref}}} [\beta_Q I^2(t) - \beta_H (T - T_{\text{amb}})].$$

The factor

$$\frac{R_p^2}{D_{s,\text{ref}}}$$

corresponds to the characteristic diffusion time previously introduced. Consequently, both the concentration field and the thermal state evolve using the same dimensionless temporal coordinate.

13.4.3 Coupling with the Diffusion Model

Although the temperature is assumed spatially uniform, it directly affects the diffusion process through the state-dependent diffusion coefficient introduced in the previous subsection.

At each integration step, the numerical solver first evaluates the current particle temperature,

$$T(\tau),$$

which is subsequently used to compute the Arrhenius correction of the solid-state diffusion coefficient,

$$D_s = D_s(C, T).$$

The updated diffusion coefficient is then employed to assemble the finite-volume diffusion operator. Consequently, concentration and temperature are solved simultaneously, producing a fully coupled thermo-diffusion model.

The complete numerical state therefore becomes

$$\mathbf{y}(\tau) = [C_0(\tau) \quad C_1(\tau) \quad \cdots \quad C_N(\tau) \quad T(\tau)]^T,$$

whose evolution is governed by the coupled system

$$\frac{d\mathbf{y}}{d\tau} = \begin{bmatrix} \mathbf{f}_C(\mathbf{C}, T) \\ f_T(T, I) \end{bmatrix}.$$

This formulation preserves the conservative finite-volume treatment of lithium diffusion while extending the reference model with a simplified yet physically meaningful thermal dynamics.

13.5 Thermal and Transport Parameters

The introduction of a state-dependent diffusion coefficient requires a number of additional physical parameters describing both the thermal dynamics of the particle and the dependence of lithium diffusivity on its operating state.

The thermal model is initialized at the ambient temperature,

$$T(0) = T_{\text{amb}} = T_{\text{ref}},$$

so that the diffusion coefficient initially coincides with its reference value,

$$D_s(C_0, T_{\text{ref}}) = D_{s,\text{ref}}.$$

As the simulation progresses, both the concentration field and the particle temperature modify the effective diffusion coefficient through the exponential relations introduced in the previous subsection.

Table 4 summarizes all the additional parameters employed in the extended reference model.

The activation energy determines the sensitivity of lithium diffusion to temperature through the Arrhenius correction, while the parameter α controls the influence of the local lithium concentration on the diffusion coefficient.

Table 4: Additional parameters introduced in the thermo-diffusion model.

Category	Parameter	Symbol	Value
Thermal	Reference temperature	T_{ref}	298.15 K
	Ambient temperature	T_{amb}	298.15 K
	Activation energy	E_a	$3.5 \times 10^4 \text{ J mol}^{-1}$
	Universal gas constant	R_g	$8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$
	Equivalent cell mass	m	0.045 kg
	Specific heat capacity	c_p	$900 \text{ J kg}^{-1} \text{ K}^{-1}$
	Equivalent electrical resistance	R_{el}	0.10 Ω
	Convective heat-transfer coefficient	h	$10 \text{ W m}^{-2} \text{ K}^{-1}$
	Equivalent heat-transfer area	A_{th}	0.01 m^2
Transport	Reference diffusion coefficient	$D_{s,\text{ref}}$	$1.213 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$
	Concentration sensitivity	α	1.0

The coefficients β_Q and β_H govern the thermal balance. The first determines the amount of heat generated by the applied current, whereas the second regulates the rate at which the particle releases heat to the surrounding environment.

The adopted values were selected so that the simulated particle experiences a moderate temperature excursion during the charging process. Consequently, the temperature remains within the typical operating range of lithium-ion batteries while still producing a measurable variation of the solid-state diffusion coefficient. This provides a more challenging reference problem for the subsequent Physics-Informed Neural Network without introducing unrealistically large thermal gradients or excessive nonlinearities.

14 Results of the Extended Reference Model

The thermo-diffusive model presented in the previous section was solved using the same numerical configuration adopted for the constant diffusivity reference model. The spatial domain was discretized with $N = 100$ finite-volume cells (101 computational nodes), while the solution was stored at 201 uniformly distributed time instants over the interval

$$0 \leq t \leq 6000 \text{ s.}$$

Unlike the previous simulations, however, the diffusion coefficient is continuously updated according to both the local lithium concentration and the particle temperature. Consequently, the transport operator changes throughout the simulation, producing a nonlinear coupled thermo-diffusion problem.

The objective of this section is twofold. First, the numerical stability of the extended model is verified. Second, the influence of the adaptive diffusion coefficient on the concentration dynamics is analysed before using the generated data as the reference dataset for the Physics-Informed Neural Network.

14.1 Temperature Evolution

Figure 8 reports the evolution of the lumped particle temperature during the complete charging and relaxation cycle.

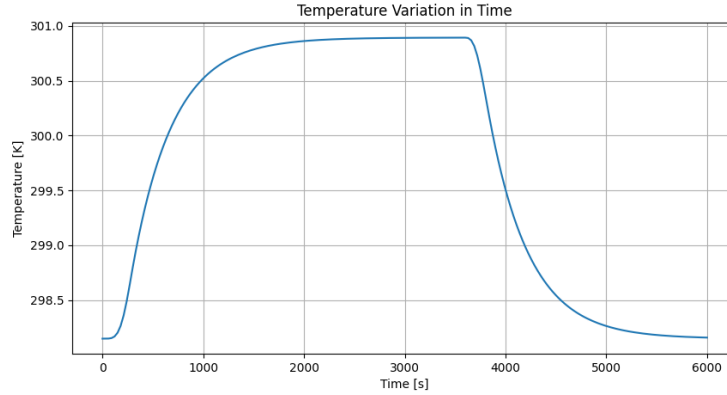


Figure 8: Temporal evolution of the particle temperature.

Initially, the particle is in thermal equilibrium with the surrounding environment,

$$T(0) = T_{\text{amb}} = 298.15 \text{ K.}$$

As the charging current increases, irreversible Joule heating becomes the dominant contribution in the thermal balance, producing a gradual temperature rise. The maximum temperature reached during the simulation is

$$T_{\text{max}} = 300.892 \text{ K,}$$

corresponding to an increase of approximately

$$\Delta T_{\text{max}} = 2.74 \text{ K}$$

with respect to the ambient temperature.

After the charging current decreases, the heat-generation term rapidly vanishes, whereas convective cooling continues to remove heat from the particle. Consequently, the temperature progressively returns towards the ambient value, reaching

$$T_{\text{final}} = 298.160 \text{ K}$$

at the end of the simulation.

The limited temperature excursion confirms that the adopted thermal parameters produce physically realistic operating conditions while still introducing a measurable variation of the diffusion coefficient through the Arrhenius correction.

14.2 Temperature Distribution

Since the thermal model is lumped, the particle temperature is assumed to be spatially uniform at every instant. The temperature map shown in Figure 9 therefore exhibits no radial gradients.

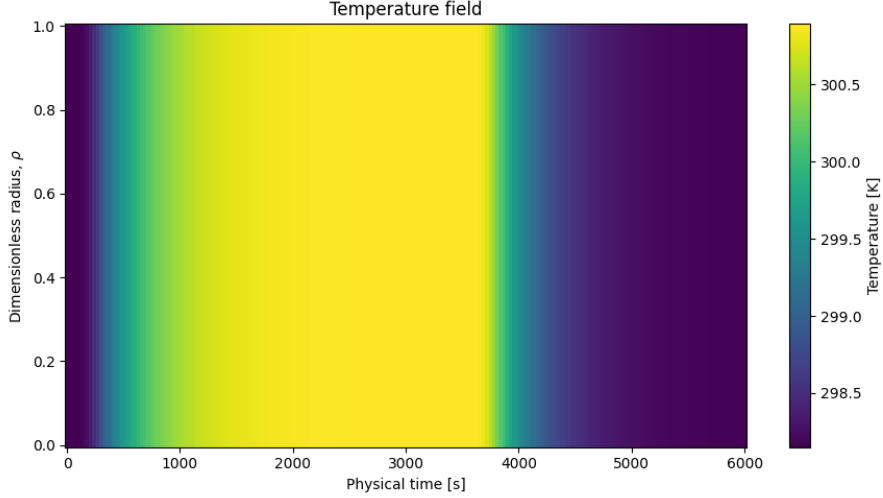


Figure 9: Spatio-temporal representation of the lumped temperature state.

Although represented over the entire radial domain for visualization purposes,

$$T(\rho, t) = T(t)$$

holds for every radial position.

The figure should therefore not be interpreted as the solution of a heat-conduction equation, but rather as a graphical representation of the single thermal state replicated along the spatial coordinate.

14.3 Adaptive Diffusion Behaviour

The increase in temperature produces a corresponding increase in the solid-state diffusion coefficient through the Arrhenius correction. Simultaneously, the progressive lithiation of the particle modifies the concentration-dependent contribution.

Consequently, the effective diffusion coefficient evolves continuously throughout the simulation according to

$$D_s = D_s(C, T).$$

Unlike the constant-diffusivity model, different regions of the particle may therefore exhibit different local transport rates.

Near the particle surface, where lithium concentration initially increases more rapidly, the concentration-dependent correction produces a larger diffusion coefficient than in the particle interior. At the same time, the global temperature increase amplifies diffusion over the entire domain.

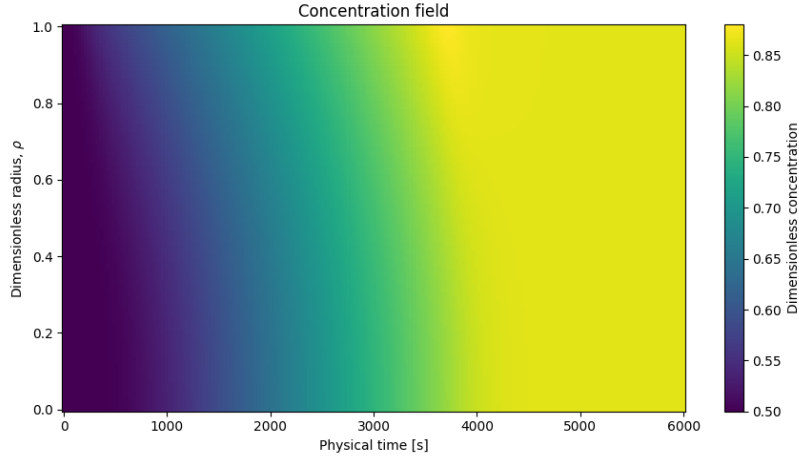


Figure 10: Concentration map in the thermo-diffusive model

The combined action of these two mechanisms accelerates the redistribution of lithium from the surface towards the particle centre, reducing concentration gradients during the charging phase.

14.4 Concentration Evolution

Despite the introduction of variable diffusivity, the qualitative behaviour of the concentration field remains consistent with the expected physical evolution.

Lithium first accumulates near the particle surface because of the prescribed intercalation flux. As diffusion progresses, the concentration front gradually propagates towards the particle centre until the entire particle approaches equilibrium during the relaxation phase.

The maximum concentration observed during the simulation is

$$C_{\max} = 0.879933,$$

and occurs at the particle surface at approximately

$$t = 3720 \text{ s}.$$

The final concentrations become

$$C_{\text{centre}} = 0.863478,$$

and

$$C_{\text{surface}} = 0.863479,$$

yielding a difference smaller than

$$10^{-6},$$

which indicates that the particle has essentially reached a uniform lithium distribution by the end of the simulation.

14.5 Mass Conservation and Numerical Stability

The conservative finite-volume discretization remains valid after the introduction of state-dependent diffusivity.

Throughout the simulation, the concentration remained bounded within the physically admissible interval,

$$0.5 \leq C \leq 0.879933,$$

while no numerical instabilities were observed.

In particular,

$$\text{NaN} = 0, \quad \text{Inf} = 0,$$

for every stored solution.

The maximum absolute mass-balance residual was

$$\boxed{4.47 \times 10^{-5}},$$

confirming that the conservative formulation preserves lithium mass despite the nonlinear diffusion coefficient.

These results demonstrate that the proposed thermo-diffusion model is numerically stable over the complete simulation interval and can therefore be employed as a reliable high-fidelity generator of synthetic training data.

14.6 Reference Dataset for PINN Training

The extended thermo-diffusive model is employed to generate the reference concentration data used throughout the subsequent PINN experiments. For each time instant and radial position, the numerical solver provides the complete concentration field together with the corresponding operating conditions, including the particle temperature.

Although the temperature is computed during every simulation, it is not included among the inputs supplied to the Physics-Informed Neural Network. The training dataset therefore consists exclusively of the spatial and temporal coordinates,

$$(\rho, \tau),$$

while the target variable is the corresponding lithium concentration,

$$C(\rho, \tau).$$

Consequently, the PINN is required to reconstruct concentration fields generated by the extended thermo-diffusive model while enforcing a simplified physical formulation based on a constant diffusion coefficient. The temperature is deliberately ignored during training, thereby creating a controlled discrepancy between the reference model used for data generation and the physical model embedded within the network.

This deliberate mismatch constitutes the core experimental setting of the present work and provides the basis for evaluating the capability of Physics-Informed Neural Networks to compensate for missing physical information through observational data.

15 Physics-Informed Neural Network with Simplified Physics

The high-fidelity thermo-diffusive finite-volume model introduced in Section 13 provides a physically consistent reference concentration field. The Physics-Informed Neural Network (PINN) developed in this work is not supplied with the complete physics used by that solver. Instead, the network is deliberately constrained by a reduced constant-diffusivity model and is simultaneously trained on concentration samples generated by the full-physics simulator.

This configuration differs from the conventional use of a PINN as a direct surrogate of a known differential equation. The objective is not to reproduce the finite-volume solution using an identical mathematical model, but to investigate whether observational data can compensate, at least partially, for deliberately missing physical information. The conventional PINN formulation combines observational, initial, boundary and differential-equation residuals in a common optimization objective [Cuomo et al., 2022; Khanra et al., 2026]. Here, the resulting prediction is therefore the outcome of a competition between two sources of information:

- the full-physics concentration data, which contain the effects of the operating current, temperature evolution and state-dependent diffusivity;
- the simplified differential equation and boundary condition, which impose constant normalized diffusivity and a constant surface flux.

This tension is intentional and constitutes the central methodological experiment of the present work.

15.1 Simplified Physical Model Enforced by the PINN

The PINN uses the dimensionless radial coordinate and time as inputs,

$$\mathbf{x} = [\rho \quad \tau]^T, \quad 0 \leq \rho \leq 1, \quad 0 \leq \tau \leq \tau_f,$$

and predicts the dimensionless solid concentration

$$\hat{C} = \hat{C}_\theta(\rho, \tau),$$

where θ denotes the trainable neural-network parameters.

The governing equation embedded in the PINN is the constant-coefficient spherical diffusion equation,

$$\boxed{\frac{\partial \hat{C}}{\partial \tau} - \frac{\partial^2 \hat{C}}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial \hat{C}}{\partial \rho} = 0.} \quad (1)$$

The initial concentration is

$$\hat{C}(\rho, 0) = C_0, \quad C_0 = 0.5, \quad (2)$$

and spherical symmetry requires

$$\left. \frac{\partial \hat{C}}{\partial \rho} \right|_{\rho=0} = 0. \quad (3)$$

At the particle surface, the PINN enforces the constant Neumann condition

$$\boxed{-\left. \frac{\partial \hat{C}}{\partial \rho} \right|_{\rho=1} = \delta_0, \quad \delta_0 = -0.2004616.} \quad (4)$$

The value δ_0 corresponds to the nominal 1C dimensionless flux evaluated using the reference diffusivity. It is held constant over the complete time interval, including the ramp-up, ramp-down and relaxation phases of the full-physics simulation.

Table 5 summarizes the deliberate mismatch between the model used to generate the data and the physics enforced during PINN training.

Table 5: Deliberate physical mismatch between the reference model and the PINN.

Modelling component	Full-physics FVM reference	Physics enforced by the PINN
Diffusion coefficient	$\tilde{D} = \tilde{D}(C, T)$	$\tilde{D} = 1$
Diffusion equation	$C_\tau = \rho^{-2} \partial_\rho (\rho^2 \tilde{D} C_\rho)$	$C_\tau = C_{\rho\rho} + 2C_\rho/\rho$
Temperature	Computed through a lumped thermal balance	Not supplied to the network
Surface flux	Ramp-constant-ramp-rest, $\delta = \delta(\tau)$	Constant, $\delta = \delta_0$
Initial condition	Uniform concentration $C_0 = 0.5$	Imposed exactly by construction
Centre symmetry	Zero radial derivative	Imposed exactly by construction

The thermal state and the adaptive diffusivity are therefore available only indirectly through the target concentration values. This choice prevents the network from simply reproducing the complete reference equations and makes it possible to evaluate the compensating role of data.

15.2 Exact Enforcement of Initial and Centre Conditions

The initial condition and the centre symmetry condition are not included as penalty terms in the optimization objective. They are embedded analytically in the network representation and are therefore satisfied for every value of the trainable parameters.

Let N_θ denote the unconstrained output of the multilayer perceptron. The final concentration prediction is constructed as

$$\boxed{\hat{C}_\theta(\rho, \tau) = C_0 + \tau N_\theta(\rho^2, \tau).} \quad (5)$$

At $\tau = 0$, Equation (5) immediately gives

$$\hat{C}_\theta(\rho, 0) = C_0,$$

independently of N_θ . The initial condition is consequently exact rather than approximately recovered through training.

The use of ρ^2 as the spatial network input produces

$$\frac{\partial \widehat{C}_\theta}{\partial \rho} = 2\rho\tau \frac{\partial N_\theta}{\partial(\rho^2)},$$

and therefore

$$\left. \frac{\partial \widehat{C}_\theta}{\partial \rho} \right|_{\rho=0} = 0.$$

This transformation removes the need for an additional centre-boundary loss and guarantees regular behaviour at the origin. Only the outer Neumann boundary condition contributes to the optimization loss. This distinction is important: no initial-condition loss and no centre symmetry loss are introduced anywhere in the training procedure.

15.3 Neural-Network Architecture

The approximation N_θ is represented by a fully connected multilayer perceptron. After the analytical input transformation, the network receives the two-dimensional vector

$$\widetilde{\mathbf{x}} = [\rho^2 \quad \tau]^T.$$

Four hidden layers are used, each containing 100 neurons with hyperbolic tangent activation. The output layer is linear and returns one scalar. The architecture is therefore

$$2 \longrightarrow 100 \longrightarrow 100 \longrightarrow 100 \longrightarrow 100 \longrightarrow 1.$$

All linear-layer weights are initialized using Xavier normal initialization with the gain associated with the tanh activation, while every bias is initialized to zero. The complete model contains 30,701 trainable parameters.

Table 6: PINN architecture and initialization.

Component	Configuration	Purpose
Raw inputs	(ρ, τ)	Space-time coordinates
Transformed inputs	(ρ^2, τ)	Exact centre symmetry
Hidden layers	4 layers, 100 neurons each	Nonlinear approximation
Activation	tanh	Smooth automatic derivatives
Output layer	1 linear neuron	Unconstrained scalar N_θ
Output transform	$C_0 + \tau N_\theta$	Exact initial condition
Weight initialization	Xavier normal	Stable initial signal scale
Bias initialization	Zero	Deterministic neutral offset
Trainable parameters	30,701	Complete parameter count

The tanh activation is particularly suitable for this problem because the PDE residual requires first-order temporal derivatives and first- and second-order radial derivatives. A smooth activation avoids the derivative discontinuities that would arise with piecewise-linear functions.

15.4 Training, Validation and Test Datasets

Three datasets are used independently during optimization.

1. **Reference-data points.** The complete full-physics finite-volume field provides tuples $(\rho, \tau, C_{\text{FVM}})$. These samples directly communicate the target high-fidelity behaviour to the network.
2. **Interior collocation points.** The spatial and temporal coordinates are extracted from the reference grid after excluding $\rho = 0$, $\rho = 1$ and $\tau = 0$. Removing the origin avoids the explicit evaluation of the singular coefficient $2/\rho$, while the centre condition remains exactly satisfied through the output transformation.
3. **Surface-boundary points.** Independent time coordinates are generated with a one-dimensional Sobol sequence and combined with $\rho = 1$. These points are used exclusively to enforce the simplified outer Neumann condition.

Each dataset is independently divided into 70% training, 15% validation and 15% test subsets using a fixed random seed. The resulting sample counts are reported in Table 7.

Table 7: Independent data, collocation and boundary dataset splits.

Dataset	Total	Training	Validation	Test
Full-physics data	20,301	14,210	3,045	3,046
Interior collocation	19,800	13,860	2,970	2,970
Surface boundary	20,301	14,210	3,045	3,046

Mini-batches of 512 samples are used during the Adam stage. The three loaders are evaluated jointly, but each loss contribution is computed from the dataset appropriate to that contribution. Thus, the same network is evaluated at data, collocation and boundary coordinates; no separate neural networks are introduced for the different terms.

15.5 Physics-Informed Objective Function

The data loss measures the discrepancy between the PINN prediction and the full-physics finite-volume concentration,

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \left| \hat{C}_\theta(\rho_i, \tau_i) - C_{\text{FVM},i} \right|^2. \quad (6)$$

At the interior collocation points, automatic differentiation is used to evaluate

$$\frac{\partial \hat{C}_\theta}{\partial \tau}, \quad \frac{\partial \hat{C}_\theta}{\partial \rho}, \quad \frac{\partial^2 \hat{C}_\theta}{\partial \rho^2}.$$

The simplified PDE residual is

$$R_f = \frac{\partial \hat{C}_\theta}{\partial \tau} - \frac{\partial^2 \hat{C}_\theta}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial \hat{C}_\theta}{\partial \rho}, \quad (7)$$

This use of automatic differentiation to evaluate governing-equation residuals at collocation points is a defining element of PINN methodology [Cuomo et al., 2022; Khanra et al., 2026].

The associated loss is

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} |R_{f,i}|^2. \quad (8)$$

At the independently generated surface points, the boundary residual is

$$R_b = - \left. \frac{\partial \hat{C}_\theta}{\partial \rho} \right|_{\rho=1} - \delta_0, \quad (9)$$

which yields

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} |R_{b,i}|^2. \quad (10)$$

The complete optimization objective is

$$\boxed{\mathcal{L} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_f \mathcal{L}_{\text{PDE}} + \lambda_b \mathcal{L}_{\text{BC}},} \quad (11)$$

with

$$\lambda_d = \lambda_f = \lambda_b = 1.$$

Equal weighting provides a transparent baseline from which the natural competition between data fidelity and simplified-physics consistency can be observed. No adaptive loss weighting is used in the present model. Most importantly, Equation (11) contains neither an initial-condition term nor a centre-boundary term, because both conditions are hard constraints.

15.6 Learning and Model-Selection Procedure

The learning procedure combines a stochastic first-order stage with an optional deterministic full-batch refinement.

15.6.1 Adam stage

Adam is used with an initial learning rate of

$$\eta_0 = 10^{-3}$$

and a maximum of 10,000 epochs. At the end of every epoch, the three loss components are independently evaluated on the validation subsets. The reported epoch losses are sample-weighted averages rather than sums of mini-batch averages.

A `ReduceLROnPlateau` scheduler monitors the total validation loss. When progress stagnates, it multiplies the learning rate by 0.5, with a patience of 200 epochs, a relative threshold of 10^{-4} , a cooldown of 50 epochs and a minimum learning rate of 10^{-6} .

Model selection is intentionally based on the validation data loss. A copy of the model parameters is retained whenever $\mathcal{L}_{\text{data, val}}$ improves. This choice reflects the main objective of the

experiment: approximating the full-physics reference field while still being regularized by simplified physical information.

If the data-validation loss fails to improve for 800 epochs, a plateau is declared and the Adam stage is terminated.

15.6.2 L-BFGS refinement

After a detected plateau, L-BFGS is applied to the complete training sets. The optimizer uses a maximum of 500 iterations, a history size of 100 and a strong-Wolfe line search. The full-batch closure recomputes the data, PDE and outer-boundary losses together with all required automatic derivatives.

The L-BFGS solution is not accepted unconditionally. Its validation data loss is compared with that of the best Adam checkpoint. If refinement worsens agreement with the full-physics observations, the best Adam parameters are restored. In the experiment reported here, this safeguard was activated and the final stored model corresponds to the best Adam state.

The final checkpoint records the model parameters, the validation losses, the physical constants, the three loss weights and the selected optimizer. Timestamped filenames prevent different experiments from overwriting one another.

Table 8: Optimization and model-selection settings.

Item	Value	Role
Adam initial learning rate	10^{-3}	Mini-batch optimization
Maximum epochs	10,000	Upper training limit
Mini-batch size	512	Adam data processing
Scheduler factor	0.5	Learning-rate reduction
Scheduler patience	200 epochs	Plateau sensitivity
Early-stopping patience	800 epochs	Switch criterion
Minimum learning rate	10^{-6}	Scheduler lower bound
L-BFGS maximum iterations	500	Full-batch refinement
L-BFGS line search	Strong Wolfe	Step selection
Selection metric	Data-validation MSE	Full-physics fidelity
Final selected optimizer	Adam	Best retained checkpoint

15.7 Numerical Diagnostics

Three diagnostic checks are evaluated during learning to detect unstable optimization before it contaminates the stored model.

1. The concentration output is checked for non-finite values and its minimum, maximum, mean and standard deviation are monitored. A warning is emitted when predictions move significantly outside the normalized physical interval.
2. The root-mean-square and maximum absolute PDE residual are monitored to identify a sudden loss of physical consistency.

3. After backpropagation, the total gradient norm and the layer-wise gradient norms are inspected for non-finite, exploding or vanishing gradients.

After training, independent test subsets are used to evaluate the relative concentration error, relative surface-concentration error, PDE RMSE, boundary-condition RMSE and synchronized inference time. Separate post-processing checks verify the exact constraints, the physical concentration interval and the integral mass balance.

16 PINN Results and Comparison with the Full-Physics Model

The final PINN was evaluated in two complementary ways. First, the randomly held-out data, collocation and boundary test subsets were used to quantify generalization within the sampled domain. Second, the model was evaluated on the complete structured 101×201 finite-volume grid to obtain directly comparable fields, profiles and integral quantities.

This distinction is useful because the test split provides an unbiased estimate at unseen samples, while the structured-grid analysis exposes where and when the model discrepancy develops.

16.1 Validation and Held-Out Test Performance

The selected checkpoint achieved a data-validation mean-square error of

$$\mathcal{L}_{\text{data, val}} = 1.4707 \times 10^{-3},$$

and a total validation loss of

$$\mathcal{L}_{\text{val}} = 3.2028 \times 10^{-3}.$$

Table 9 summarizes the independent test-set results. The concentration error is measured against the full-physics finite-volume targets, whereas the PDE and boundary residuals are computed with respect to the simplified physics enforced by the PINN.

Table 9: PINN performance on the independent test subsets.

Metric	Value	Interpretation
Full-field relative L^2 error	4.9633×10^{-2} (4.9633%)	Agreement with held-out FVM data
Surface relative L^2 error	7.4243×10^{-2} (7.4243%)	Agreement at $\rho = 1$
Simplified-PDE RMSE	1.9074×10^{-2}	Interior physics residual
Constant-flux BC RMSE	3.6957×10^{-2}	Outer-boundary residual
Total inference time	4.5469 ms	3,046 held-out data samples on MPS
Inference time per sample	1.4927 μ s	Synchronized forward evaluation

A relative full-field error close to 5% is a meaningful result in the present setting. The network is not being tested against data generated by its own PDE: the targets contain a time-varying current, thermal dynamics and a concentration- and temperature-dependent diffusivity that are all absent from the PINN formulation. The result therefore shows that the observational term successfully transfers a substantial part of the missing behaviour into the learned approximation.

The larger surface error is physically interpretable. The surface is the location where the imposed constant PINN flux conflicts most directly with the ramped and subsequently vanishing reference flux. Errors are partially smoothed as lithium diffuses towards the particle interior, explaining why the centre and full-field measures are lower than the surface measure.

16.2 Standalone PINN Prediction

Figure 11 shows the concentration field predicted by the PINN over the complete space–time domain.

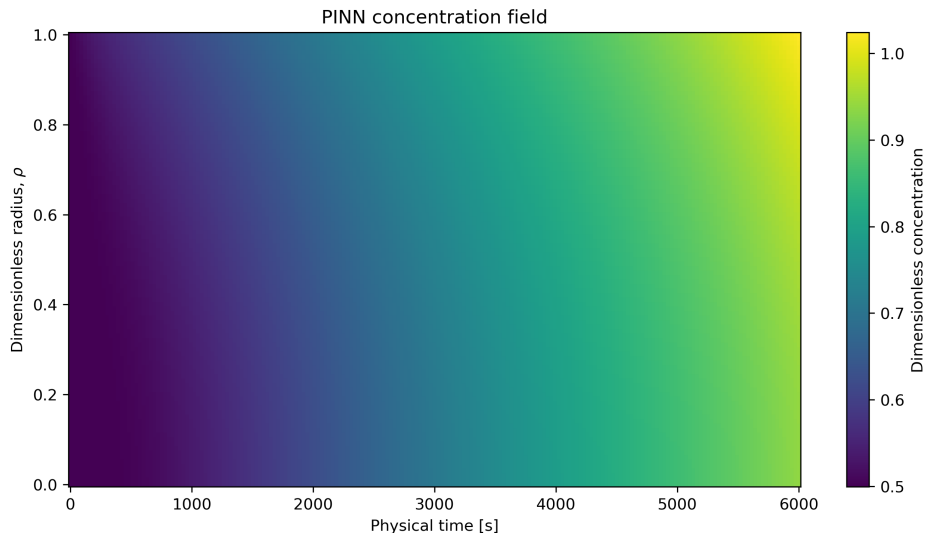


Figure 11: PINN concentration field.

The PINN correctly reproduces the initial uniform concentration and the development of a positive radial gradient, with larger concentration at the particle surface. Unlike the full-physics reference solution, however, the PINN concentration continues to increase during the final part of the simulation. This behaviour follows directly from the constant-flux condition: the simplified model continues to insert lithium even after the reference current has ramped down to zero.

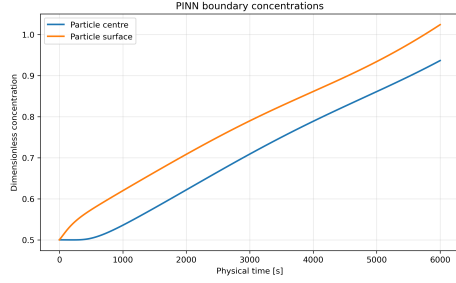
The boundary and volume-averaged concentrations are reported in Figure 12.

The surface concentration responds more rapidly than the centre concentration, as expected for diffusion driven from the external boundary. The volume average evolves smoothly and remains free from oscillations or discontinuities. The late-time monotonic increase is not a numerical instability; it is the deterministic consequence of the simplified boundary input.

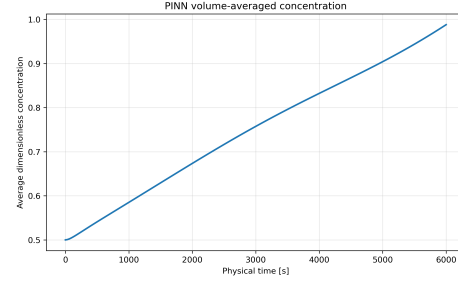
Figure 13 further illustrates that the network produces smooth radial profiles and exactly satisfies the initial uniform state.

At late times, the profiles retain a non-zero surface-to-centre difference, whereas the full-physics solution becomes almost spatially uniform after removal of the applied current. This discrepancy will be examined explicitly in the direct comparison.

The simplified-model mass-balance residual and the constant-flux Guo–White comparison are included in Figure 14.



(a) Centre and surface concentrations.



(b) Volume-averaged concentration.

Figure 12: Standalone temporal predictions of the PINN.

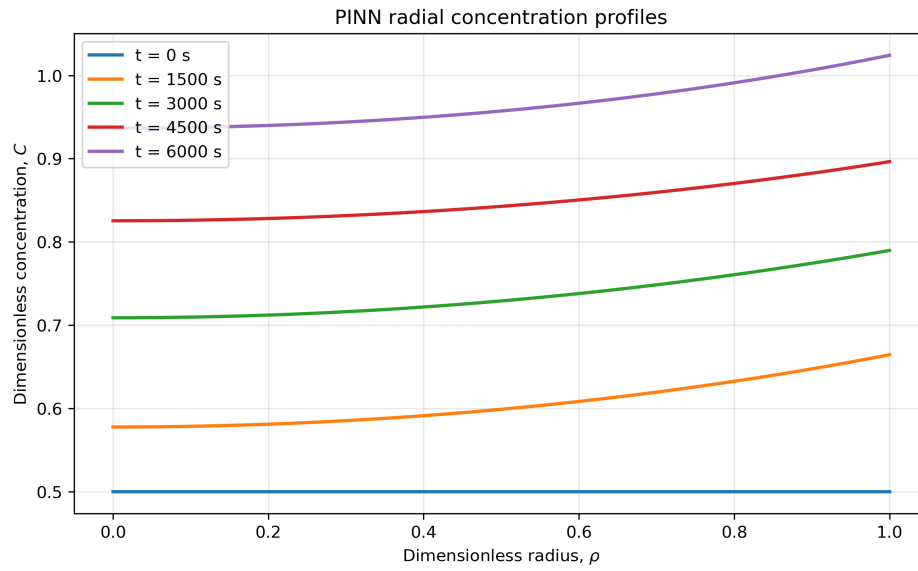


Figure 13: PINN radial concentration profiles at selected physical times.

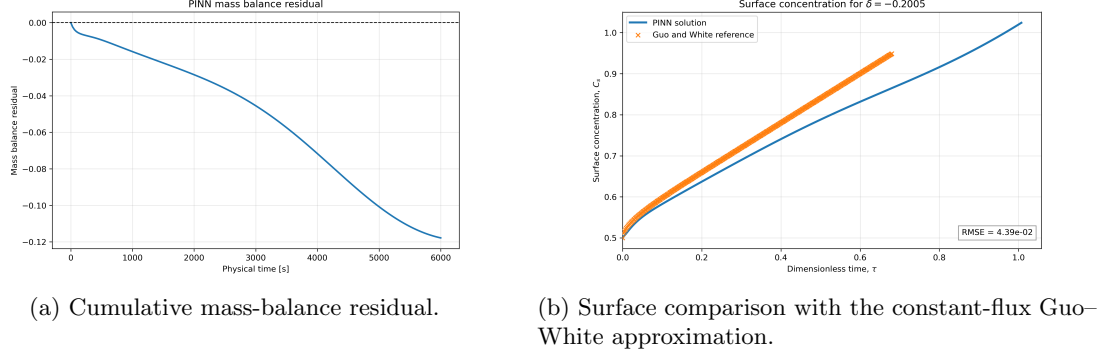


Figure 14: Physics-oriented diagnostics of the standalone PINN.

These diagnostics must be interpreted together with the supervised objective. The learned field is not a pure solution of the simplified constant-flux problem: the data loss pulls the prediction towards the full-physics response. A non-zero simplified-physics mass residual is therefore evidence of the trade-off made by the optimizer, rather than evidence that the finite-volume reference data violate conservation.

16.3 Direct Full-Field Comparison

For direct comparison, the PINN and FVM solutions were evaluated on the same 101×201 grid. Identical concentration limits were used for both colour maps. This shared colour normalization is essential: if each field were plotted using its own minimum and maximum, the two maps would appear artificially similar and the PINN overshoot would be visually hidden.

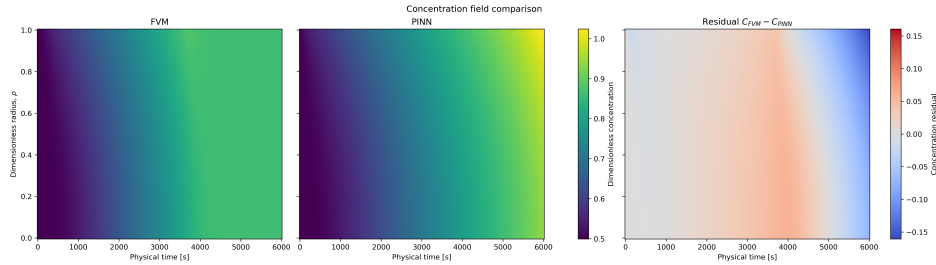


Figure 15: Full-physics FVM field, PINN field and pointwise residual $C_{FVM} - C_{PINN}$. A common colour scale is used for the two concentration maps.

The reference field reaches an almost uniform plateau after the current is removed. In contrast, the PINN continues to evolve because its constant surface flux remains active. The residual map is smooth and strongly structured in time. It remains small during the early loading stage, changes sign around the transition to relaxation and becomes largest close to the particle surface at late times. This pattern is a direct signature of model-form error rather than unstructured neural noise.

The quantitative dense-grid errors are reported in Table 10.

Table 10: PINN errors with respect to the full-physics FVM solution on the complete structured grid.

Quantity	Relative L^2 error	RMSE	MAE	Maximum absolute error
Complete field	0.050674	0.037757	0.027721	0.160606
Particle centre	0.045458	0.033251	0.026263	0.073184
Particle surface	0.068609	0.052938	0.034088	0.160606
Volume average	0.053791	0.040645	0.028642	0.124484

The 5.07% relative error computed on the complete grid is close to the 4.96% error obtained on the randomly held-out test samples. This agreement indicates that the random split does not conceal a substantial generalization failure. It also confirms that the comparison plots and the independent test routine describe the same overall model quality.

The surface remains the most demanding output, with a dense-grid relative error of 6.86% and a maximum absolute error of 0.1606. The largest discrepancy occurs at the end of the simulation, when the FVM has relaxed to approximately 0.8635 while the PINN surface prediction reaches approximately 1.0241. The centre error remains lower because the interior response is delayed and spatial diffusion attenuates the direct effect of the boundary mismatch.

16.4 Integral and Boundary Quantities

Figure 16 compares the quantities most directly related to lithium inventory and boundary behaviour.

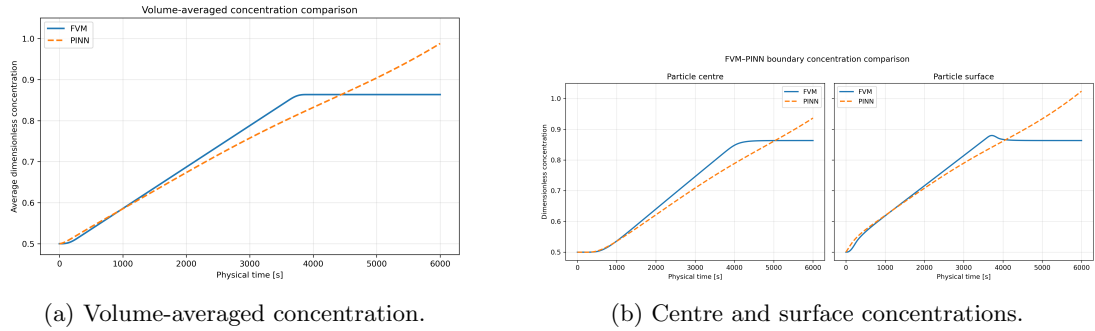


Figure 16: Direct comparison of integral and boundary concentration quantities.

The volume-average curves are close during the first part of the loading phase. The reference average then approaches a constant value as the current is removed, while the PINN average continues to increase. A similar trend is observed at both boundaries. The reference surface concentration displays a local maximum during ramp-down followed by relaxation, whereas the constant-flux PINN cannot reproduce this turning point exactly.

This observation identifies the time-dependent boundary input as the dominant source of late-time error. Adaptive diffusivity also affects the curvature and radial redistribution, but the persistent increase after the reference current reaches zero is specifically attributable to the simplified constant-flux assumption.

Radial profiles and the surface residual are reported in Figure 17.

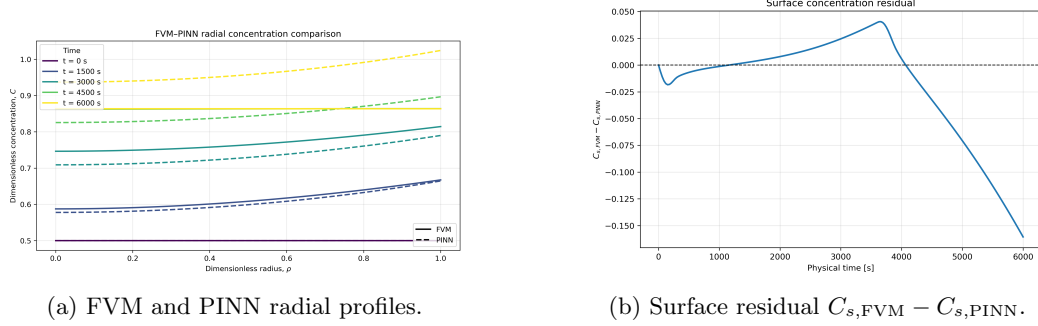


Figure 17: Spatial comparison and temporal development of the surface discrepancy.

The residual changes gradually rather than oscillating rapidly. This is important for the intended hybrid extension: a smooth, phase-dependent error is more amenable to a compact corrective model than high-frequency or apparently random errors.

16.5 Error Distribution over the Operating Phases

To connect prediction error with the imposed operating conditions, the full-field discrepancy was evaluated separately during ramp-up, constant-current operation, ramp-down and rest. The resulting metrics are shown in Table 11.

Table 11: Full-field PINN error over the different operating phases.

Operating phase	Relative L^2 error	RMSE	Physical interpretation
Ramp-up	0.009335	0.004683	Initial response captured accurately
Constant current	0.033957	0.022696	Moderate accumulated mismatch
Ramp-down	0.059622	0.050293	Boundary inputs begin to diverge
Rest	0.062203	0.053651	PINN constant flux remains active

The ramp-up relative error is below 1%, demonstrating that the network captures the initial thermo-diffusive response very accurately even though neither temperature nor adaptive diffusivity is provided as an input. The error rises to approximately 3.4% during the constant-current stage and to approximately 6% during ramp-down and rest.

This monotonic relation between operating-phase mismatch and prediction error is one of the most informative results of the experiment. The network performs best when the simplified constant flux is close to the reference operating condition and deteriorates when the two physical descriptions diverge. The behaviour therefore remains physically interpretable and supports the conclusion that the remaining error is primarily model-form error.

16.6 Physical Consistency Checks

The PINN and full-physics FVM fields were subjected to the same post-processing checks. The volume average was evaluated as

$$\overline{C}(\tau) = \frac{\int_0^1 C(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho}, \quad (12)$$

and the cumulative mass-balance residual was defined consistently as

$$R_{\text{mass}}(\tau) = \overline{C}(\tau) - \left[C_0 - 3 \int_0^\tau \delta(s) ds \right]. \quad (13)$$

For the FVM, $\delta(\tau)$ is the ramp-constant-ramp-rest profile. For the PINN, $\delta(\tau) = \delta_0$ is the constant flux actually used in its loss function. Thus, both residuals use the same mathematical definition but are evaluated against the boundary law belonging to the corresponding model.

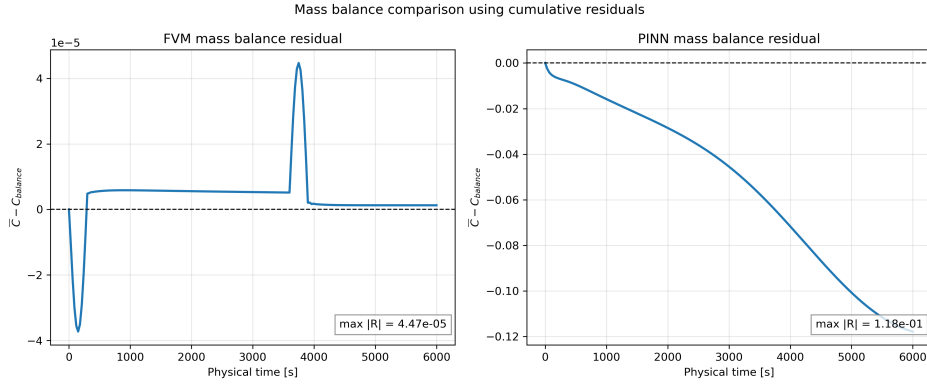


Figure 18: FVM and PINN cumulative mass-balance residuals. Each model is evaluated using its corresponding prescribed surface flux.

The numerical checks are summarized in Table 12.

Table 12: Physical and hard-constraint checks on the structured grid.

Check	FVM	PINN	Comment
Minimum concentration	0.500000	0.499530	Both remain close to the initial lower level
Maximum concentration	0.879933	1.024085	PINN exceeds the normalized upper bound
Maximum $ R_{\text{mass}} $	4.47×10^{-5}	1.1783×10^{-1}	FVM conservative; PINN balances data and simplified physics
Initial-condition maximum error	—	0	Exact hard constraint
Centre-gradient maximum error	—	0	Exact hard constraint

The exact zero errors associated with the initial state and centre symmetry verify that the analytical output transformation behaves as intended. These conditions do not deteriorate during optimization and do not compete with the three soft loss terms.

The PINN maximum concentration of 1.0241 represents a limited but physically relevant overshoot. It occurs near the surface at late time, where the simplified model still applies an insertion flux after the full-physics model has entered relaxation. This result should not be hidden by rescaling the comparison plots. It identifies a clear regime in which a corrective strategy is needed.

The larger PINN mass residual also has a precise interpretation. A pure solution of the constant-flux PDE would follow its own integral balance, whereas the supervised full-physics data tend towards a plateau. The trained network adopts a compromise between these incompatible targets. The residual therefore quantifies the effect of combining deliberately inconsistent data and physics, while the much smaller FVM residual confirms that the reference dataset itself remains conservative.

16.7 Assessment of the Simplified-Physics Strategy

The results support four main conclusions.

1. **The dominant diffusion behaviour is learned.** The PINN reproduces the initial condition, the direction and smoothness of radial transport, the delayed centre response and the main increase in particle lithium content. A global relative error of approximately 5% is obtained despite the deliberately incomplete governing model.
2. **Data compensate for a significant part of the missing physics.** During ramp-up, the relative field error remains below 1%, even though the reference temperature and variable diffusivity are never provided to the network. The observational term therefore conveys information that cannot be generated by the simplified PDE alone.
3. **The remaining error is structured and physically explainable.** The largest discrepancy appears at the surface and after the reference current begins to decrease. This is exactly where the constant-flux approximation becomes least representative. The residual is consequently linked to identifiable missing mechanisms rather than to numerical instability or random overfitting.
4. **The model retains useful physical structure.** The hard initial and symmetry conditions are satisfied to machine precision, while the PDE and boundary residuals remain finite and controlled. The model is therefore more than an unconstrained interpolator, even though data fidelity is deliberately prioritized during checkpoint selection.

Within the objective of this study, this constitutes a good result. A nearly exact PINN would leave little systematic discrepancy for the next stage and would make it difficult to study hybrid correction. Conversely, a poorly performing base model would force a subsequent network to learn the complete solution rather than a correction. The present PINN occupies a useful intermediate regime: it captures the dominant field with approximately 5% global error, while preserving a smooth and physically localized residual associated with the intentionally omitted physics.

The result should nevertheless be interpreted as a baseline rather than as a final high-fidelity surrogate. The late-time overshoot, surface error and mass-balance discrepancy prevent the standalone PINN from replacing the extended finite-volume model over the complete operating cycle. Their value lies in revealing where the simplified description ceases to be sufficient.

16.8 Transition to the Implemented Residual-Learning Extension

The structured discrepancy identified above is subsequently learned by a dedicated data-driven model. This separation between a misspecified PINN and a discrepancy network is consistent with corrective strategies developed for model-form error in PINNs [Zou et al., 2024]. For the PINN-based corrective branch, the supervised target is

$$r_C(\rho, \tau) = C_{\text{FVM}}(\rho, \tau) - \widehat{C}_\theta(\rho, \tau). \quad (14)$$

and the corrected prediction is defined as

$$C_{\text{hybrid}}(\rho, \tau) = \widehat{C}_\theta(\rho, \tau) + \widehat{r}_\phi(\rho, \tau), \quad (15)$$

where $\widehat{r}_\phi$ denotes the learned correction. The selected PINN checkpoint is frozen before the residual dataset is generated and is not retrained during this stage. The additional network therefore approximates only the part of the full-physics response that the reduced PINN does not represent.

The comparative analysis explains why residual learning is suitable in this case. The residual map is smooth, its magnitude changes systematically with the operating phase, and its largest values are localized near the surface and during relaxation. Consequently, the correction task is substantially more structured than learning the full concentration field from scratch. The simplified PINN can retain the dominant diffusion behaviour and exact analytical constraints, while the residual model focuses specifically on the effects omitted from the reduced physics.

The detailed architecture, data flow, numerical results and physical checks of this implemented post-training correction are reported in Section 21. This branch must not be confused with the standalone FVM–FNN hybrid model introduced next: the latter corrects a simplified numerical solver and returns a complete radial profile, whereas the Hybrid-PINN corrects a frozen coordinate-based PINN one point at a time.

17 Hybrid Physics–ML Modelling through Residual Learning

The simplified-physics PINN investigated in the previous section combines observational data and reduced governing equations within a single optimization problem. The present section develops one realization of the corrective stage by adopting an independently executable FVM–FNN hybrid strategy. A complete, independently executable simplified finite-volume model is first used to approximate the dominant concentration dynamics. A feedforward neural network is then trained exclusively on the discrepancy between this baseline and the high-fidelity thermo-diffusive reference solution. Related hybrid frameworks likewise learn model-form residuals or missing operators while retaining an approximate physics-based core [Garg et al., 2022; Ghnatios and Chinesta, 2025].

This formulation is distinct from the post-training Hybrid-PINN in Section 21. In the present section the baseline is a simplified finite-volume solver and the network output is a 101-node radial correction; in the later extension the baseline is a frozen PINN and the residual network outputs one scalar correction for each coordinate pair.

The resulting model has the additive structure

$$\boxed{\mathbf{C}_{\text{hybrid}}(\tau) = \mathbf{C}_{\text{simpl}}(\tau) + \widehat{\Delta \mathbf{C}}_\phi(\tau)}, \quad (16)$$

where $\mathbf{C}_{\text{simpl}}$ denotes the radial concentration profile obtained from the simplified finite-volume model and $\widehat{\Delta \mathbf{C}}_\phi$ is the correction produced by the neural network with parameters ϕ .

This construction differs from a purely data-driven surrogate. The simplified solver remains responsible for time integration, spherical geometry, conservation, the imposed operating current and the dominant diffusion process. The neural network is assigned only the remaining model-form

discrepancy generated by the neglected state dependence of the diffusion coefficient and by the omitted thermal dynamics. Such state-aware integration of physics-based models and feedforward networks has also been explored for lithium-ion battery modelling [Tu et al., 2024]; related grey-box battery models replace unknown transport dependencies with trainable neural components [Brucker et al., 2023].

17.1 Hybrid Modelling Objective

The high-fidelity reference field is denoted by

$$\mathbf{C}_{\text{ref}}(\tau_k) = [C_{\text{ref}}(\rho_0, \tau_k) \quad \cdots \quad C_{\text{ref}}(\rho_N, \tau_k)]^T,$$

whereas the solution of the simplified model is

$$\mathbf{C}_{\text{simpl}}(\tau_k) = [C_{\text{simpl}}(\rho_0, \tau_k) \quad \cdots \quad C_{\text{simpl}}(\rho_N, \tau_k)]^T.$$

The supervised target assigned to the correction network is the pointwise radial residual

$$\boxed{\Delta \mathbf{C}(\tau_k) = \mathbf{C}_{\text{ref}}(\tau_k) - \mathbf{C}_{\text{simpl}}(\tau_k)}. \quad (17)$$

Consequently, a perfect residual prediction would recover the complete reference field exactly. In practice, the role of the neural network is to identify the systematic relationship between the internal state of the simplified model, the applied current and the radial concentration error.

The methodology therefore follows three sequential operations:

1. execute the simplified finite-volume model using the known operating current;
2. use selected state variables of the simplified model as the input of a feedforward neural network;
3. add the predicted radial discrepancy to the simplified concentration profile.

The complete computational structure is shown in Figure 19.

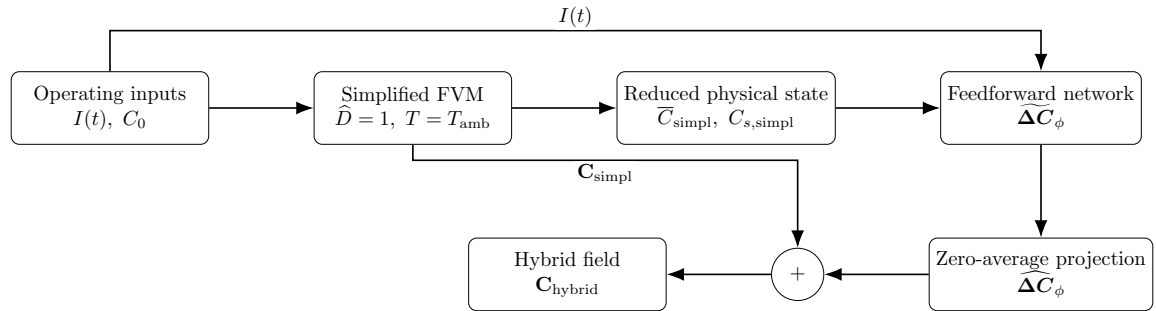


Figure 19: Architecture of the residual hybrid model. The simplified finite-volume solution is corrected by a mass-conserving radial discrepancy predicted from reduced physical states and current.

17.2 Simplified Physics-Based Baseline

The simplified component is obtained from the same spherical finite-volume framework used by the reference simulator, but it is executed with a constant solid-state diffusion coefficient. The dimensionless governing equation is therefore

$$\boxed{\frac{\partial C_{\text{simpl}}}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C_{\text{simpl}}}{\partial \rho} \right)}. \quad (18)$$

The initial condition and centre symmetry condition are

$$C_{\text{simpl}}(\rho, 0) = C_0, \quad C_0 = 0.5, \quad (19)$$

$$\left. \frac{\partial C_{\text{simpl}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (20)$$

At the particle surface, the simplified model retains the known time-dependent operating flux,

$$\boxed{- \left. \frac{\partial C_{\text{simpl}}}{\partial \rho} \right|_{\rho=1} = \delta(\tau)}. \quad (21)$$

This point distinguishes the present hybrid baseline from the constant-flux model enforced in the PINN. The simplified finite-volume model receives the same ramp–constant–ramp–rest current profile used by the reference simulation. Therefore, the boundary excitation is not part of the model-form discrepancy learned by the feedforward network.

The thermal state is removed by imposing

$$\boxed{T(\tau) = T_{\text{ref}} = T_{\text{amb}} = 298.15 \text{ K}, \quad \frac{dT}{d\tau} = 0.} \quad (22)$$

As a result, the normalized diffusion coefficient remains

$$\boxed{\hat{D}_{\text{simpl}} = 1} \quad (23)$$

throughout the simulation. The simplified model consequently omits both the concentration dependence and the Arrhenius temperature correction contained in the reference diffusivity $\hat{D}(C, T)$.

Table 13 summarizes the resulting model-form mismatch.

The simplified baseline therefore remains a physically executable and conservative model. It does not reproduce the complete reference transport rate, but it retains the correct geometry, lithium inventory, operating input and dominant diffusion mechanism.

17.3 Physical and Numerical Parameters

The reference and simplified simulations use identical geometric, electrochemical and numerical parameters. Only the diffusivity mode and thermal evolution differ. The principal settings are summarized in Table 14.

Both concentration fields are consequently available on the same structured grid,

Table 13: Physical content of the reference and simplified models used for residual learning.

Modelling component	Thermo-diffusive reference	Simplified FVM baseline
Spatial domain	Spherical particle	Spherical particle
Numerical method	Conservative FVM	Conservative FVM
Radial grid	101 nodes	101 nodes
Diffusivity	$\hat{D} = \hat{D}(C, T)$	$\hat{D} = 1$
Temperature	Lumped evolving state $T(\tau)$	Constant, $T = T_{\text{amb}}$
Applied current	Operating profile $I(t)$	Same operating profile $I(t)$
Surface flux	$\delta = \delta(\tau)$	Same $\delta = \delta(\tau)$
Initial concentration	Uniform, $C_0 = 0.5$	Uniform, $C_0 = 0.5$
Centre symmetry	Zero radial derivative	Zero radial derivative

Table 14: Physical, operating and numerical parameters used to generate the hybrid-modelling datasets.

Quantity	Symbol	Value
Particle radius	R_p	8.5×10^{-6} m
Active surface area	S_p	1.167 m ²
Reference diffusivity	$D_{s,\text{ref}}$	1.213×10^{-14} m ² s ⁻¹
Maximum concentration	$c_{s,\text{max}}$	5.1410×10^4 mol m ⁻³
Faraday constant	F	96487 C mol ⁻¹
Initial normalized concentration	C_0	0.5
Nominal current	I_{1C}	1.656 A
Current ramp duration	t_r	300 s
End of constant-current stage	t_{load}	3600 s
Final time	t_f	6000 s
Reference and ambient temperature	$T_{\text{ref}}, T_{\text{amb}}$	298.15 K
Radial intervals	N	100
Stored time instants	N_τ	201
Absolute ODE tolerance	–	10 ⁻⁸
Relative ODE tolerance	–	10 ⁻⁶
Time integrator	–	BDF

$$(N_\tau, N_\rho) = (201, 101),$$

containing

$$201 \times 101 = 20,301$$

space-time concentration values for each model.

17.4 State-Aware Residual Representation

The correction network does not receive the radial coordinate as an input. Instead, every evaluation predicts the complete radial discrepancy vector at a given time. The network input is

$$\mathbf{x}_k = \begin{bmatrix} \overline{C}_{\text{simpl}}(\tau_k) \\ C_{s,\text{simpl}}(\tau_k) \\ I(t_k) \end{bmatrix}, \quad (24)$$

where

$$\overline{C}_{\text{simpl}}(\tau) = \frac{\int_0^1 C_{\text{simpl}}(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho} \quad (25)$$

is the volume-averaged concentration and

$$C_{s,\text{simpl}}(\tau) = C_{\text{simpl}}(1, \tau) \quad (26)$$

is the surface concentration.

These quantities provide a compact description of the state of the simplified particle. The average concentration represents the total lithium inventory, whereas the surface concentration contains information about the boundary response and the radial concentration gradient. Their difference was not introduced as an additional input, because it is an exact linear combination of the two retained concentration features and would therefore be redundant.

The current is supplied directly to the network because the same particle state may evolve differently under loading, ramp-down and rest. In the present dataset, $I(t)$ is time dependent and cannot be absorbed into a constant bias.

The network output is

$$\widetilde{\Delta C}_\phi(\mathbf{x}_k) \in \mathbb{R}^{101}, \quad (27)$$

with one output component associated with each ordered radial node. Thus,

$$\widetilde{\Delta C}_{\phi,i} \approx C_{\text{ref}}(\rho_i, \tau_k) - C_{\text{simpl}}(\rho_i, \tau_k), \quad i = 0, \dots, 100.$$

The radial grid is metadata defining the meaning and order of the output neurons. The network is therefore tied to the 101-node discretization used during training.

17.5 Feedforward Neural-Network Architecture

The residual mapping is represented by a fully connected feedforward neural network. Three hidden layers are employed, each containing 32 neurons with rectified-linear-unit activation. The final layer is linear and returns the complete radial correction vector.

The architecture is

$$3 \longrightarrow 32 \longrightarrow 32 \longrightarrow 32 \longrightarrow 101.$$

For an input $\mathbf{x}$, the network evaluation can be written as

$$\mathbf{z}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1), \quad (28)$$

$$\mathbf{z}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{z}_1 + \mathbf{b}_2), \quad (29)$$

$$\mathbf{z}_3 = \text{ReLU}(\mathbf{W}_3 \mathbf{z}_2 + \mathbf{b}_3), \quad (30)$$

$$\widetilde{\Delta \mathbf{C}}_\phi = \mathbf{W}_4 \mathbf{z}_3 + \mathbf{b}_4. \quad (31)$$

The output activation is linear because the concentration discrepancy may be positive or negative. The complete network contains 5,573 trainable parameters.

Table 15: Feedforward correction-network architecture.

Component	Configuration	Role
Inputs	3	$\bar{C}_{\text{simpl}}, C_{s,\text{simpl}}, I$
Hidden layers	3 layers, 32 neurons each	Residual approximation
Hidden activation	ReLU	Nonlinear state mapping
Output layer	101 linear neurons	Complete radial correction
Radial coordinate	Not supplied	Encoded by output ordering
Trainable parameters	5,573	Complete parameter count

No temperature input is supplied to the correction network. The thermal influence contained in the reference concentration field is therefore learned only through its systematic effect on the residual. Accordingly, the present model predicts concentration but does not reconstruct the temperature history.

17.6 Dataset Construction and Splitting

The simplified and reference datasets are first converted from their flattened space-time representation to matrices whose rows correspond to time instants and whose columns correspond to radial nodes. Before constructing the learning problem, three consistency checks are performed:

1. the reference and simplified dimensionless time grids must coincide;
2. the radial grids must contain the same ordered coordinates;
3. the current profiles must agree at every stored time.

The resulting input matrix has shape

$$\mathbf{X} \in \mathbb{R}^{201 \times 3},$$

while the residual target matrix has shape

$$\mathbf{Y} = \mathbf{C}_{\text{ref}} - \mathbf{C}_{\text{simpl}} \in \mathbb{R}^{201 \times 101}.$$

A complete radial profile is treated as one sample. Radial nodes belonging to the same time instant are never distributed across different subsets. The 201 profiles are randomly permuted using the fixed seed 26 and divided into 70% training, 15% validation and 15% test subsets.

Table 16: Dataset dimensions and profile-wise random split.

Subset	Profiles	Input shape	Target shape
Training	140	140×3	140×101
Validation	30	30×3	30×101
Test	31	31×3	31×101
Total	201	201×3	201×101

The training loader shuffles the 140 profiles at every epoch, whereas validation and test loaders preserve their deterministic order. A mini-batch size of 32 is used for all subsets.

It is important to distinguish this split from validation on a new operating experiment. All three subsets originate from the same continuous current trajectory. The held-out test therefore measures interpolation at unseen time instants within the sampled operating cycle, not extrapolation to a different current profile, initial condition or ambient temperature.

17.7 Residual-Learning Objective and Optimization

The feedforward network is trained using the mean-square discrepancy between the true and predicted residual profiles,

$$\mathcal{L}_{\Delta C}(\phi) = \frac{1}{N_s N_\rho} \sum_{k=1}^{N_s} \sum_{i=0}^{N_\rho} \left[\Delta C_i(\tau_k) - \widetilde{\Delta C}_{\phi,i}(\mathbf{x}_k) \right]^2. \quad (32)$$

The additive residual target is therefore a model-form-error representation rather than a second surrogate of the complete state, in line with residual-based grey-box formulations [Garg et al., 2022].

Here, N_s denotes the number of profiles in the corresponding subset and $N_\rho = 101$ is the number of radial nodes. No PDE, boundary-condition or temperature loss is added: physical information enters through the state variables and through the independently executed simplified solver.

Adam is used with a fixed learning rate

$$\eta = 10^{-3}$$

and a maximum of 10,000 epochs. At the end of each epoch, the sample-weighted validation MSE is evaluated. Whenever this loss improves, a copy of the model parameters is retained. If no improvement is observed for 800 consecutive epochs, training is terminated and the best validation checkpoint is restored.

Table 17: Optimization and model-selection settings for the residual network.

Item	Value
Optimizer	Adam
Learning rate	10^{-3}
Mini-batch size	32 profiles
Maximum epochs	10,000
Early-stopping patience	800 epochs
Selection metric	Validation residual MSE
Final activation	Linear
Execution device	Apple MPS

17.8 Mass-Conserving Correction

The reference and simplified concentration models use the same initial lithium inventory and the same imposed surface flux. Their volume-averaged concentrations must therefore coincide up to numerical integration error, even though their radial distributions differ. Accordingly, the physically admissible correction should satisfy

$$\boxed{\frac{\int_0^1 \Delta C(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho} = 0.} \quad (33)$$

This condition is imposed during inference through a deterministic projection. Let w_i denote normalized spherical trapezoidal weights,

$$w_i = \frac{\omega_i \rho_i^2}{\sum_{j=0}^N \omega_j \rho_j^2}, \quad \sum_{i=0}^N w_i = 1,$$

where ω_i are the one-dimensional trapezoidal quadrature weights. The volume average of the raw network correction is

$$\overline{\Delta C}_\phi = \sum_{i=0}^N w_i \widetilde{\Delta C}_{\phi,i}. \quad (34)$$

The correction applied to the simplified field is then

$$\boxed{\widehat{\Delta C}_{\phi,i} = \widetilde{\Delta C}_{\phi,i} - \overline{\Delta C}_\phi.} \quad (35)$$

Because subtracting a constant does not alter radial gradients, this operation removes the artificial change in lithium inventory without modifying the gradient structure learned by the network. By construction,

$$\sum_{i=0}^N w_i \widehat{\Delta C}_{\phi,i} = 0.$$

The final corrected profile is therefore

$$C_{\text{hybrid},i} = C_{\text{simpl},i} + \widehat{\Delta C}_{\phi,i}. \quad (36)$$

The network is trained on the unprojected residual target, while the mass-conserving projection is applied during test evaluation and during generation of the complete corrected dataset.

The reference and simplified models have the same initial lithium inventory and are subjected to the same surface flux. Their difference therefore concerns the radial redistribution of lithium rather than its total amount. Nevertheless, the approximate residual predicted by the FNN may contain a small non-zero average component, which would artificially add or remove lithium when applied to the simplified solution. The zero-average projection removes this uniform component before the residual is added. Consequently, the correction can move lithium between radial regions while preserving the total lithium inventory predicted by the conservative finite-volume model.

17.9 Reconstruction of Derived Concentration Quantities

After the complete concentration field has been corrected, the surface and volume-averaged quantities stored in the output dataset must be updated consistently. The hybrid surface concentration is extracted directly from the last radial node,

$$C_{s,\text{hybrid}}(\tau_k) = C_{\text{hybrid}}(1, \tau_k). \quad (37)$$

The hybrid volume average is recomputed from the corrected profile,

$$\bar{C}_{\text{hybrid}}(\tau_k) = \sum_{i=0}^N w_i C_{\text{hybrid},i}(\tau_k). \quad (38)$$

The output file consequently preserves the original structured space–time dataset while replacing three mutually consistent fields:

- the complete radial concentration;
- the surface concentration;
- the volume-averaged concentration.

Time, radius and current remain unchanged. This reconstruction prevents the dataset from containing simplified state columns that are inconsistent with the corrected radial field.

17.10 Evaluation Metrics

Let $e_{k,i}$ denote the concentration error relative to the high-fidelity target. For the simplified and hybrid models,

$$e_{k,i}^{\text{simpl}} = C_{\text{simpl},i}(\tau_k) - C_{\text{ref},i}(\tau_k),$$

$$e_{k,i}^{\text{hybrid}} = C_{\text{hybrid},i}(\tau_k) - C_{\text{ref},i}(\tau_k).$$

The principal metrics are

$$\text{MSE} = \frac{1}{N_s N_\rho} \sum_{k,i} e_{k,i}^2, \quad (39)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (40)$$

$$\text{MAE} = \frac{1}{N_s N_\rho} \sum_{k,i} |e_{k,i}|, \quad (41)$$

$$e_{\text{max}} = \max_{k,i} |e_{k,i}|. \quad (42)$$

The relative error reduction is defined using the RMSE,

$$\boxed{\text{RER} = \frac{\text{RMSE}_{\text{simpl}} - \text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simpl}}} \times 100\%}. \quad (43)$$

The same definitions are applied independently to the surface and volume-averaged concentrations. Physical diagnostics additionally monitor the minimum and maximum predicted concentration, the number of violations of $0 \leq C \leq 1$, and the cumulative mass-balance residual.

18 Hybrid Model Results

The final residual model was evaluated on a test set containing 31 radial concentration profiles, corresponding to approximately 15% of the 201 available time-indexed profiles, and subsequently executed over the complete structured grid to generate the corrected hybrid dataset. The results presented in this section describe the standalone hybrid model. A dedicated comparison with the PINN and with additional reference figures is intentionally deferred to a subsequent section.

18.1 Training Convergence and Computational Cost

Training was allowed to proceed for at most 10,000 epochs. Early stopping was activated after 800 consecutive epochs without an improvement in validation loss, terminating the run at epoch 5,881. The best retained checkpoint achieved

$$\boxed{\mathcal{L}_{\Delta C, \text{val}} = 8.3743 \times 10^{-8}}. \quad (44)$$

The complete training stage required 63.07 s on the Apple MPS device. The synchronized corrective inference over all 31 held-out profiles required 7.429 ms, corresponding to approximately

0.2397 ms/profile.

Each profile contains 101 corrected radial values. The reported inference time includes the feedforward evaluation and the mass-conserving projection, but it does not include execution of the simplified finite-volume solver. It therefore quantifies only the additional online cost introduced by the data-driven correction.

Table 18: Training and inference summary for the selected hybrid checkpoint.

Quantity	Value
Maximum training epochs	10,000
Completed epochs	5,881
Early stopping activated	Yes
Best validation MSE	8.3743×10^{-8}
Training time	63.07 s
Test profiles	31
Corrective inference time	7.429 ms
Inference time per profile	0.2397 ms
Execution device	Apple MPS

18.2 Held-Out Test Performance

The held-out test MSE after the mass-conserving projection was

$$\boxed{\mathcal{L}_{\text{test}} = 6.9869 \times 10^{-8}.} \quad (45)$$

The corresponding RMSE is

$$\sqrt{\mathcal{L}_{\text{test}}} = 2.6433 \times 10^{-4}.$$

Table 19 reports the complete held-out field metrics. The simplified-model values are included as the baseline required to quantify the effect of residual learning.

Table 19: Concentration-profile errors on the 31 held-out time instants.

Metric	Simplified FVM	Hybrid model
MSE	7.0949×10^{-5}	6.9869×10^{-8}
RMSE	8.4231×10^{-3}	2.6433×10^{-4}
MAE	6.0552×10^{-3}	1.4840×10^{-4}
Maximum absolute error	2.1558×10^{-2}	2.0037×10^{-3}

The correction decreases the full-profile test RMSE by

$$\boxed{\text{RER}_C = 96.86\%}. \quad (46)$$

The surface concentration exhibits a similarly large improvement, as reported in Table 20.

Table 20: Surface and volume-average concentration errors on the held-out profiles.

Quantity and metric	Simplified FVM	Hybrid model	RMSE reduction
Surface RMSE	8.7886×10^{-3}	3.3645×10^{-4}	96.17%
Surface MAE	7.1190×10^{-3}	1.2979×10^{-4}	—
Volume-average RMSE	1.0378×10^{-6}	1.0378×10^{-6}	—
Volume-average MAE	9.3480×10^{-7}	9.3479×10^{-7}	—

The near-identical volume-average errors are expected rather than problematic. Both physical models receive the same current and surface flux, and the zero-average projection prevents the network from altering the corresponding lithium inventory. The purpose of the correction is therefore to recover the radial redistribution and surface response produced by state-dependent diffusivity, not to modify the total amount of lithium inside the particle.

No test prediction violated the normalized physical interval

$$0 \leq C \leq 1.$$

18.3 Complete Structured-Grid Prediction

After test evaluation, the selected checkpoint was applied to all 201 states of the simplified simulation. This operation generated the complete corrected concentration field used for post-processing and visualization.

Because the complete grid contains the training, validation and test profiles together, the metrics in Table 21 describe reconstruction over the sampled trajectory and must not be interpreted as an independent generalization estimate.

Table 21: Dense-grid concentration errors over the complete sampled operating trajectory.

Model	Relative L^2 error	RMSE	MAE	Maximum absolute error
Simplified FVM	1.1016×10^{-2}	8.2082×10^{-3}	5.5855×10^{-3}	2.1888×10^{-2}
Hybrid model	4.2024×10^{-4}	3.1312×10^{-4}	1.6676×10^{-4}	2.1476×10^{-3}

Over the complete trajectory, the hybrid correction reduces the RMSE by 96.19%. Table 22 reports the errors of selected derived quantities.

Table 22: Hybrid errors for selected quantities on the complete structured grid.

Quantity	RMSE	MAE	Maximum absolute error
Complete field	3.1312×10^{-4}	1.6676×10^{-4}	2.1476×10^{-3}
Particle centre	5.5977×10^{-4}	3.3695×10^{-4}	2.1337×10^{-3}
Particle surface	2.1881×10^{-4}	1.2105×10^{-4}	1.8207×10^{-3}
Volume average	9.4936×10^{-7}	8.1103×10^{-7}	1.6431×10^{-6}

The complete hybrid field remains within

$$\boxed{0.498575 \leq C_{\text{hybrid}} \leq 0.880228,} \quad (47)$$

and therefore satisfies the normalized physical bounds. Unlike the initial condition of the PINN, the initial hybrid profile is not imposed as a hard constraint on the neural correction. The maximum initial profile error is nevertheless limited to

$$3.6410 \times 10^{-4}.$$

18.4 Standalone Hybrid Concentration Field

Figure 20 shows the complete corrected concentration field.

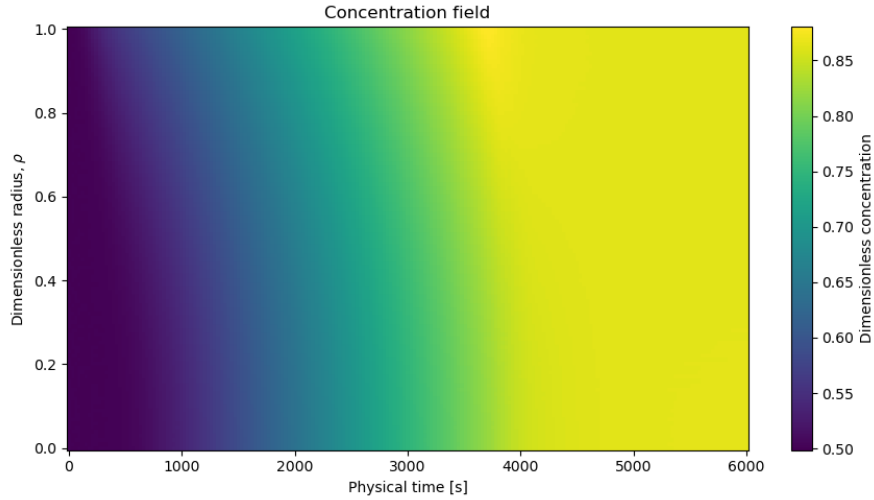


Figure 20: Concentration field generated by the residual hybrid model.

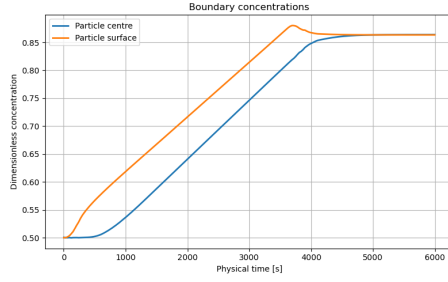
The hybrid field preserves the smooth spatial and temporal behaviour of the finite-volume baseline. During the loading stage, lithium enters through the particle surface and generates a positive radial gradient. After ramp-down, this gradient progressively disappears and the field approaches a spatially uniform plateau. No high-frequency radial oscillations or discontinuities are visible in the corrected solution.

The boundary and volume-averaged concentration histories are reported in Figure 21.

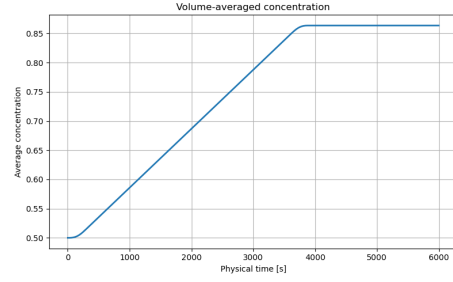
The particle surface responds first to the applied current and reaches a maximum near the end of the loading phase. During relaxation, the surface concentration decreases while the centre continues to rise until the profile becomes nearly uniform. The volume average increases according to the imposed lithium flux and remains constant after the current reaches zero.

Selected radial profiles are shown in Figure 22 together with the cumulative mass-balance residual.

At 1,500 and 3,000 s, the concentration increases monotonically from the centre towards the surface. At 4,500 and 6,000 s, the profiles are almost spatially uniform because the applied current has vanished and the particle has entered the relaxation stage.

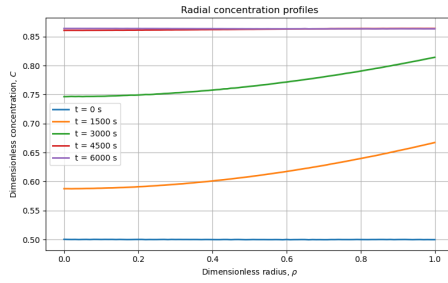


(a) Particle-centre and surface concentrations.

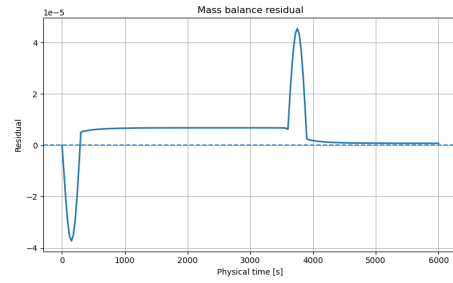


(b) Volume-averaged concentration.

Figure 21: Boundary and integral quantities reconstructed from the corrected concentration field.



(a) Radial profiles at selected physical times.



(b) Cumulative mass-balance residual.

Figure 22: Spatial evolution and conservation diagnostic of the standalone hybrid model.

18.5 Mass Conservation and Physical Consistency

The cumulative mass-balance residual is evaluated as

$$R_{\text{mass}}(\tau) = \overline{C}_{\text{hybrid}}(\tau) - \left[C_0 - 3 \int_0^\tau \delta(s) ds \right]. \quad (48)$$

The dimensionless flux appearing in this diagnostic is the known operating input common to the reference and simplified simulations. No reference concentration is used to construct the hybrid mass-balance curve.

The maximum absolute residual over the complete simulation is

$$\boxed{\max_{\tau} |R_{\text{mass}}(\tau)| = 4.5528 \times 10^{-5}.} \quad (49)$$

This value is consistent with the numerical quadrature and finite-volume integration error of the original concentration models. The deterministic zero-average projection prevents the neural network from introducing a larger inventory error.

The main physical checks are summarized in Table 23.

Table 23: Physical consistency checks for the complete hybrid field.

Check	Result	Interpretation
Minimum concentration	0.498575	Within normalized bounds
Maximum concentration	0.880228	Within normalized bounds
NaN values	0	Numerically finite field
Infinite values	0	Numerically finite field
Physical-bound violations	0	$0 \leq C \leq 1$
Maximum mass residual	4.5528×10^{-5}	Lithium inventory preserved
Maximum initial-profile error	3.6410×10^{-4}	Small soft-constraint discrepancy

18.6 Assessment and Present Scope

The results support four principal observations.

1. **Residual learning is effective within the sampled trajectory.** The held-out full-profile RMSE decreases by 96.86%, while the surface RMSE decreases by 96.17%. The network therefore learns a compact correction for the radial redistribution omitted by the constant-diffusivity model.
2. **The simplified solver remains physically useful.** It supplies the correct operating current, spherical geometry, lithium inventory and dominant diffusion behaviour. The neural component is not required to reconstruct the complete solution from time and radius alone.
3. **Mass conservation can be retained explicitly.** The zero-average correction makes the role of the network physically specific: it changes the distribution of lithium inside the particle without changing its total amount.

4. **The online correction is computationally light.** One feedforward evaluation produces all 101 radial corrections, requiring approximately 0.24 ms per held-out profile on the tested hardware.

The present accuracy must nevertheless be interpreted within the scope of the experiment. The random split contains unseen time profiles but not unseen operating cycles. Adjacent training and test states belong to the same smooth trajectory, and the reported result primarily quantifies interpolation. Validation under different current profiles, initial concentrations and thermal conditions is required before claiming operating-condition generalization.

The correction network is also discretization specific because its 101 outputs correspond directly to the adopted radial nodes. A different mesh would require retraining or an alternative coordinate-based output representation.

Finally, the model currently predicts only the concentration field. The temperature influences the learned discrepancy through the reference targets, but it is not reconstructed as an independent state. The identification of an interpretable data-driven thermal dynamics is issued in a further section.

19 Comparison of the Simplified, PINN and FVM–FNN Hybrid Models

This section compares three modelling strategies against the same full-physics finite-volume reference field. The later post-training Hybrid-PINN is intentionally excluded here because it corrects the PINN itself and is assessed directly against its frozen baseline in Section 21. The present comparison includes:

- the **simplified finite-volume model**, based on constant solid-state diffusivity and an isothermal particle;
- the **physics-informed neural network**, which represents the concentration as a continuous function of (ρ, τ) and is constrained by the constant-diffusivity equation and a constant surface flux;
- the **FVM–FNN hybrid residual model**, which corrects the radial profile predicted by the simplified finite-volume model using the states $\bar{C}_{\text{simpl}}$, $C_{s,\text{simpl}}$ and the operating current $I(t)$.

The reference, simplified and hybrid fields were read directly from their stored datasets. The PINN field was regenerated from the selected checkpoint without additional training. All predictions were evaluated on the same structured grid containing 101 radial nodes and 201 time instants. The comparison script and the resulting numerical metrics are stored separately from the training routines, so that the figures can be reproduced without repeating either neural-network optimization.

19.1 Scope and Fairness of the Comparison

Although the three predictions are compared with the same target, the PINN and hybrid model do not receive exactly the same information. This distinction is necessary for a correct interpretation of the results. The PINN adopts the fixed dimensionless flux

$$\delta(\tau) = \delta_0 \simeq -0.2005, \quad (50)$$

throughout its time domain. Conversely, the simplified model used by the hybrid framework follows the same ramp–constant–ramp–rest operating current imposed on the reference simulator. Therefore, after the current starts to decrease, the PINN and reference problems possess different boundary inputs, whereas the simplified and hybrid models retain the correct input history.

Furthermore, the PINN is a coordinate-based representation: it takes (ρ, τ) as input and produces one concentration value. The hybrid network receives reduced state information generated by a numerical model and returns the complete 101-node discrepancy profile. The PINN is therefore mesh independent after training, while the present hybrid output is tied to the selected radial grid. Conversely, the hybrid model benefits from a stronger physics-based prior and from direct access to the operating current.

Consequently, the following analysis compares the models as they were actually formulated in this project. It demonstrates the practical effect of the two strategies under the selected operating cycle, but it must not be interpreted as a universal ranking between PINNs and hybrid residual learning.

19.2 Common Error Measures

For a predicted quantity C_{pred} and the corresponding reference C_{ref} , the relative L^2 error, root-mean-square error and mean absolute error were calculated as

$$\varepsilon_{L^2} = \frac{\|C_{\text{pred}} - C_{\text{ref}}\|_2}{\|C_{\text{ref}}\|_2}, \quad (51)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_{\text{pred},i} - C_{\text{ref},i})^2}, \quad (52)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |C_{\text{pred},i} - C_{\text{ref},i}|. \quad (53)$$

These quantities were evaluated for the complete concentration field, the particle centre, the particle surface and the volume-averaged concentration. Table 24 reports the resulting dense-grid metrics.

The simplified model already provides a reasonable approximation, with a complete-field relative error of 1.10%. Its largest error is found at the particle centre, where the concentration response is most strongly affected by the accumulated error in radial transport. The PINN obtains a complete-field error of 5.07%. This larger value is consistent with the deliberately mismatched constant-flux boundary condition and is not an indication of unstable optimization.

The hybrid model reduces the complete-field RMSE from 8.2082×10^{-3} to 3.1312×10^{-4} . The corresponding error reduction relative to the simplified baseline is

$$\text{RER}_{\text{hybrid}} = \left(1 - \frac{\text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simpl}}}\right) 100 = 96.19\%. \quad (54)$$

Table 24: Errors of the three predictive models with respect to the full-physics FVM reference on the common 101×201 grid.

Model	Quantity	Relative L^2	RMSE	MAE	Maximum absolute error
Simplified	Complete field	0.011016	0.008208	0.005586	0.021888
	Particle centre	0.016695	0.012211	0.009360	0.021888
	Particle surface	0.010792	0.008327	0.006371	0.015136
	Volume average	1.2565×10^{-6}	9.4944×10^{-7}	8.1108×10^{-7}	1.6437×10^{-6}
PINN	Complete field	0.050674	0.037757	0.027721	0.160606
	Particle centre	0.045458	0.033251	0.026263	0.073184
	Particle surface	0.068609	0.052938	0.034088	0.160606
	Volume average	0.053791	0.040645	0.028642	0.124484
FVM-FNN hybrid	Complete field	4.2024×10^{-4}	3.1312×10^{-4}	1.6676×10^{-4}	0.002148
	Particle centre	7.6528×10^{-4}	5.5977×10^{-4}	3.3695×10^{-4}	0.002134
	Particle surface	2.8358×10^{-4}	2.1881×10^{-4}	1.2105×10^{-4}	0.001821
	Volume average	1.2564×10^{-6}	9.4936×10^{-7}	8.1103×10^{-7}	1.6431×10^{-6}

Equivalently, the complete-field RMSE is reduced by a factor of 26.2 relative to the simplified model and by a factor of 120.6 relative to the current PINN implementation. The surface concentration displays the same behaviour: its RMSE decreases from 8.3272×10^{-3} for the simplified model and 5.2938×10^{-2} for the PINN to 2.1881×10^{-4} for the hybrid model.

The dense-grid measures characterize reconstruction over the operating trajectory used to generate the available datasets; they are not an out-of-distribution test under a new current profile. Both learned models were exposed to samples belonging to this space-time domain. For the hybrid model, the independently held-out set of 31 complete profiles provides a separate interpolation check: its concentration RMSE was 2.6433×10^{-4} , corresponding to a 96.86% reduction relative to the simplified-model RMSE of 8.4231×10^{-3} . The agreement between held-out and complete-grid errors supports temporal interpolation within the current experiment, while generalization to different currents, initial concentrations or ambient conditions remains to be evaluated.

Figure 23 summarizes these RMSE values on a logarithmic scale.

The volume-average result requires separate interpretation. Both the simplified and hybrid averages have an RMSE close to 9.49×10^{-7} . The simplified model already receives the correct operating flux, so its total lithium inventory is governed accurately even when the radial distribution is imperfect. In addition, the zero-average projection applied to the learned correction deliberately prevents the FNN from altering this inventory. The unchanged average is therefore an intended physical property of the hybrid formulation, not a failure of the network to learn the residual.

19.3 Concentration-Field Comparison

Figure 24 compares the complete reference, PINN and hybrid concentration fields. A single colour scale, computed over all three fields, is used for every panel. This prevents independent normalization from hiding the PINN late-time overshoot.

The reference field increases during the active current phase and reaches an almost uniform plateau after current removal. The hybrid field reproduces both the loading and relaxation stages, and its difference from the reference is not visible at the scale of the map. By contrast, the PINN field continues to increase during the final part of the simulation. Its maximum concentration is 1.0241, compared with 0.8799 for the reference and 0.8802 for the hybrid model. The continued increase is the direct consequence of Eq. (50): the simplified PINN boundary condition remains

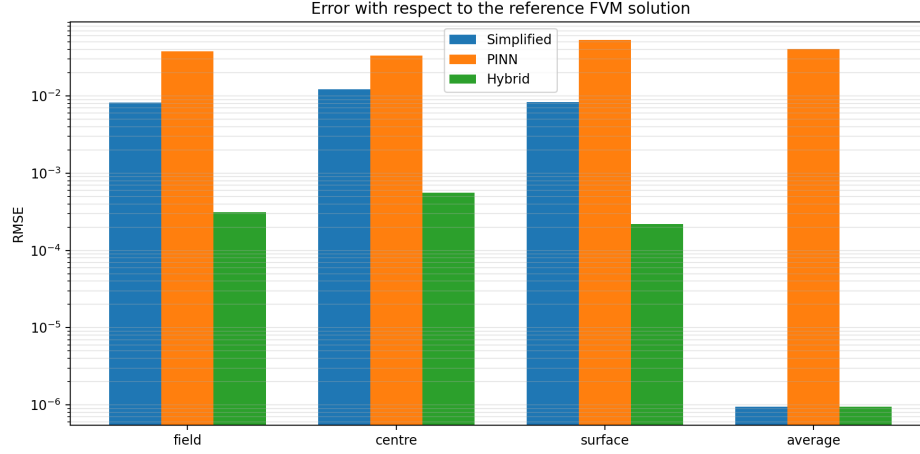


Figure 23: RMSE of the simplified, PINN and hybrid predictions with respect to the full-physics FVM reference. The logarithmic vertical scale is required because the hybrid errors are between one and two orders of magnitude smaller.

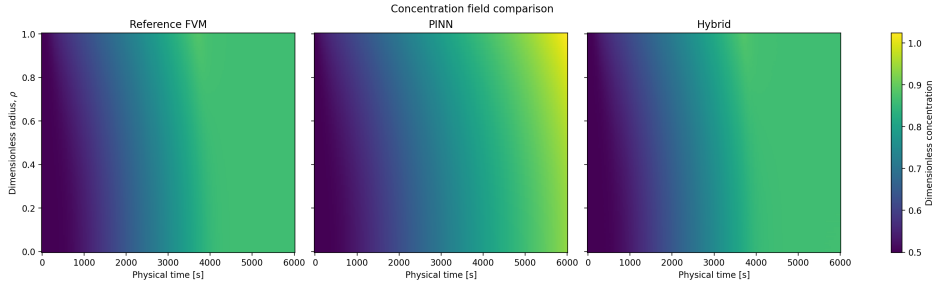


Figure 24: Concentration fields produced by the full-physics FVM reference, the PINN and the hybrid residual model. All panels use the same colour normalization.

active while the reference and hybrid models have entered the zero-current relaxation phase.

The volume-averaged histories are shown in Figure 25. The simplified and hybrid curves overlap the reference almost exactly because they share the correct flux history and integral balance. The PINN follows the reference during the initial interval but diverges progressively after the two boundary inputs become different.

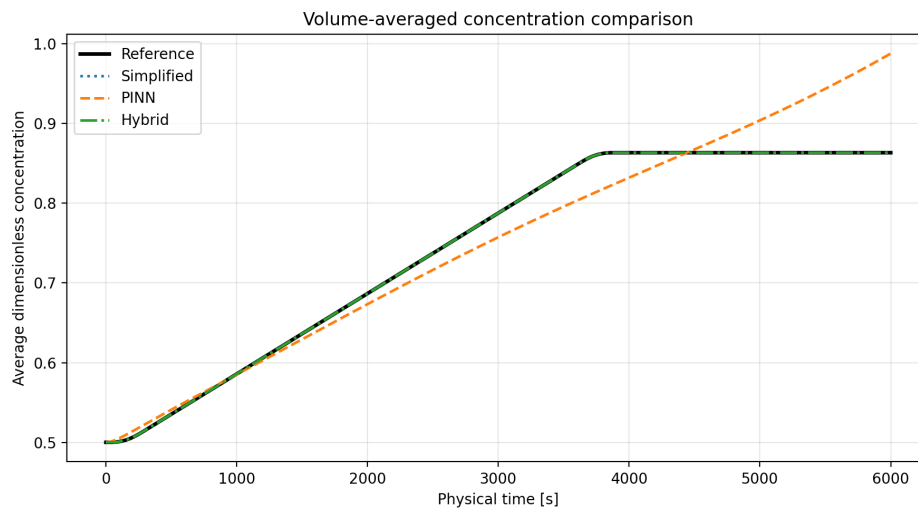


Figure 25: Volume-averaged concentration predicted by the reference, simplified, PINN and hybrid models.

19.4 Boundary and Radial-Profile Comparison

The histories at the particle centre and surface are reported in Figure 26.

At the surface, the reference concentration reaches a local maximum during ramp-down and subsequently relaxes towards the volume average. The constant-diffusivity simplified model overestimates this maximum, while the hybrid correction reconstructs both its magnitude and the following relaxation. At the centre, the simplified response is delayed relative to the reference; this discrepancy is again removed almost completely by the residual model.

The PINN is reasonably close during the first part of the trajectory, but it cannot reproduce the final plateau because neither the current transition nor the thermal and adaptive-diffusivity states are included in its input. Its largest boundary error is observed at the surface, where the prescribed flux acts directly.

Figure 27 provides the corresponding spatial interpretation at five selected times.

At 1500 and 3000 s, all models reproduce the expected positive radial gradient. The simplified model slightly overestimates the curvature, whereas the PINN underestimates the overall concentration level. At 4500 s, the reference profile is already close to spatial equilibrium. The hybrid model reproduces this transition, while the PINN retains a non-zero surface-to-centre gradient. At 6000 s, the reference, simplified and hybrid fields are almost uniform; the PINN still predicts both increasing concentration and an appreciable radial gradient.

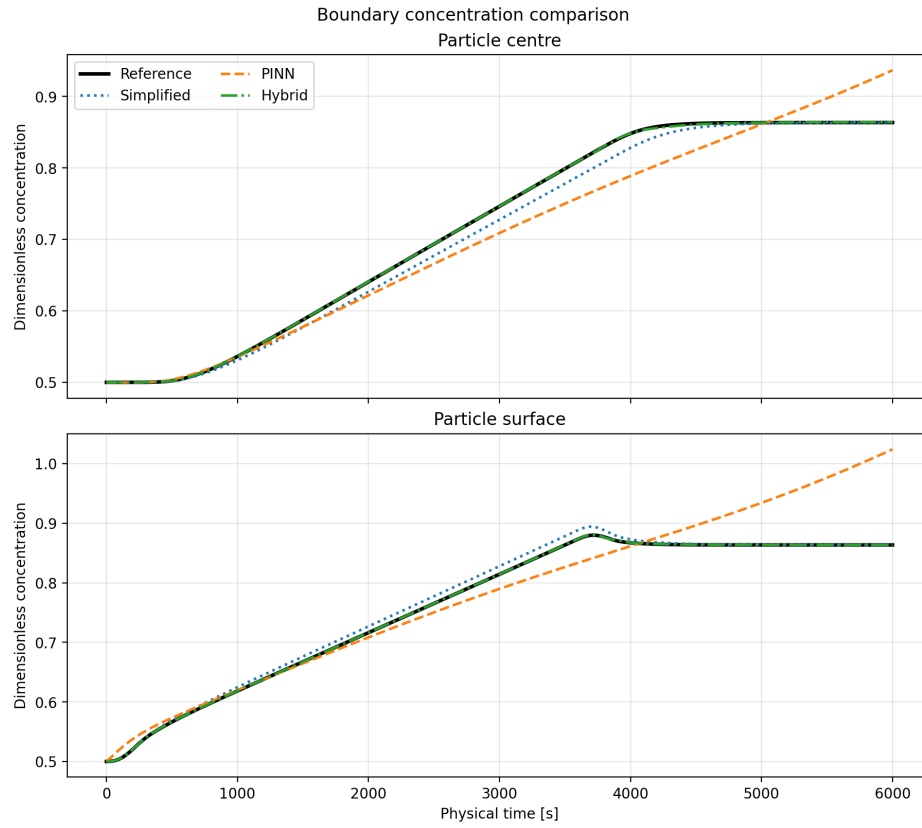


Figure 26: Particle-centre and particle-surface concentration histories for all considered methods.

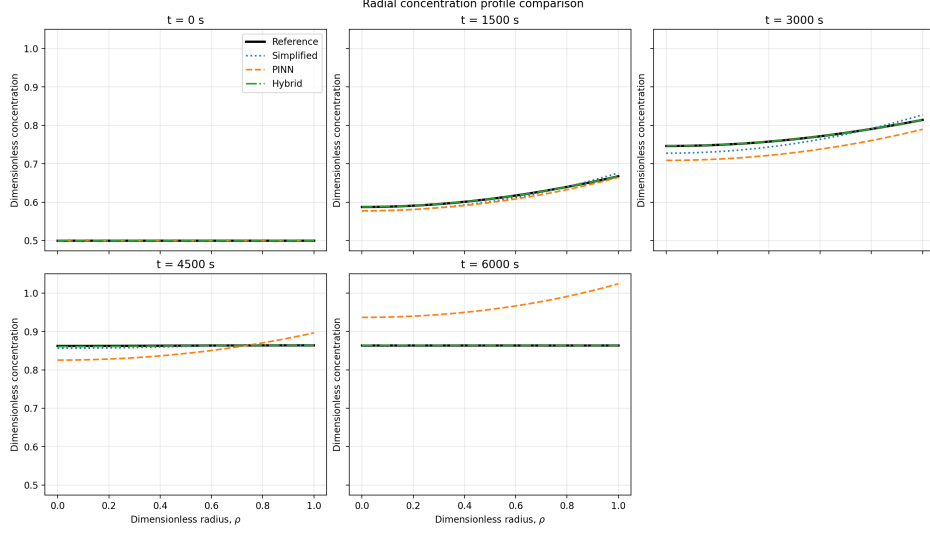


Figure 27: Radial concentration profiles at selected physical times. The reference and hybrid curves are nearly superposed throughout the operating cycle.

19.5 Mass Conservation and Physical Bounds

The cumulative mass-balance residual was evaluated as

$$R_{\text{mass}}(\tau) = \overline{C}(\tau) - \left[\overline{C}(0) - 3 \int_0^\tau \delta(s) ds \right]. \quad (55)$$

For the reference, simplified and hybrid models, $\delta(\tau)$ is the operating ramp–constant–ramp–rest profile. For the PINN, $\delta(\tau) = \delta_0$ is the constant flux actually enforced during training. Thus each prediction is checked against its own prescribed boundary law.

The maximum absolute residuals are summarized in Table 25.

Table 25: Mass-balance and concentration-range checks.

Model	Minimum C	Maximum C	Maximum $ R_{\text{mass}} $
Reference	0.500000	0.879933	4.47×10^{-5}
Simplified	0.500000	0.894176	4.55×10^{-5}
PINN	0.499530	1.024085	1.1783×10^{-1}
FVM–FNN hybrid	0.498575	0.880228	4.55×10^{-5}

The reference, simplified and hybrid residuals are all of order 10^{-5} and are dominated by numerical quadrature and time discretization. The zero-average correction prevents the FNN from modifying the conservative evolution supplied by the simplified model. The hybrid concentration exhibits a small minimum undershoot of 1.43×10^{-3} relative to the normalized lower level 0.5. This occurs in the corrected initial profile and remains small compared with the field error; nevertheless, it shows that exact initial and box constraints are not hard-wired into the present residual

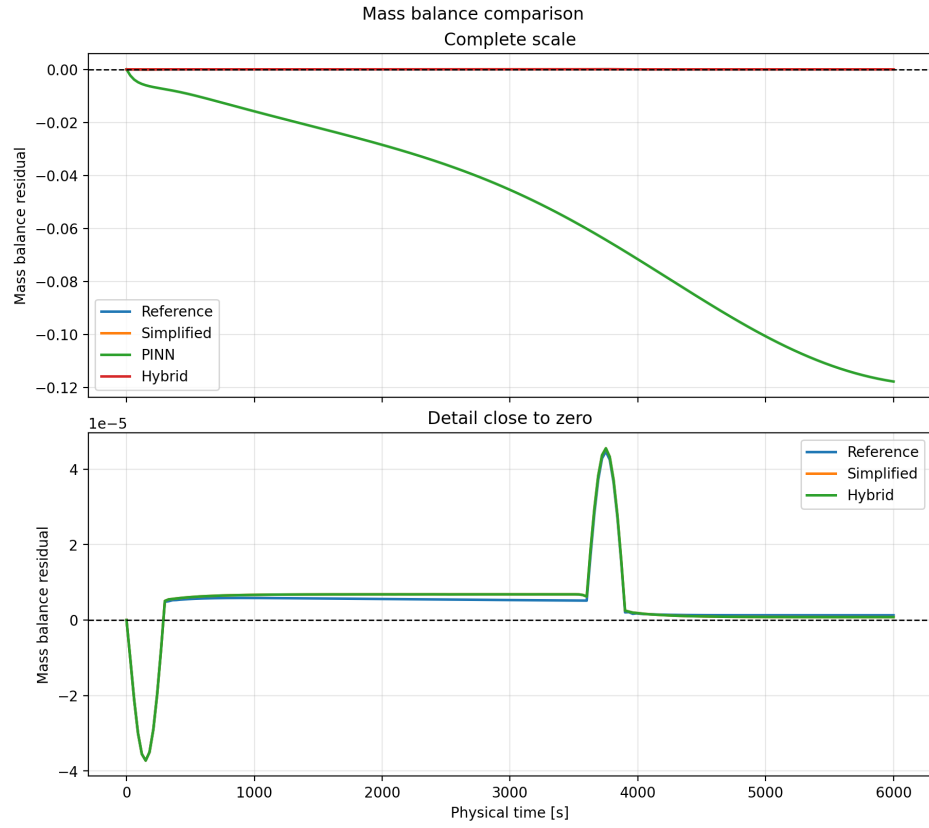


Figure 28: Cumulative mass-balance residuals. The upper panel shows all methods on the complete scale; the lower panel resolves the reference, simplified and hybrid residuals close to zero.

architecture.

The larger PINN mass residual reflects the compromise between two incompatible training objectives: matching data generated by the time-dependent reference current while approximately satisfying a constant-flux PDE. It should therefore be interpreted as a consequence of the deliberate model-form mismatch, rather than as evidence that the reference data violate conservation.

19.6 Model Size and Computational Interpretation

Table 26 summarizes the main structural differences between the learned models.

Table 26: Structural comparison of the PINN and FVM-FNN hybrid correction network.

Property	PINN	FVM-FNN residual network
Network input	(ρ, τ)	$(\bar{C}_{\text{simpl}}, C_{s,\text{simpl}}, I)$
Network output	One concentration value	101-node discrepancy profile
Hidden architecture	Four layers, 100 neurons each	Three layers, 32 neurons each
Activation	$\tanh$	ReLU
Trainable parameters	30,701	5,573
Physics contribution	Differential residuals in the loss	Online simplified FVM state
Surface input	Constant flux	Time-dependent operating current
Spatial representation	Continuous coordinate model	Fixed output grid
Hard constraints	Initial condition and centre symmetry	Zero-average residual projection

The PINN checkpoint required 4.55 ms to evaluate 3,046 held-out coordinate samples on MPS, corresponding to approximately $1.49 \mu\text{s}$ per coordinate. The hybrid corrective branch required 7.43 ms for 31 complete test profiles, or 0.240 ms per 101-node profile. These timings are not directly comparable. The PINN timing is a pointwise neural evaluation, whereas the hybrid timing excludes the cost of advancing the simplified finite-volume model that supplies its state. An end-to-end deployment benchmark would therefore need to include the numerical baseline and evaluate both methods for the same requested output grid and time horizon.

The smaller hybrid network is nevertheless significant. Because the simplified model already supplies the dominant diffusion behaviour, the FNN only needs to represent a smooth discrepancy. The PINN must instead represent the complete concentration mapping while simultaneously balancing data, PDE and boundary losses.

19.7 Robustness to Variations in the Simplified Model

The robustness of the hybrid model was evaluated by varying the constant solid-phase diffusion coefficient employed by the simplified model while keeping the reference dataset and the trained FNN unchanged. Therefore, no additional training or fine-tuning was performed during this analysis.

This experiment does not represent a change in the physical system described by the reference model. Instead, it measures the sensitivity of the hybrid model to parameter uncertainty and model mismatch in its simplified physics-based component. In particular, the constant diffusion coefficient was varied within the range approximately covered by the adaptive coefficient $D_s(C, T)$ of the reference model.

The relative error reduction was computed as

$$\text{RER} = \frac{\text{RMSE}_{\text{simplified}} - \text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simplified}}} \times 100. \quad (56)$$

Consequently, a positive RER indicates that the FNN correction improves the simplified model, whereas a negative value indicates that the correction increases the prediction error.

Table 27: Performance of the frozen hybrid model for different values of the constant solid-phase diffusion coefficient.

Case	$D_s [\text{m}^2 \text{s}^{-1}]$	Simplified RMSE	Hybrid RMSE	RER [%]
Nominal	1.213×10^{-14}	8.423×10^{-3}	2.643×10^{-4}	96.86
Variant 0	1.35×10^{-14}	5.394×10^{-3}	2.722×10^{-3}	49.55
Variant 1	1.55×10^{-14}	2.703×10^{-3}	5.812×10^{-3}	-114.97
Variant 2	1.75×10^{-14}	2.618×10^{-3}	8.217×10^{-3}	-213.87
Variant 3	1.87×10^{-14}	3.467×10^{-3}	9.420×10^{-3}	-171.72
Variant 4	2.01×10^{-14}	4.579×10^{-3}	1.065×10^{-2}	-132.51

The nominal hybrid model provides a substantial improvement over the simplified model, reducing the concentration RMSE by approximately 96.86%. A positive improvement is also retained for Variant 0, for which the diffusion coefficient differs moderately from the value used during training. In this case, the hybrid model still reduces the error by approximately 49.55%.

For larger variations of D_s , however, the FNN correction becomes detrimental. From Variant 1 onward, the hybrid RMSE exceeds the error of the corresponding simplified model, producing negative RER values. The largest relative deterioration occurs for Variant 2, where the hybrid error is more than three times the simplified-model error.

This behaviour follows directly from the residual-learning formulation. The FNN was trained to approximate the discrepancy associated with the nominal simplified model,

$$\Delta C_{D_{s,0}}(\rho, t) = C_{\text{ref}}(\rho, t) - C_{\text{simplified}}(\rho, t; D_{s,0}), \quad (57)$$

where

$$D_{s,0} = 1.213 \times 10^{-14} \text{ m}^2 \text{s}^{-1}. \quad (58)$$

When the constant diffusion coefficient is changed, the correction required by the new simplified model becomes

$$\Delta C_{D_s}(\rho, t) = C_{\text{ref}}(\rho, t) - C_{\text{simplified}}(\rho, t; D_s). \quad (59)$$

In general,

$$\Delta C_{D_s}(\rho, t) \neq \Delta C_{D_{s,0}}(\rho, t). \quad (60)$$

The frozen FNN nevertheless continues to generate a correction learned from the nominal configuration. Therefore, when the modified simplified model is already closer to the reference solution, the nominal FNN tends to overcorrect its concentration field.

This effect is particularly evident around $D_s = 1.55\text{--}1.75 \times 10^{-14} \text{ m}^2 \text{s}^{-1}$, where the uncorrected simplified model reaches its lowest errors. The best result for the simplified model is obtained at approximately

$$D_s = 1.75 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}, \quad (61)$$

which is close to the median value, $1.746 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$, sampled by the adaptive diffusion law of the reference model. In this region, applying the correction learned for the lower nominal coefficient causes a significant overcorrection.

The volume-averaged concentration error remains practically unchanged after the FNN correction for all the investigated variants. This is an expected consequence of enforcing a zero-volume-average residual:

$$3 \int_0^1 \Delta C(\rho, t) \rho^2 d\rho = 0. \quad (62)$$

The hybrid model consequently modifies the radial distribution of lithium while preserving the total lithium content represented by the simplified model. The close agreement between the simplified and hybrid errors in the average concentration therefore confirms that the imposed conservation constraint is operating as intended.

The surface-concentration results follow the same trend observed for the complete concentration field. Moderate variations close to the nominal configuration can still be corrected successfully, whereas larger variations cause the nominal FNN to increase the surface-concentration error.

Overall, these results demonstrate that the proposed residual model has a local domain of validity around the simplified model used during training. The experiment should therefore be interpreted as a parameter-sensitivity and model-mismatch analysis, rather than as evidence of generalization to new operating conditions. The excellent nominal accuracy does not imply that the same correction can be transferred unchanged to substantially different parameterizations of the physics-based model.

To obtain a hybrid model that is robust to significant variations of physical parameters or forcing terms, such as the solid-phase diffusion coefficient D_s or the intercalation flux, the training dataset must include multiple values of these quantities. The FNN should additionally be conditioned on D_s , on the dimensionless intercalation flux, or on another equivalent quantity capable of identifying the current physical configuration. Most importantly, extending the validity range in this manner requires the FNN to be retrained. A network trained only on the nominal configuration cannot, in general, provide a reliable residual correction for substantially different physical coefficients or boundary fluxes without additional training.

19.8 Overall Comparative Assessment

The comparison leads to the following conclusions.

1. **The simplified model is already conservative and captures the dominant dynamics.** Its complete-field relative error is approximately 1.10%, and its volume average is essentially exact because the correct operating flux is imposed.
2. **The PINN learns a smooth physically meaningful field, but its late-time response is limited by the constant-flux assumption.** The 5.07% field error and the surface overshoot are concentrated where the simplified boundary condition diverges most strongly from the reference operating cycle.

3. **Residual learning provides the best concentration reconstruction in the present setting.** The hybrid model reduces the simplified-model field RMSE by 96.19% and reproduces the centre, surface and radial profiles throughout loading and relaxation.
4. **The conservation projection is effective.** The hybrid correction changes the radial distribution without changing the lithium inventory inherited from the simplified model, maintaining a mass-balance residual of order 10^{-5} .
5. **The comparison does not establish universal method superiority.** The hybrid model receives the correct current profile and reduced physical states, while the PINN is intentionally trained with a constant boundary flux. A PINN conditioned on the operating current, or trained with the same time-dependent boundary input, would constitute a different and more symmetric experiment.

For the objective considered here—recovering the full-physics concentration field from an inexpensive but imperfect particle model—the hybrid residual strategy offers the most favourable balance between accuracy, physical interpretability and network size. Its main current limitations are the fixed 101-node output, dependence on the training operating regime, absence of an explicit temperature state and the lack of exact box and initial-condition constraints.

Temperature is deliberately excluded from the graphical comparison. Only the full-physics reference model currently reconstructs the lumped temperature history; neither the PINN nor the hybrid FNN predicts $T(t)$. The hybrid concentration field can reproduce the effect that temperature-dependent diffusivity has on lithium redistribution, but this does not make temperature an identifiable output. Recovering the thermal history requires a separate measured-temperature model or a dedicated equation-discovery stage and cannot be inferred uniquely from the present concentration correction alone.

20 Data-Driven Reconstruction of the Thermal Dynamics: SINDYc and Feedforward Neural Networks

The hybrid concentration model developed in the previous sections corrects the concentration field predicted by a simplified Single Particle Model. However, the simplified model assumes a constant diffusion coefficient evaluated at the reference concentration and ambient temperature. Its thermal dynamics are therefore omitted and the particle temperature remains equal to the ambient value.

To reconstruct the missing temperature evolution without supplying the thermal balance equation used internally by the reference solver, two data-driven models were investigated. The first is Sparse Identification of Nonlinear Dynamics with Control (SINDYc), which searches for an explicit and parsimonious differential equation in a prescribed library of candidate functions. The second is a feedforward neural network (FFN), which directly approximates the temperature derivative as a nonlinear function of the instantaneous thermal state and applied current. SINDYc extends the sparse identification of nonlinear dynamics framework by including external forcing and control variables [Brunton et al., 2016a,b].

The comparison is intended to separate two different modelling properties. SINDYc favours interpretability and structural discovery when the relevant candidate functions are present in the library. The FFN requires no explicit choice of candidate terms and is therefore more flexible, but it does not automatically recover a sparse or physically interpretable equation. Both models

are trained and evaluated using the same collection of thermal trajectories and are reconstructed through the same numerical rollout procedure.

20.1 Thermal state and controlled dynamics

The temperature deviation from the ambient condition is adopted as the scalar state,

$$\theta(t) = T(t) - T_{\text{amb}}, \quad T_{\text{amb}} = 298.15 \text{ K}, \quad (63)$$

while the applied current is treated as an external control input,

$$u(t) = I(t). \quad (64)$$

Both data-driven models therefore seek an approximation of the unknown controlled dynamics

$$\dot{\theta}(t) = f(\theta(t), I(t)). \quad (65)$$

The use of θ rather than the absolute temperature improves both conditioning and physical interpretation. In particular, a negative term proportional to θ represents relaxation towards the ambient condition, for which $\theta = 0$.

20.2 Generation of multiple thermal trajectories

The original reference dataset contains one row for every time–radius pair. Because the reference model adopts a lumped thermal description, temperature and current are independent of the radial coordinate and are repeated for all radial nodes. For each simulation, redundant radial entries were removed by retaining one sample per physical time and sorting the resulting signals in ascending temporal order:

$$\mathcal{D}_T^{(j)} = \left\{ t_k^{(j)}, T_k^{(j)}, I_k^{(j)} \right\}_{k=1}^{N_t}, \quad j = 0, \dots, 7. \quad (66)$$

Eight full-reference simulations were generated by varying the peak current and the durations of the ramp-up, constant-current, ramp-down and cooling stages. Each trajectory contains $N_t = 601$ unique temporal samples. The main characteristics and assigned roles are reported in Table 28.

Table 28: Thermal trajectories used for identification, validation and out-of-sample testing.

Dataset	Role	t_f [s]	$I_{\max}$ [A]	$T_{\max}$ [K]
0	Training	7200	1.2420	299.693
1	Training	6000	1.6560	300.892
2	Training	6000	1.6560	300.891
3	Training	5200	2.0700	302.429
4	Training	4800	2.4840	304.294
5	Validation	6400	1.4904	300.371
6	Interpolation test	5000	2.2356	303.138
7	Extrapolation test	4500	2.7324	305.557

Datasets 0–4 are used jointly for model identification or FFN training. Dataset 5 is used exclusively for SINDYc threshold selection and FFN early stopping. Dataset 6 is an interpolative

test because its peak current and thermal excursion lie inside the envelope covered by the training trajectories. Dataset 7 is an extrapolative test because its peak current, $I_{\max} = 2.7324$ A, exceeds the largest training value, 2.4840 A, and produces a temperature beyond the observed training range. Neither test trajectory is used for parameter estimation, model selection or normalization.

20.3 SINDYc formulation and sparse regression

For a nonlinear system influenced by an external input, SINDYc approximates the unknown dynamics as a sparse linear combination of candidate nonlinear functions:

$$\dot{\mathbf{X}} = \mathbf{\Theta}(\mathbf{X}, \mathbf{U})\mathbf{\Xi}, \quad (67)$$

where $\mathbf{\Theta}$ is the candidate-function library and $\mathbf{\Xi}$ contains the coefficients to be identified. Sparsity is imposed under the assumption that only a small subset of the candidate terms is required to describe the underlying dynamics [Brunton et al., 2016a].

For the present scalar state and control input, a second-order polynomial library was adopted:

$$\mathbf{\Theta}(\theta, I) = \begin{bmatrix} 1 & \theta & I & \theta^2 & \theta I & I^2 \end{bmatrix}. \quad (68)$$

This library can represent linear heat removal, current-dependent heat generation and possible second-order coupling, while remaining sufficiently small to preserve interpretability. Temperature derivatives were estimated using a second-order finite-difference method, with endpoint samples excluded from the sparse regression.

The coefficients were identified through Sequentially Thresholded Least Squares (STLSQ), with

$$\alpha_{\text{STLSQ}} = 10^{-8} \quad (69)$$

and column normalization enabled. The ridge parameter α_{STLSQ} improves numerical conditioning, while sparsity is controlled primarily through the threshold λ .

The following threshold values were evaluated:

$$\lambda \in \{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}. \quad (70)$$

For each candidate, a single SINDYc model was fitted jointly to training trajectories 0–4 and integrated over validation trajectory 5. A candidate was accepted when its validation RMSE was no greater than 1.15 times the minimum validation RMSE. Among the acceptable candidates, the model with the smallest number of active terms was retained; validation RMSE was used as the secondary criterion.

All examined thresholds produced the same two-term structure and a validation temperature RMSE of 1.4640×10^{-4} K. The first threshold, $\lambda = 10^{-3}$, was retained by the tie-breaking rule. The resulting multi-trajectory equation is

$$\boxed{\frac{d\theta}{dt} = -2.4685252 \times 10^{-3}\theta + 2.4685255 \times 10^{-3}I^2}. \quad (71)$$

Equivalently,

$$\frac{dT}{dt} = -2.4685252 \times 10^{-3}(T - T_{\text{amb}}) + 2.4685255 \times 10^{-3}I^2. \quad (72)$$

The first term represents effective heat rejection to the environment, whereas the second represents current-induced heat generation. The recovered equation is therefore both sparse and physically interpretable, even though the thermal balance employed by the reference model was not supplied to the algorithm.

20.4 Feedforward neural-network model

The FFN provides a black-box alternative to sparse equation identification. It approximates the same controlled vector field through

$$\hat{\theta} = \mathcal{N}_{\phi}(\theta, I), \quad (73)$$

where ϕ denotes the trainable parameters. The network has two inputs, two hidden layers containing 32 neurons each, hyperbolic-tangent activation functions and one linear output. Its architecture can be written as

$$2 \longrightarrow 32 \longrightarrow 32 \longrightarrow 1, \quad (74)$$

corresponding to 1185 trainable parameters.

The derivative targets are constructed independently for every trajectory:

$$\dot{\theta}_k^{(j)} \approx \text{gradient} \left(\theta^{(j)}, t^{(j)} \right)_k, \quad (75)$$

using a second-order numerical gradient. The two input variables and derivative target are standardized. To prevent information leakage, both input and target scalers are fitted only on the union of training trajectories 0–4 and are subsequently applied without refitting to validation and test trajectories.

The FFN is trained by minimizing the mean-squared error in standardized derivative space,

$$\mathcal{L}_{\text{FFN}} = \frac{1}{N_{\text{tr}}} \sum_{n=1}^{N_{\text{tr}}} \left| \tilde{\theta}_n - \mathcal{N}_{\phi} \left(\tilde{\theta}_n, \tilde{I}_n \right) \right|^2. \quad (76)$$

Optimization is performed with Adam, a learning rate of 10^{-3} , a batch size of 512 and a maximum of 2000 epochs. The parameters associated with the lowest validation loss are restored, and training is stopped if the validation loss does not improve for 200 consecutive epochs.

No thermal balance, sparse-library structure or explicit physics residual is included in Eq. (76). The quantity denoted as physical derivative MSE in the numerical implementation is the derivative error after inverse scaling, expressed in $\text{K}^2 \text{s}^{-2}$; it is a diagnostic metric rather than an additional physics-informed loss.

20.5 Common rollout and temperature reconstruction

Pointwise derivative accuracy does not by itself guarantee an accurate temperature trajectory. Consequently, both identified vector fields are evaluated through an autonomous rollout driven only by the prescribed current profile:

$$\frac{d\hat{\theta}}{dt} = \hat{f} \left(\hat{\theta}(t), I(t) \right), \quad \hat{\theta}(0) = T(0) - T_{\text{amb}}. \quad (77)$$

Here, $\hat{f}$ is either the sparse SINDYc equation or the trained FFN. The discrete current samples are linearly interpolated during integration. Both models are integrated with LSODA using relative and absolute tolerances of 10^{-8} and 10^{-10} , respectively. The FFN receives the current predicted state at every solver evaluation; the reference temperature is used only to set the initial condition and to calculate the final metrics.

Because the thermal model is lumped, the reconstructed scalar trajectory is replicated over the radial grid:

$$\hat{T}(\rho_i, t_k) = \hat{T}(t_k), \quad i = 1, \dots, N_r. \quad (78)$$

The resulting maps are visual representations of a spatially uniform thermal state and should not be interpreted as predictions of radial heat conduction.

20.6 Evaluation metrics

The trajectory error is characterized through RMSE, MAE and maximum absolute error. Because the absolute temperature is dominated by the ambient offset, the relative L^2 error is evaluated on the temperature deviation:

$$\varepsilon_{L^2, \theta} = \frac{\|\hat{\theta} - \theta_{\text{ref}}\|_2}{\|\theta_{\text{ref}}\|_2}. \quad (79)$$

Additional diagnostics include the peak-temperature error, peak-time error, RMSE during maximum-current operation, cooling-stage RMSE, final-temperature error, and RMSE and MAE of the temperature derivative. Training or identification time, rollout inference time and model complexity are also reported. The SINDYc complexity is the number of active library terms, whereas the FFN complexity is the number of trainable parameters.

20.7 Interpolative test results

Table 29 reports the results on unseen trajectory 6. Both models reconstruct the temperature with small absolute errors. The FFN reaches an RMSE of 6.91×10^{-3} K and a maximum error of 1.16×10^{-2} K, demonstrating accurate interpolation. SINDYc is nevertheless more accurate for every reported metric, with a temperature RMSE approximately 20.3 times smaller than that of the FFN.

Table 29: Thermal-model metrics on the interpolative test trajectory.

Metric	SINDYc	FFN
Temperature RMSE [K]	3.3981×10^{-4}	6.9088×10^{-3}
Temperature MAE [K]	2.4985×10^{-4}	5.0773×10^{-3}
Maximum error [K]	8.7669×10^{-4}	1.1593×10^{-2}
Relative L^2 on θ	1.0263×10^{-4}	2.0866×10^{-3}
Peak-temperature error [K]	6.2663×10^{-6}	9.0167×10^{-3}
Peak-time error [s]	0	0
Peak-current RMSE [K]	4.1724×10^{-4}	9.7556×10^{-3}
Cooling RMSE [K]	2.2675×10^{-4}	6.5103×10^{-4}
Final-temperature error [K]	7.9096×10^{-5}	1.2358×10^{-3}
Derivative RMSE [K s^{-1}]	3.0900×10^{-6}	2.2267×10^{-5}
Derivative MAE [K s^{-1}]	1.2570×10^{-6}	1.6928×10^{-5}

The zero peak-time error indicates that the predicted and reference maxima occur at the same sampled time. It should be interpreted relative to the temporal resolution of the dataset and not as proof that the continuous peak times are exactly identical.

20.8 Extrapolative test results

Trajectory 7 evaluates the models beyond the maximum current and temperature covered by the training data. The resulting metrics are given in Table 30. SINDYc maintains approximately the same relative accuracy observed in interpolation. Its relative L^2 error changes from 1.0263×10^{-4} to 1.0844×10^{-4} , an increase of only approximately 5.7%.

The FFN remains stable and produces a physically plausible rollout, but its temperature RMSE increases from 6.91×10^{-3} to 7.95×10^{-2} K, a factor of approximately 11.5. The largest discrepancy occurs during the high-current heating stage, where the FFN slightly underestimates the temperature rise. During cooling, the prediction returns close to the reference trajectory.

Table 30: Thermal-model metrics on the extrapolative test trajectory.

Metric	SINDYc	FFN
Temperature RMSE [K]	4.9426×10^{-4}	7.9459×10^{-2}
Temperature MAE [K]	4.0328×10^{-4}	5.4771×10^{-2}
Maximum error [K]	1.0331×10^{-3}	1.3522×10^{-1}
Relative L^2 on θ	1.0844×10^{-4}	1.7432×10^{-2}
Peak-temperature error [K]	7.0909×10^{-5}	1.1227×10^{-1}
Peak-time error [s]	0	0
Peak-current RMSE [K]	5.9842×10^{-4}	1.2084×10^{-1}
Cooling RMSE [K]	4.0332×10^{-4}	5.0187×10^{-3}
Final-temperature error [K]	8.1410×10^{-5}	8.2387×10^{-4}
Derivative RMSE [K s^{-1}]	2.0568×10^{-6}	2.3533×10^{-4}
Derivative MAE [K s^{-1}]	1.1682×10^{-6}	1.5873×10^{-4}

The direct trajectory comparisons are shown in Figure 29. The three curves are nearly indistinguishable in interpolation. In extrapolation, the SINDYc prediction remains superimposed on the reference, while the small FFN underprediction is visible around the maximum-temperature region.

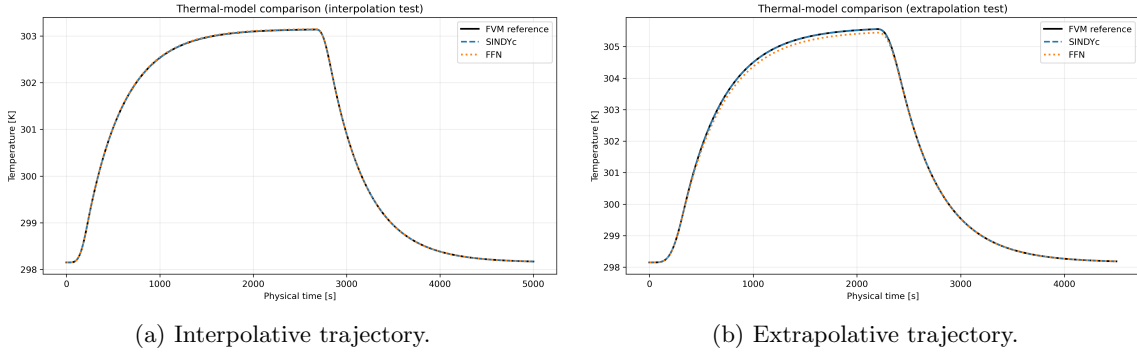


Figure 29: Temperature trajectories reconstructed by SINDYc and the FFN compared with the full-reference thermal state.

The temperature maps reconstructed through SINDYc are shown in Figure 30. In both cases, the absence of radial variation is the expected consequence of the lumped thermal formulation.

The interpolative trajectory reaches approximately 303.14 K, whereas the extrapolative trajectory reaches approximately 305.56 K because of its larger applied current. Despite this higher thermal excursion, the SINDYc rollout remains almost indistinguishable from the reference trajectory, as quantified by the sub-millikelvin errors in Table 30. The maps therefore illustrate generalization in time and operating conditions, not the emergence of a spatial temperature distribution.

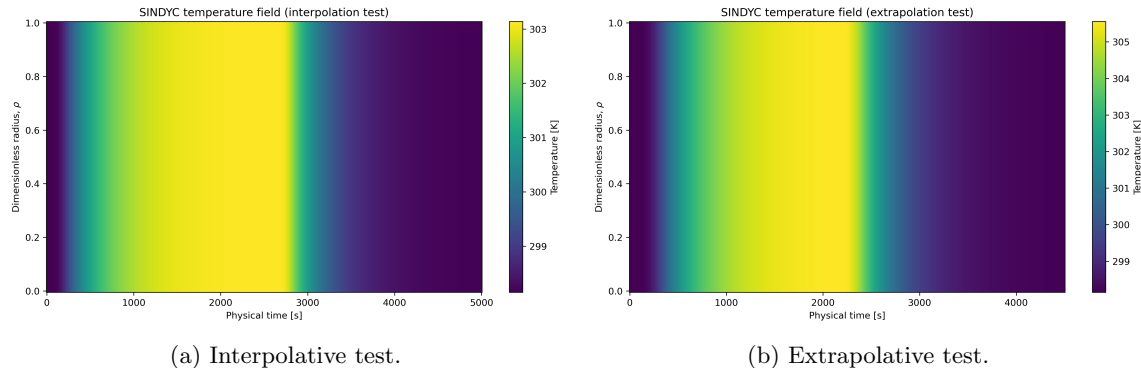


Figure 30: Spatially uniform temperature maps reconstructed by integrating the sparse thermal equation identified through SINDYc.

The corresponding FFN temperature maps are reported in Figure 31. Their uniformity along ρ follows from Eq. (78); only their temporal evolution is predicted by the network.

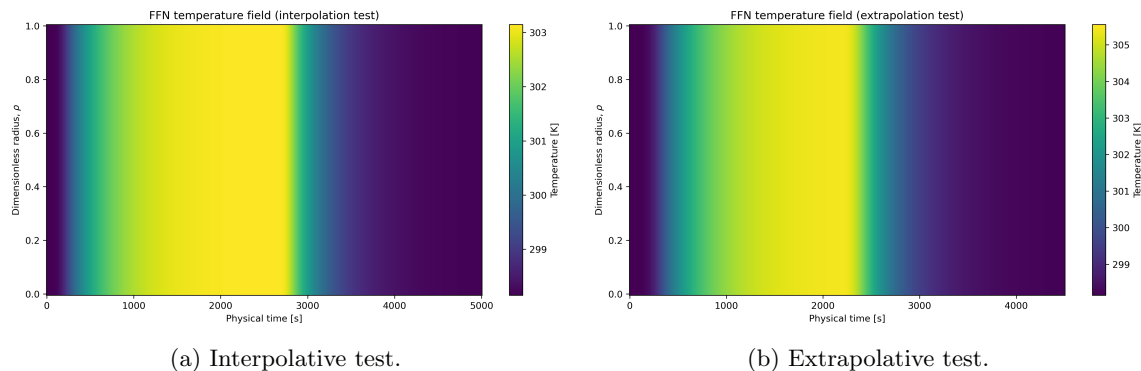


Figure 31: Spatially uniform temperature maps reconstructed through the FFN thermal rollout.

20.9 Computational cost and model complexity

Table 31 compares the measured computational costs. Identification or training was repeated when producing the two test outputs; the reported range therefore reflects the two observed runs. SINDYc requires approximately 5.2 s for sparse identification, whereas FFN training requires approximately 34 s. The sparse model also provides the faster rollout.

Table 31: Computational cost and complexity of the thermal models.

Quantity	SINDYc	FFN
Identification/training time [s]	5.21–5.23	33.87–33.98
Interpolation rollout time [s]	1.034	2.427
Extrapolation rollout time [s]	1.294	2.873
Active terms/trainable parameters	2	1185

These timings are implementation- and hardware-dependent. SINDYc evaluates a two-term algebraic expression on the CPU, whereas the FFN was executed on the Apple MPS device and repeatedly called by an adaptive scalar ODE solver. For a network of this size, device-transfer and kernel-launch overhead can represent a substantial part of the FFN rollout time. The measured values therefore describe the present implementation and should not be interpreted as a universal timing comparison between sparse models and neural networks.

20.10 Comparative assessment and limitations

The multi-trajectory experiment provides a substantially stronger assessment than a reconstruction performed on the same trajectory used for identification. Both models are now tested on completely held-out current profiles, and the extrapolative trajectory explicitly leaves the training range.

For the current toy problem, SINDYc provides the most effective thermal model. It is more accurate in both test regimes, generalizes almost unchanged beyond the training current, requires fewer computational resources and yields an explicit equation containing only heat-removal and Joule-heating terms. This outcome is consistent with the simple structure of the reference thermal balance and with the fact that the second-order polynomial library contains the correct terms θ and I^2 .

The FFN should nevertheless not be regarded as unsuccessful. Its interpolation errors are of millikelvin order, and even in extrapolation its maximum error remains below 0.136 K. Moreover, the FFN does not require the user to enumerate candidate functions in advance. This flexibility may become advantageous when the thermal dynamics include higher-dimensional states, non-polynomial effects, hysteresis or partially unknown couplings that cannot be represented by a compact predefined library.

Conversely, the present results do not establish that SINDYc will retain its advantage for more complex systems. Sparse identification depends on the quality of derivative estimates and on the inclusion of appropriate functions in the candidate library. When the required terms are absent, the resulting equation can be sparse but structurally misspecified. A broader conclusion on the relative merits of the two approaches would therefore require noisy data, additional operating conditions and reference thermal models of increasing complexity.

Within the present scope, SINDYc is retained as the preferred thermal extension because it recovers a compact, interpretable and highly generalizable equation. The FFN is retained as a complementary black-box benchmark that quantifies the accuracy achievable without prior selection of a function library. Together, the two approaches clarify the trade-off between structural interpretability and representational flexibility in the data-driven reconstruction of physics omitted from the simplified electrochemical model.

21 Post-Training Residual Correction of the Physics-Informed Neural Network

21.1 Motivation and Position within the Modelling Framework

The standalone PINN in Sections 15 and 16 provides a continuous approximation $\hat{C}_\theta(\rho, \tau)$ of the concentration field while retaining the exact initial condition and centre symmetry. Its remaining error is nevertheless systematic: it grows when the constant surface flux imposed in the simplified PINN differs from the time-dependent operating flux of the thermo-diffusive reference model.

This section introduces a post-training residual correction of that PINN. The selected PINN checkpoint is loaded, placed in evaluation mode and frozen. Its parameters are never updated during the corrective stage. A separate feedforward neural network is then trained to predict the concentration discrepancy. Correcting a misspecified PINN with an additional neural representation of its discrepancy has direct precedent in the model-correction literature [Zou et al., 2024]:

$$\Delta C_{\text{ref}}(\rho, \tau) = C_{\text{FVM}}(\rho, \tau) - \hat{C}_\theta(\rho, \tau). \quad (80)$$

The resulting Hybrid-PINN is

$$C_{\text{HP}}(\rho, \tau) = \hat{C}_\theta(\rho, \tau) + \widehat{\Delta C}_\phi(\rho, \tau, \hat{C}_\theta(\rho, \tau)). \quad (81)$$

This separation makes the workflow modular. The expensive PINN optimization is not repeated when the residual model is trained, tested or replaced. It also preserves the standalone PINN as an unchanged baseline, enabling a direct comparison between the uncorrected and corrected fields. More generally, physics-guided hybrid models seek this division of labour: the mechanistic component supplies structure, whereas the learning component absorbs systematic inadequacy [Jia et al., 2020; Garg et al., 2022].

The present Hybrid-PINN is different from the FVM–FNN hybrid model of Section 17. The latter corrects a simplified finite-volume solver and returns an entire radial vector. The present model instead corrects a frozen, coordinate-based PINN and returns one scalar residual for each (ρ, τ) pair.

21.2 Frozen PINN Baseline and Residual Dataset

The residual-learning script restores the selected PINN checkpoint and applies

$$\text{requires_grad}(\theta) = \text{False}. \quad (82)$$

The reference concentration dataset contains the common $101 \times 201 = 20\,301$ structured space–time points. The same 70%–15%–15% pointwise split and fixed random seed used for the PINN data are retained. For each subset, the frozen PINN is evaluated without gradient tracking and the residual samples are assembled as

$$\mathbf{x}_{\text{res}} = [\rho, \tau, \hat{C}_\theta(\rho, \tau)], \quad (83)$$

$$y_{\text{res}} = C_{\text{FVM}}(\rho, \tau) - \hat{C}_\theta(\rho, \tau). \quad (84)$$

The PINN concentration is included as a state-aware feature. Therefore, the residual network is not asked to infer the baseline response again from the coordinates alone; it receives the value that must be corrected at the same point.

Training batches are shuffled, whereas validation and test batches retain deterministic ordering. The adopted batch size is 512. Because the split is pointwise and all subsets belong to the same operating trajectory, the held-out subset measures interpolation within the available trajectory. It does not establish generalization to unseen current profiles, initial concentrations or thermal conditions.

21.3 Residual-Network Architecture and Optimization

The residual network is a compact fully connected neural network. Its effective input is

$$\left[\rho^2, \tau, \widehat{C}_\theta\right], \quad (85)$$

followed by four 32-neuron hidden transformations with hyperbolic-tangent activation and a scalar linear output. The architecture contains 3,329 trainable parameters, compared with 30,701 parameters in the frozen PINN. Their roles are compared in Table 32.

Table 32: Structural comparison of the frozen PINN baseline and the trainable residual FFN.

Property	Frozen PINN	Residual FFN
Effective input	(ρ^2, τ)	$(\rho^2, \tau, \widehat{C}_\theta)$
Output	Concentration C_P	Signed correction $\widehat{\Delta C}_\phi$
Hidden architecture	Four layers, 100 neurons each	Four layers, 32 neurons each
Activation	tanh	tanh
Parameters	30,701, frozen	3,329, trainable
Training objective	Data, PDE and boundary terms	Residual MSE and soft box penalty

If the raw network output is denoted by $\mathcal{N}_\phi$, the applied residual is

$$\widehat{\Delta C}_\phi = \tau \mathcal{N}_\phi \left(\rho^2, \tau, \widehat{C}_\theta \right). \quad (86)$$

The multiplicative factor τ forces the correction to vanish at the initial time. The use of ρ^2 , together with the identical transformation used by the PINN, makes the radial derivative of the corrected field vanish at the particle centre.

The network is optimized with Adam using an initial learning rate of 10^{-3} , a maximum of 3,000 epochs and early-stopping patience of 300 epochs. The retained parameters are those associated with the smallest validation objective. The residual-fitting term is

$$\mathcal{L}_{\text{res}} = \frac{1}{N} \sum_{i=1}^N \left[\widehat{\Delta C}_{\phi,i} - \Delta C_{\text{ref},i} \right]^2. \quad (87)$$

To discourage normalized concentrations outside the physical interval $[0, 1]$, a soft box-constraint penalty is added to the raw corrected prediction $C_{\text{raw}} = \widehat{C}_\theta + \widehat{\Delta C}_\phi$:

$$\mathcal{L}_{\text{box}} = \frac{1}{N} \sum_{i=1}^N [\text{ReLU}(-C_{\text{raw},i})^2 + \text{ReLU}(C_{\text{raw},i} - 1)^2]. \quad (88)$$

The optimized objective is

$$\mathcal{L}_{\text{HP}} = \mathcal{L}_{\text{res}} + \mathcal{L}_{\text{box}}. \quad (89)$$

The residual output is signed: it can either increase or decrease the PINN concentration. A rectifier is therefore used only to measure violations in the box penalty; it is not applied to $\widehat{\Delta C}_\phi$ itself.

This penalty is differentiable almost everywhere and leaves admissible predictions unchanged. It is nevertheless a soft term and does not mathematically guarantee the bounds. No hard clipping is applied to the residual or to the final Hybrid-PINN concentration.

21.4 Inherited and Non-Inherited Physical Structure

The transformations in the two neural components imply

$$C_{\text{HP}}(\rho, 0) = C_0, \quad \left. \frac{\partial C_{\text{HP}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (90)$$

These two properties are inherited exactly from the architecture. Other physical properties are not inherited automatically. In particular, pointwise residual fitting does not enforce:

- global lithium conservation under the reference flux history;
- the full-physics flux condition at $\rho = 1$;
- the variable-diffusivity thermo-diffusive PDE;
- strict concentration bounds for every possible input.

These quantities are therefore evaluated as diagnostic checks after training. They are not added to the residual-network loss in the present implementation. This distinction is essential: agreement with the checks is empirical evidence of physical consistency, not a hard guarantee supplied by the corrective architecture.

21.5 Direct Comparison with the Standalone PINN

The numerical values below correspond to the retained reference run. After residual training, the train, validation and test coordinates are reassembled in their original structured-grid order. The dense-grid errors therefore quantify reconstruction over the complete available trajectory; they must not be read as independent out-of-distribution test errors.

Table 33 compares the frozen PINN and Hybrid-PINN with the full-physics FVM field.

The relative full-field error decreases from 5.0674×10^{-2} to 9.5238×10^{-4} . Defining

$$\text{RER}_{\text{HP}} = \left(1 - \frac{\varepsilon_{L^2, \text{HP}}}{\varepsilon_{L^2, \text{PINN}}} \right) 100, \quad (91)$$

Table 33: Dense-grid errors of the standalone PINN and the post-training Hybrid-PINN with respect to the full-physics reference.

Model	Quantity	Relative L^2	RMSE	MAE	Maximum absolute error
PINN	Complete field	5.0674×10^{-2}	3.7757×10^{-2}	2.7721×10^{-2}	1.6061×10^{-1}
	Particle centre	4.5458×10^{-2}	3.3251×10^{-2}	2.6263×10^{-2}	7.3184×10^{-2}
	Particle surface	6.8609×10^{-2}	5.2938×10^{-2}	3.4088×10^{-2}	1.6061×10^{-1}
	Volume average	5.3791×10^{-2}	4.0645×10^{-2}	2.8642×10^{-2}	1.2448×10^{-1}
Hybrid-PINN	Complete field	9.5238×10^{-4}	7.0962×10^{-4}	3.9662×10^{-4}	1.2054×10^{-2}
	Particle centre	6.1086×10^{-4}	4.4682×10^{-4}	3.6112×10^{-4}	1.1427×10^{-3}
	Particle surface	2.9153×10^{-3}	2.2495×10^{-3}	1.2578×10^{-3}	1.2054×10^{-2}
	Volume average	7.4503×10^{-4}	5.6295×10^{-4}	3.8110×10^{-4}	2.4472×10^{-3}

the reduction is 98.12%. Equivalently, the relative error is reduced by a factor of approximately 53.2. The largest remaining discrepancy is at the particle surface, which is also the location where the omitted time-dependent boundary physics acts directly.

Table 34 reports the full-field error over the four operating phases.

Table 34: Full-field error by operating phase before and after the post-training PINN correction.

Phase	PINN		Hybrid-PINN	
	Relative L^2	RMSE	Relative L^2	RMSE
Ramp-up	9.5419×10^{-3}	4.7915×10^{-3}	4.6664×10^{-3}	2.3433×10^{-3}
Constant current	3.4402×10^{-2}	2.3050×10^{-2}	7.1251×10^{-4}	4.7739×10^{-4}
Ramp-down	5.9515×10^{-2}	5.0262×10^{-2}	9.0556×10^{-4}	7.6476×10^{-4}
Rest	6.2203×10^{-2}	5.3651×10^{-2}	4.7456×10^{-4}	4.0932×10^{-4}

The correction is particularly effective during ramp-down and rest, where the standalone PINN continues to respond to its constant-flux boundary condition while the reference current decreases and then vanishes. The centre, surface and volume-average histories, together with representative radial profiles, are shown in Figure 33.

21.6 Physical-Consistency Diagnostics

21.6.1 Concentration Bounds and Exact Constraints

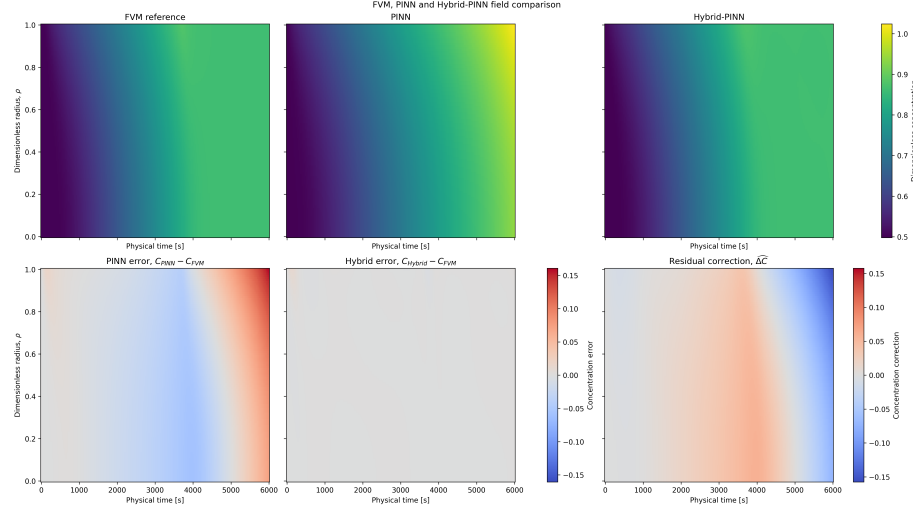
The corrected concentration range on the complete grid is

$$0.497915 \leq C_{\text{HP}} \leq 0.876149. \quad (92)$$

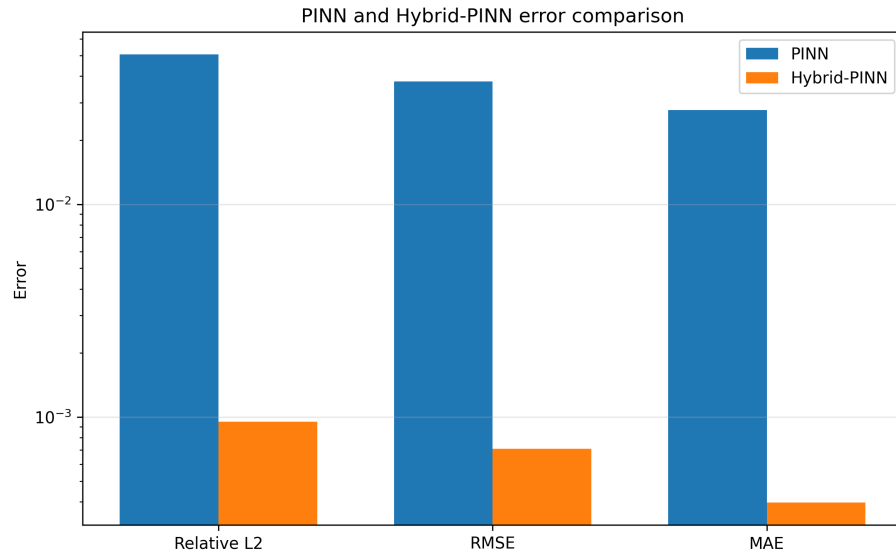
It therefore remains inside the normalized physical interval $[0, 1]$ for the retained run. The standalone PINN instead reaches 1.024085. The Hybrid-PINN result is a successful empirical outcome of the residual fit and soft box penalty, not an unconditional guarantee for unseen inputs.

The maximum initial-condition error and centre derivative are both zero to the reported numerical precision:

$$\max_{\rho} |C_{\text{HP}}(\rho, 0) - C_0| = 0, \quad \max_{\tau} \left| \frac{\partial C_{\text{HP}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (93)$$

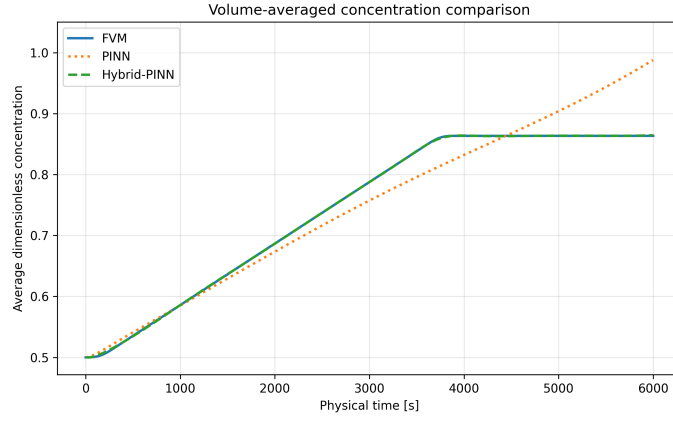


(a) Full concentration fields.

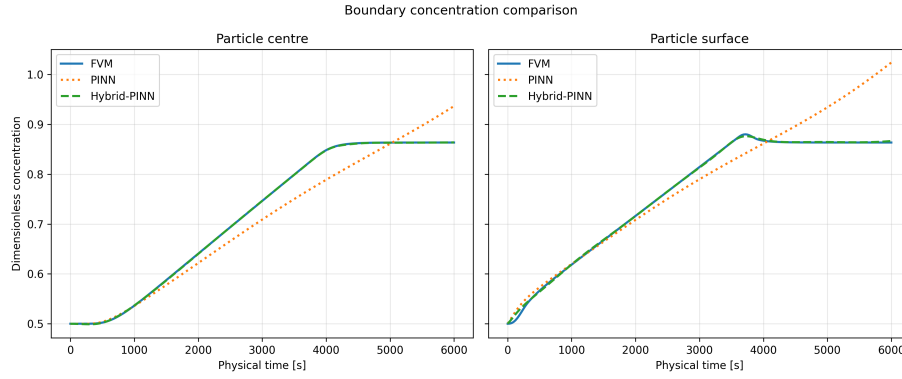


(b) Error metrics.

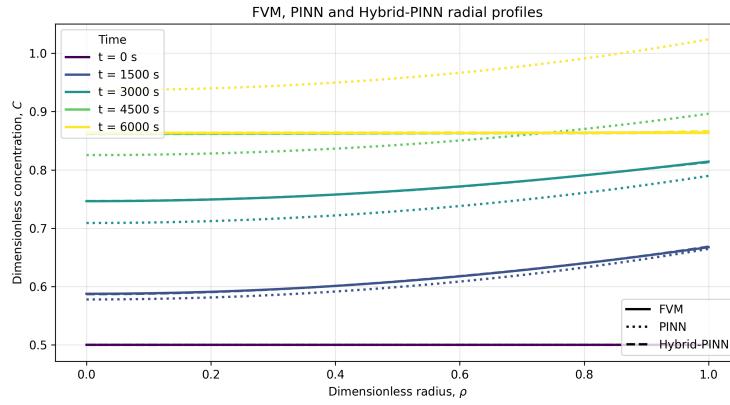
Figure 32: Direct comparison of the frozen PINN and post-training Hybrid-PINN with the full-physics reference model.



(a) Volume-averaged concentration.



(b) Centre and surface concentrations.



(c) Radial profiles at selected times.

Figure 33: Integral, boundary and radial comparison of the PINN and Hybrid-PINN predictions with the reference solution.

These two checks confirm the exact architectural properties in Eq. (90).

21.6.2 Global Mass Balance

For all three fields, mass consistency is evaluated against the time-dependent dimensionless flux stored in the full reference dataset:

$$\overline{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho, \quad (94)$$

$$\overline{C}_{\text{expected}}(\tau) = C_0 - 3 \int_0^\tau \delta(s) ds, \quad (95)$$

$$R_M(\tau) = \overline{C}(\tau) - \overline{C}_{\text{expected}}(\tau). \quad (96)$$

Table 35 reports the resulting checks.

Table 35: Mass-balance diagnostics evaluated using the full-reference time-dependent flux.

Field	Maximum $ R_M $	RMSE	Final residual	Relative mass-change error
FVM reference	4.4670×10^{-5}	1.0115×10^{-5}	1.2627×10^{-6}	3.4739×10^{-6}
PINN	1.2448×10^{-1}	4.0642×10^{-2}	1.2448×10^{-1}	3.4248×10^{-1}
Hybrid-PINN	2.4182×10^{-3}	5.5686×10^{-4}	1.0487×10^{-3}	2.8852×10^{-3}

The residual correction reduces the mass discrepancy by approximately two orders of magnitude relative to the PINN, even though conservation is not imposed in the FFN loss. The maximum residual remains larger than the 10^{-4} diagnostic tolerance and the Hybrid-PINN therefore does not pass the strict mass-balance check. This result must be reported as an empirical improvement rather than exact conservation.

21.6.3 Full-Physics Surface-Flux Condition

The full thermo-diffusive boundary condition is evaluated using

$$\widehat{D}(C, T) = \exp[\alpha(C - C_0)] \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad (97)$$

with $\alpha = 1$, $E_a = 35\,000 \text{ J mol}^{-1}$, $R_g = 8.314462618 \text{ J mol}^{-1}\text{K}^{-1}$ and $T_{\text{ref}} = 298.15 \text{ K}$. The residual is

$$R_{\text{BC}}(\tau) = -\widehat{D}(C_{\text{HP},s}, T) \left. \frac{\partial C_{\text{HP}}}{\partial \rho} \right|_{\rho=1} - \delta(\tau). \quad (98)$$

Automatic differentiation is used for the surface gradient, while temperature and dimensionless flux are read from the full reference dataset. The global RMSE is 2.4816×10^{-2} , the maximum absolute residual is 1.1902×10^{-1} , and the final residual is -3.7961×10^{-2} . The phase-resolved values are reported in Table 36.

The concentration correction is therefore highly accurate without exactly recovering the full surface differential law. This is consistent with its training objective, which contains concentration targets but no surface-flux residual.

Table 36: Hybrid-PINN full-physics surface-flux residual by operating phase.

Phase	RMSE	Maximum $ R_{BC} $
Ramp-up	7.1101×10^{-2}	1.1902×10^{-1}
Constant current	1.4760×10^{-2}	3.8614×10^{-2}
Ramp-down	3.4270×10^{-2}	4.5953×10^{-2}
Rest	2.1912×10^{-2}	3.9015×10^{-2}

21.6.4 Full and Simplified PDE Consistency

The Hybrid-PINN is evaluated against the full reference equation

$$R_{\text{PDE}}^{\text{full}} = \frac{\partial C_{\text{HP}}}{\partial \tau} - \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C_{\text{HP}}, T) \frac{\partial C_{\text{HP}}}{\partial \rho} \right], \quad (99)$$

and, separately, against the constant-diffusivity equation used by the standalone PINN:

$$R_{\text{PDE}}^{\text{simpl}} = \frac{\partial C_{\text{HP}}}{\partial \tau} - \frac{\partial^2 C_{\text{HP}}}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial C_{\text{HP}}}{\partial \rho}. \quad (100)$$

The 19,800 points satisfying $0 < \rho < 1$ and $\tau > 0$ are used. The centre and surface are excluded from the direct evaluation to avoid the apparent spherical-coordinate singularity and to keep the boundary checks separate. Results are summarized in Table 37.

Table 37: Hybrid-PINN differential-equation consistency diagnostics.

Residual	Phase	RMSE	Maximum absolute value
Full physics	Complete interior	5.9839×10^{-2}	5.3836×10^{-1}
	Ramp-up	1.2435×10^{-1}	5.3836×10^{-1}
	Constant current	5.0012×10^{-2}	2.6283×10^{-1}
	Ramp-down	5.5133×10^{-2}	3.1563×10^{-1}
	Rest	6.1003×10^{-2}	3.2896×10^{-1}
Simplified physics	Complete interior	1.3705×10^{-1}	4.9865×10^{-1}
	Ramp-up	1.1351×10^{-1}	4.9865×10^{-1}
	Constant current	1.6677×10^{-1}	3.8466×10^{-1}
	Ramp-down	1.9448×10^{-1}	2.8531×10^{-1}
	Rest	5.3412×10^{-2}	2.7771×10^{-1}

Neither residual is a training constraint for the FFN. The full-physics residual is smaller in global RMSE than the simplified residual, which is compatible with the correction moving the concentration field toward the reference dynamics. However, the non-zero values prevent claiming that the Hybrid-PINN is a direct solution of the full PDE. They quantify differential consistency of a data-corrected surrogate.

21.6.5 Numerical Regularity

All Hybrid-PINN concentrations, evaluated derivatives and PDE residuals are finite. The maximum derivative magnitudes over the interior evaluation set are

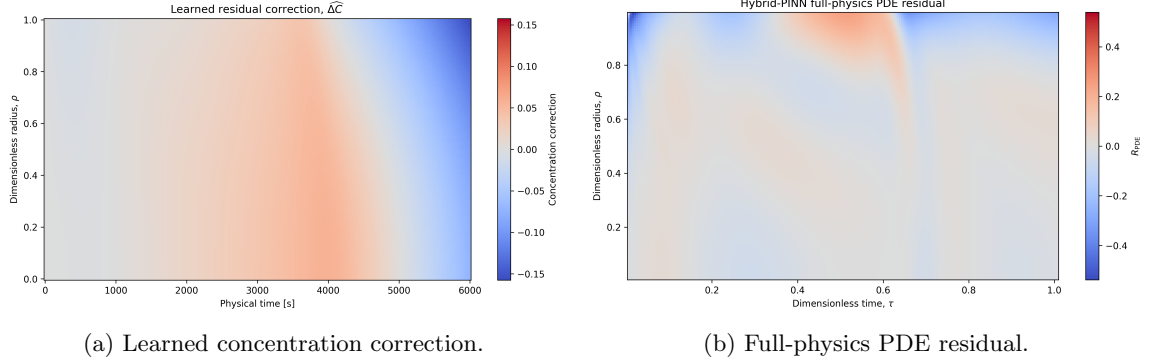


Figure 34: Space–time maps of the Hybrid-PINN residual correction and its a posteriori full-physics differential residual.

$$\max \left| \frac{\partial C_{\text{HP}}}{\partial \rho} \right| = 0.17963, \quad \max \left| \frac{\partial C_{\text{HP}}}{\partial \tau} \right| = 0.92585, \quad \max \left| \frac{\partial^2 C_{\text{HP}}}{\partial \rho^2} \right| = 1.1283. \quad (101)$$

These values are not independent physical constraints, but the absence of NaN, infinite derivatives and numerical blow-up confirms that the smooth tanh correction does not introduce an obvious instability on the evaluated grid.

21.7 Computational Cost and Corrected Dataset

Table 38 reports the synchronized timings of the retained run.

Table 38: Timing of the post-training Hybrid-PINN correction pipeline.

Operation	Time [s]
Frozen PINN inference for residual-dataset construction	0.173452
Residual-network training	147.040652
Residual inference and full-grid correction	0.033015
Total correction pipeline	147.247118

The timing excludes the original PINN training because that model is loaded from an existing checkpoint. It also excludes plotting and the a posteriori physical checks. Consequently, the reported total measures the additional cost required to construct the residual data, train the FFN and generate the corrected field.

The corrected dataset preserves the physical quantities stored by the reference FVM— ρ , τ , physical time, current, intercalation flux, dimensionless flux and temperature—and appends:

- the reference and PINN concentrations;
- the exact reference residual and learned residual correction;

- the Hybrid-PINN concentration;
- the full-physics and simplified-physics PDE residuals at the admissible interior points.

This stored dataset decouples visualization and comparison from training. Hybrid-PINN plots can therefore be regenerated without retraining either neural model.

21.8 Assessment and Limitations

The post-training correction achieves the intended numerical objective. It reduces the full-field relative error by 98.12%, removes the late-time concentration overshoot, reconstructs centre, surface and volume-average quantities, and preserves the initial and centre conditions exactly. The result demonstrates that a frozen PINN can be treated as a reusable physics-informed baseline rather than retrained as part of a monolithic hybrid script.

The physical checks also refine the interpretation of this accuracy. The corrected field remains within the normalized concentration bounds and improves mass balance substantially, but conservation is not exact. Similarly, the full surface-flux and PDE residuals remain non-zero because they are diagnostic quantities rather than terms in the FFN objective. The Hybrid-PINN should therefore be described as a high-accuracy, data-corrected surrogate with inherited hard constraints, not as a network that exactly solves the complete thermo-diffusive problem.

Finally, the residual model is trained and evaluated on samples from one operating trajectory. The random pointwise split supports interpolation inside that trajectory but does not demonstrate transfer to a new current history. A stronger generalization study would require several independent operating cycles and trajectory-wise splitting, with entire current profiles reserved for validation and testing. Within the scope of the present experiment, the Hybrid-PINN provides a clear and quantified extension of the standalone simplified-physics PINN.

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Appendices

A Algorithm[1] – Constant diffusion coefficient

Algorithm 1: Reference finite-volume solver

Input: Physical parameters, operating profile, mesh size N , stored time instants N_τ , solver tolerances.

Output: Reference concentration field, validation quantities and PINN dataset.

1. Read the physical and numerical parameters.
2. Compute $t_D = R_p^2/D_s$ and generate the dimensionless grids ρ and τ .
3. Build the finite-volume mesh and initialize $C(\rho, 0) = C_0$.
4. Define the applied current profile $I(t)$ and compute

$$j(t) = -\frac{I(t)}{FS_p}, \quad \delta(\tau) = \frac{R_p j}{D_s c_{s, \max}}.$$

5. Assemble the semi-discrete finite-volume system

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}).$$

6. Integrate the ODE system using the BDF solver.
7. Extract $C_s(\tau)$ and $\bar{C}(\tau)$.
8. Verify the mass balance and physical bounds.
9. Construct the reference dataset

$$(\rho, \tau, t, I, j, \delta, C).$$

10. Generate the PINN training dataset and the validation figures.
-

B Algorithm[2] – Thermo-diffusive model

Algorithm 2: Thermo-diffusive finite-volume solver

Input: Physical, thermal and numerical parameters, operating profile, mesh size N , stored time instants N_τ , solver tolerances.

Output: Reference concentration field, temperature evolution, validation quantities and PINN dataset.

1. Read the physical, thermal and numerical parameters.
2. Compute the diffusion time

$$t_D = \frac{R_p^2}{D_{s,\text{ref}}},$$

generate the dimensionless grids (ρ, τ) , build the finite-volume mesh and initialize

$$\mathbf{y}(0) = [C_0, \dots, C_0, T_{\text{ref}}]^T.$$

3. Define the applied current profile $I(t)$ and compute the intercalation and dimensionless fluxes.
4. At each ODE evaluation:

- (a) recover $t = \tau t_D$;
- (b) compute $D_s(C, T)$ and normalize it,

$$\hat{D}_i = \frac{D_{s,i}}{D_{s,\text{ref}}};$$

- (c) interpolate the interface diffusivities,

$$\hat{D}_{i+\frac{1}{2}} = \frac{\hat{D}_i + \hat{D}_{i+1}}{2};$$

- (d) assemble the concentration and thermal equations.

5. Integrate the coupled thermo-diffusive system using the BDF solver.
6. Extract

$$C(\rho, \tau), C_s(\tau), \overline{C}(\tau), T(\tau).$$

7. Verify mass conservation and physical bounds.
8. Construct the reference dataset

$$(\rho, \tau, t, I, j, \delta, C, T),$$

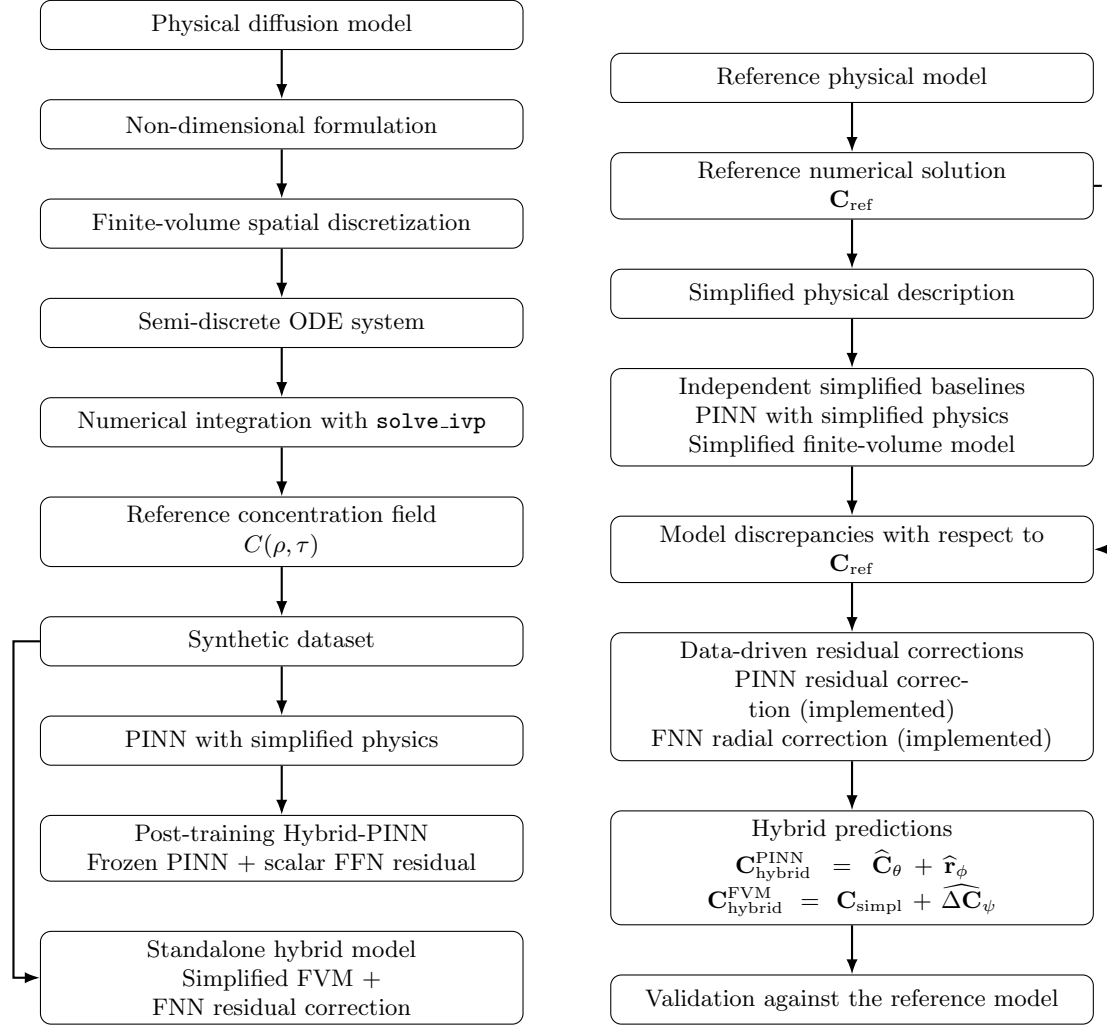
then generate the PINN dataset using only

$$(\rho, \tau, C),$$

discarding the temperature.

9. Generate the validation figures.
-

C Flowchart



(a) Reference-data generation and its use in the simplified-physics PINN and in the two residual-learning extensions.

(b) Common residual-learning methodology for the implemented Hybrid-PINN and the independently implemented standalone FVM-FNN hybrid model.

Figure 35: Overview of the numerical and methodological workflows adopted in the project. The left diagram describes the generation of the reference dataset and the resulting learning branches. The right diagram shows the common additive residual-learning structure of the implemented Hybrid-PINN and the independently implemented standalone FVM-FNN hybrid model.